

International Patenting at the European Patent Office (EPO):
Aggregate and Sectoral Filings

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Abstract:

This study analyzes the behavior of patent filings at the European Patent Office (EPO), and conducts forecasting exercises using the models of patenting behavior in the EPO. The study examines the behavior of total EPO patents as well as patents disaggregated by mode of filing, technological sector, and selected patent families. The study finds that research and development (R&D), patent protection levels, and patent filing fees, are significant factors explaining the demand for EPO patents. These factors not only motivate the propensity to file patents but also affect inventive efforts. In terms of forecasting performance, the study finds that a dynamic model augmented with R&D generally performs best (based on root mean squared percentage errors (RMSPE) as measures of forecast accuracy). The study illustrates with some sample forecasts for individual source countries.

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Introduction

Nations trade and invest physical capital in each other's markets. Indeed, the international economy has become much more interdependent through these trade and investment linkages. Less well understood and appreciated, however, is the increased interdependence due to the diffusion of technological ideas among nations. The international patent system and institutions governing intellectual property rights help support a formal *marketplace* for knowledge capital. Yet, the volume, direction, and underlying determinants of international patent flows have not been the subject of much inquiry.

International patent filings help indicate where technologies originate and to where they spread. Of course, international patenting data imperfectly capture the international flow of new knowledge capital, since not all new discoveries are patented; some are kept as trade secrets and some are not patentable (for example, mathematical ideas). The technologies that are patented, however, tend to form a pool of industrially applicable (or marketable) innovations and to have been perceived as profitable to patent.

Thus far, few studies exist that seek to explain patenting behavior. These studies generally treat the nation as the unit of analysis, focusing on its growth of patenting and on whether this growth is innovation driven or due to the strengthening of patent laws. What has not been addressed is the global breadth of patenting activities. Moreover, in examining patenting activities from national perspectives, the existing literature pays little or no attention to regional or multilateral patenting systems (among a bloc of nations), such as that of the *European Patent Office (EPO)*. Such systems are, in fact, relevant to explaining the worldwide growth and spread of patenting.

Two factors motivate this study. First, studying patenting from a regional or multilateral perspective is that it helps shed light on the role that institutions play in determining patenting behavior. The development of the EPO or the *Patent Cooperation Treaty (PCT)* has fundamentally changed the way inventors obtain patent protection, yet these institutional developments have not been formally incorporated into existing work. This study contributes to the literature by focusing on the EPO. This study examines the determinants of EPO patent filings and develops models for forecasting EPO filings.

A second motivation is that, for national and regional patent offices alike, the extent of patenting activity has implications for internal workload (processing applications, conducting searches and examinations, and so forth) and patent office revenues (which are determined, among other things, by the volume of filings and official fees). A better understanding of the underlying determinants of the *demand for patents* could better assist organizations like the EPO to price its services, project revenues, and make operational decisions. Improved projection of patenting demand could be useful in any work-sharing or revenue-sharing arrangements with national offices or with supranational offices. For instance, trends in Euro-direct filings versus Euro via PCT filings would be useful for coordination and workload planning between the EPO and the World Intellectual Property Office (WIPO) which administers the PCT.

This study is organized into two parts. Part I focuses on an analysis of patenting behavior in the EPO and Part II on forecasting EPO patent filings (and related filings). The purpose of Part I is to take a comprehensive look at the demand for EPO patents before developing in Part II a compact, more practical set of models for conducting forecasts. In Part I, the first section provides a brief literature review. We review studies based on survey data and statistical data, and critique

different approaches to modelling patenting behavior. Section 2 provides a theoretical model of international patenting. It derives an equation that can be estimated econometrically. The equation explains total EPO patenting (from various source countries to the EPO). Section 3 describes the data and some sample statistics. Section 4 presents the empirical results and analyses.

In Part II, we conduct some forecasting exercises with the basic patenting model. Three kinds of patent filings will be the subject of forecasts: first, *aggregate* patent filings at the EPO. By aggregate, we mean the sum of filings across technological fields. Furthermore, we will consider the breakdown of these filings by mode of filing; that is, whether a filing is a direct EPO filing or indirect (via the PCT system), and whether it is a priority filing or subsequent filing. The second kind of filing to be forecasted will be *sectoral* filings. By sector, we mean by field of technology (classified according to the EPO's Joint Cluster divisions). We will also consider these disaggregated filings by mode of filing (direct vs. indirect, first vs. subsequent filings). The third (and last) type of filing to be forecasted will be *patent family filings*. A patent family "is a group of patent filings that claim the priority of a single filing, including the original priority forming filing itself and any subsequent filings made throughout the world."¹ A distinct set of priority forming filings is used to index the set of patent families. From the data set on international patent family filings, we can select those families that contain a subsequent filing at the EPO (and/or other types of 'Blocs'). Section 1 of Part II provides an overview of the forecasting methodology. Sections 2, 3, and 4 are devoted separately to forecasts of aggregate filings, sectoral filings, and patent family filings respectively.

¹ Trilateral Statistical Report, 2002 edition, p. 2. First filings are applications that do not claim the priority of any previous filing, while subsequent filings constitute all other applications. The latter are usually made within one year of the first filings.

Finally, a concluding part will summarize the main results in both Parts I and II, and discuss some extensions for further study. The appendix provides technical details on the construction of a patent rights index and of the stocks of research and development (R&D) capital.

Overall, this study finds that EPO patenting is significantly driven by R&D activities, and that forecast accuracy is generally improved through the use of R&D along with dynamic terms representing lagged patenting. Forecast performance does vary somewhat, as one would expect, by technological field, by mode of filing, and by type of patent family. For the most part, however, the forecasts of autoregressive models augmented with R&D are generally quite close to actual patent filings (for example, 90% of actual filings or better).

Part I: Analysis

1. Literature Review

Relative to the literature at large on the economics of the patent system, there has been very little empirical work to date on the determinants of patenting, and none with a specific focus on regional patenting systems, such as that of the European Patent Office (EPO). There are studies on the impacts of the patent system (on innovation, trade, productivity, and welfare), but not very much on what drives patenting behavior.

In this paper, we identify fourteen studies which can be categorized in the following way. First, one set of studies is based on firm level *surveys* (interviewing managers as to why firms patent and as to how important patents and patent laws are to the firms); the second set is based on

statistical data sources (conducting regression analyses on patent data in order to infer the factors that influence patenting). The second set of studies are of two kinds. The older studies are based on *ad-hoc* models of patenting behavior, while the more recent ones are based on *optimizing models*. Under the ad-hoc approach, researchers postulate plausible factors that drive patenting behavior. (Often the variables they posit are similar to those derived more rigorously using theoretical microfoundations (of profit-maximizing firms, and so forth); however, there are subtle differences which do impact importantly upon the econometric estimation and interpretation of the results.) Under the optimizing approach, researchers derive a testable model using a decision-theoretic framework of innovation and patenting; that is, they study utility or profit-maximizing agents, who weigh the costs and benefits of their actions. The outcome of utility or profit maximization is a decision-rule (to patent or not to patent), which is then tested on the data. Finally, among the studies adopting the optimizing approach, there are two varieties: the first kind focuses on patenting behavior (i.e. *end in itself*), the second kind estimates a model of patenting but uses it to analyze other economic phenomena (i.e. *means to an end*). Given this way of organizing the literature, we shall discuss the ‘state of knowledge’ thus far on the determinants of patenting and point out some gaps in understanding.

A. Survey-Based Studies

The conventional wisdom is that firms demand patent protection in order to safeguard their intangible assets, which are easy to copy and distribute at nearly zero marginal cost (without other producers needing to incur any of the ‘sunk’ development costs). Infringement and imitation work

to dissipate the gains to firms and thereby (ex ante) reduce their incentives to innovate. Recent surveys have challenged head on whether patent protection is necessary to stimulate investment in invention and commercialization. The Levin et. al. (1987) survey of U.S. firms' patenting behavior reports findings which have generated much controversy – namely that firms do not, in general, regard patent protection as very important to protecting their competitive advantage (and thus to appropriating the returns to their investments). The idea is that firms have various alternative means (other than patenting) for appropriating the rewards to their innovations; for example, trade secrecy, lead time, reputation, sales and service effort, and moving quickly down the learning curve. Patent protection ranked low among these alternative means of appropriation. The study therefore questions previous understanding of what motivates patenting.

However, the result varies by industry. For sectors producing inorganic or organic chemicals, drugs, and plastic materials, patents as an instrument for appropriation rated high. These sectors are in general highly R&D intensive. Mansfield (1994) concurs in finding that patent protection is very important in pharmaceutical and chemical industries, moderately important in petrochemicals, machinery, and fabricated metals, and not important in the rest of his industrial sample.² Mansfield based his survey on 100 U.S. multinational firms in 12 industries in various countries (mostly developing).

The question then is, if a patent is not important as an instrument for appropriating the returns to innovation, why do firms patent (and patent a lot)? The survey by Cohen et. al. (1997) reports that firms have various reasons to patent – as a means to block rivals from patenting related inventions,

² Here, the term “important” refers to the impression that an invention would not have been developed without patent protection.

as strategic bargaining chips (in cross-licensing agreements), as a means to measure internal performance (of the firms' scientists and engineers), and so forth. Thus these various "other" factors are what primarily determines (or motivates) patenting, rather than the protection of their R&D investment returns.

Some criticisms can be made of these survey analyses. First, it would be useful to update the sectors under study to incorporate new industries which have emerged since the surveys were conducted. The biotechnology and software industries may, for example, provide interesting perspectives on the rationale for and importance of patenting. Secondly, the responses of firms (or their attitudes towards patents) may have been influenced by the patent regime in place. It would be useful to separate these two out. Thirdly, the responses of interviewees may not be fully comparable. One person's rating of 9 out of 10 may differ from another's. There is no anchor in the way that ratings are scaled. Thus it is difficult to tell whether the responses reflect differences in firm behavior or random errors. Finally, while the surveys are very time-consuming and commendable work, the information is based on U.S. firms' experiences. A similar comprehensive study for Europe and Asia, and so forth, would shed more light on patenting behavior (such as why firms patent globally and if so, why they choose certain routes (e.g. EPO, PCT, etc.)). Also, working with U.S. data alone may not produce as much variation in certain kinds of data. For example, the strength of patent regimes varies much more internationally than domestically, particularly since firms largely face similar laws within a nation (with the exception of laws governing patentable material, which may exclude certain industries from obtaining a patent). Thus, a study focusing only on the U.S. biases the results about the importance and determinants of patenting. Where patent regimes are very strong (as in the U.S.), patentees may take certain things for granted or find the

marginal valuation of patent rights to be lower (than in a different environment).

B. Statistical Data-Based Studies

B1. Ad-Hoc Approaches

Early work by Schiffel and Kitti (1978) was motivated by the fact that foreign patenting in the U.S., during the period 1963-73, grew at a faster pace than U.S. patenting abroad. This seemed to have created concerns about the loss of U.S. technological leadership, a conclusion which the authors challenged. Several equations were estimated: first, a regression of foreign patent applications in the U.S. on foreign gross exports to the U.S. and on domestic patent applications (i.e. by the foreigners in their own respective countries); secondly, a regression of foreign patent applications in Japan on foreign gross exports to Japan and on domestic patent applications; and thirdly, domestic patent applications on R&D expenditures and dummy variables for patent law changes (such as the introduction of deferred examination and changes in filing fees). These equations were estimated *separately* for nine countries (U.S., U.K., Germany, Switzerland, Sweden, Netherlands, Canada, and Japan) using time-series data. Hence a major problem with this study is that there were too few observations (approximately 12 per country). The patent data or the R&D data are not deflated or divided by something else (such as GDP or work force) to control for country size. Thus the issues raised and the contribution of this paper were novel, but the empirical methods crude. Nonetheless, the general findings are that gross exports tend to be an important explanatory factor. Thus, the rise in foreign filings in the U.S. reflected increased world trading opportunities,

and not a reduction in U.S. inventiveness vis-a-vis foreigners. The home (domestic) patent variable was assumed to be important because it reflects the size or ‘pool’ of innovations from which foreigners can file for patents abroad.³ However, this variable is of mixed statistical significance. The R&D variable is an important factor to some countries, but not to others; moreover, contemporaneous R&D is more important to patenting than a cumulative sum of the previous three years of R&D, which is somewhat surprising given that innovations are often the result of cumulative effort.

Bosworth (1980) improves upon the limited observations problem of Schiffel and Kitti by working with cross-sectional data. The dependent variable is U.S. outward patenting (in 50 countries). Bosworth adds destination GDP, destination GDP per capita, and the number of U.S. corporate subsidiaries in the destination as additional explanatory variables. GDP is used to measure market size, and GDP per capita the purchasing power of destination agents. The number of U.S. subsidiaries is used to measure the extent of technology transfer (but it would have been preferable to use investment data or asset position – that is, to use *value* data rather than *count* data to measure the extent of foreign direct investment). Bosworth also tries to measure the state of patent laws in the destination by gathering information on six features (duration of patent rights, date at which patent life begins, whether there is examination for novelty, compulsory licensing, years of non-working before a patent right is revoked or compulsory licensing issued, and whether the Lisbon amendment was adopted). These six features are entered individually as dummy variables. The

³ However, domestic patent applications do not represent the universe of domestic inventions; usually, the former are a subset of the latter. And in some countries (i.e. Canada) patent applications abroad to a particular country (say to the U.S.) exceed domestic patent applications.

explanatory variables are all significant except for the six patent regime features, which would suggest that patent rights do not affect U.S. patenting abroad. This is at odds with the strong advocacy U.S. firms have shown towards international intellectual property law reform. Later work (as described below) which improves upon the measurement of patent regimes do show the importance of patent rights to international patenting behavior (including U.S. patenting abroad). Bosworth (1984) repeats the analysis for patenting flows into and out of the U.K., and finds qualitatively similar results (the main difference being that U.S. patenting abroad is less dependent on market size and more on multinational presence abroad).

Slama (1981) fits a gravity model (used popularly in international trade) to international patenting data (for 27 countries during the period 1967-78 (i.e. pre-EPO era)). The dependent variable is cross-country patenting as a function of the GNPs and populations of the country of origin and destination, the geographic distance between (capital cities of) countries, and dummy variables for regional trade membership (such as the *European Community*, *European Free Trade Area*, and so forth). A key finding is that regional trade areas create positive preference, in that members engage in more bilateral patenting than would otherwise be the case. The Slama study, though, does not incorporate innovation-related determinants, such as R&D, scientists and engineers, and so forth.

B2. Optimizing Approaches

The first few “optimizing” models of patenting behavior set out to address a larger issue, such as the sources of productivity growth, the welfare effects of global patent harmonization, and the state of the technology gap between developed and developing countries. This paper builds upon

these models of patenting behavior. Eaton and Kortum (1996), for instance, develop a decision-theoretic model of patenting and embed it in a ‘quality ladders’ model of economic growth.⁴ They then fit the joint model of productivity and patenting to OECD data and find that substantive differences in productivity are due to impediments in the diffusion of technology. The kinds of impediments that are responsible can be discerned from the determinants of patenting. Among the critical determinants are: geographic distance, intellectual property rights, supply of R&D scientists and engineers, human capital in the destination countries (measured by the average years of schooling), destination country imports as a percentage of destination GDP, and the productivity gap between the country of origin (of patents) and the destination. For example, a country would not get access to patentable technology (and would thereby lag behind in productivity) if the level of human capital is too low (making it difficult to adopt new technology from abroad), if their degree of trade with the rest of the world is low, if they are located in more distant regions, or if their intellectual property protection is too weak to attract foreign patents. On the other hand, a country’s ability to catch up is enhanced if it receives patentable technology from a more technologically advanced nation (which is why the list of explanatory variables includes the ‘productivity gap’ between origin and destination). Overall, this study finds that every OECD country (except for the U.S.) obtains more than 50% of its productivity growth from patentable ideas that originated abroad.

McCalman (2001) extends the Eaton-Kortum model to analyze the impact of the *Trade-Related Intellectual Property Rights Agreement (TRIPS)* on the division of the welfare gains among countries. While this is the main focus of his study, McCalman also estimates a patenting equation.

⁴ Such a growth model was developed by Grossman and Helpman (1991) to explain innovations which improve upon existing goods as opposed to innovations which expand the variety of goods.

A key extension is to allow for *legal barriers* to the patenting of otherwise profitable inventions (for example, working requirements or exclusions from patenting (because the inventions pertain to plant and animal varieties, and so forth)). The estimates of the patenting equation are qualitatively similar to those of Eaton and Kortum. As predicted, the restrictions on patenting proportionally reduce the fraction of inventions that are patented in a destination country.

Park (2001) studies the extent to which international technology gaps, as measured by total factor productivities (TFPs), can be explained by differences in patent protection levels and patenting across countries. The focus is on whether international patent reform helps narrow technology gaps. To address this, Park derives a model where TFP is a function of domestic and foreign patent knowledge stocks and an auxiliary model where domestic and foreign patenting are functions of the strength of patent regimes (along with other control variables). The other control variables that matter to patenting are the destination's market size, the destination's ability to imitate foreign innovations, and the cost of patenting in a destination.

Some critical differences among these studies are that Eaton and Kortum use a different measure of patent protection than McCalman and Park. Eaton and Kortum use the Rapp and Rozek (1990) measure which is based on a cross-sectional estimate as of 1984. On patenting costs, Park uses a different measure than McCalman and Eaton and Kortum. The latter two studies use Helfgott's (1993) data, which are available for one year and are from the perspective of U.S. (English-speaking) inventors. The same cost figures cannot be assumed to apply to inventors from other countries; for example, the same high cost of translation in Japan would not apply to Japanese patent applicants. Moreover, in the McCalman and Eaton and Kortum models, the patent protection variable is interacted (nonlinearly) with the patent cost to destination GDP ratio, making it difficult

to separate out the effects of costs.

In other studies, the focus of attention is the trend in patenting itself. Kortum and Lerner (1999) observe an ‘explosion’ in U.S. patenting (domestically and abroad) and examine several hypotheses that might explain that. The two critical competing hypotheses are the *pro-patent* hypothesis and the *fertile technology* hypothesis. According to the pro-patent (or friendly court) hypothesis, changes in the legal regime precipitated the increase in patenting (for example, the establishment of a specialized appellate court (namely, the *Court of Appeals of the Federal Circuit (or CAFC)*) which appeared to render decisions favorable to patent holders, upholding patent validity decisions or reversing invalidity rulings). This increased the incentive to acquire patent rights. According to the fertile technology (or increased inventiveness) hypothesis, firms have become more productive and the management of R&D more efficient. Hence, what we are observing is an increase in the supply of ideas – not all of which are patentable, but nonetheless leading to a rise in patent applications. In a sense, the two hypotheses are not altogether separable. To the extent that strengthened patent rights stimulate R&D, the regime changes might have led as well to increased innovation potential. Secondly, increased R&D efficiency and innovation potential might have been the reason the courts ruled more favorably to patent rights holders; patents awarded to higher quality technologies would less likely be ruled as invalid. Thus, it is not clear that we are discussing two distinct hypotheses.

Nonetheless, the authors set up a horse race to see which hypothesis better explains the data. Their conclusion is that it is more likely to be the fertile technology hypothesis, based on the fact that if the pro-patent argument were true, we would have observed an ‘explosive’ increase in both domestic and foreign patent applications in the U.S. Instead, the explosive increase occurs primarily

with U.S. patenting at home and abroad, which is more consistent with the hypothesis that U.S. inventors were more prolific. One slight problem, which the authors note, is that while U.S. patenting grew, the U.S. R&D to GDP ratio grew at a slower rate. They therefore conclude that patenting was not driven by a boom in R&D but by better *management* of R&D.

Interestingly, in this study, the explanatory variables are all categorical variables (i.e. dummy variables – alone or interacted with others); for example, dummies for destination, source, year, home country, and so forth.⁵ It would be useful to incorporate more explicit variables (such as market size, patent rights, filing costs, and so forth). Moreover, the authors seem to treat the shift in legal regime as a one-shot affair (namely with the formation of the CAFC in 1982) to test the pro-patent hypothesis. They do not study the dynamic evolution of legal changes (say with the help of an explicit variable that measures ongoing changes in the legal regime), in which case they might have picked up the effects of landmark court cases which affect patenting in software and computer-related industries or of international institutions (such as the PCT and EPO systems) which create economies of scale in patent filings.

Rafiquzzaman and Whewell (1998) reproduce the Kortum and Lerner analysis for Canadian patenting behavior at home and abroad. They too find an increase in the growth rate of Canadian patenting, and likewise attribute it to fertile technology factors. They cite the explosion of new firms in the high-tech sectors (software, biotechnology, and communications technology) and the application of information technology to the R&D process which increases the productivity of R&D. The fact that there is a similar increase in patenting in other countries should shift some focus away

⁵ This methodology is extended in Eaton et. al. (1999) to study patenting behavior at the sectoral level.

from nation-specific factors (such as the U.S. CAFC) to some common international factors.

A micro-level study by Hall and Ziedonis (2001) challenges the hypothesis that U.S. firms patented more because they were more inventive. Using a sample of U.S. semiconductor firms, the authors find that the motive for, or determinant of, patenting is strategic: to pre-empt “hold-ups” or blocking if rivals own key patents. The argument is that if a firm could own critical patents itself, it could better negotiate with others who have rights to technologies that the firm might need. The authors argue that recent legal changes put firms in a situation where they need to patent for this purpose. The legal changes broadened patent scope and facilitated entry by specialized firms. In an environment of cumulative innovation (such as in the semi-conductor industry), the possibilities for patent hold-up are greater. Firms cannot afford not to acquire patents while others are amassing vast patent portfolios. Thus, it is the filing of these vast patent portfolios that may account for the explosion in patenting in recent years.⁶

Gallini et. al. (2003) also contribute to the ongoing empirical analysis of what determines patenting. The authors estimate the patenting model of Eaton and Kortum using a different measure of patent protection (namely, the index in Park (2001)). They estimate the model using OECD data, while providing a Canadian patenting perspective. The authors argue that patent protection can have both an ‘innovation effect’ (by stimulating R&D) and an ‘increased filing effect’ (by raising monopoly rents). To focus on the filing effect (or on patent propensity per se), the authors use as

⁶ In their estimated equation for patenting (namely, a Poisson model), the authors find that, over time, capital per worker is the more significant explanatory variable (to explain patenting), which is consistent with the idea that firms with large sunk costs in complex manufacturing facilities face greater litigation costs and have a vested interest in strategic patenting. Meanwhile, R&D per worker declines in significance, over time, as an explanatory variable. This result, they argue, does not appear to be consistent with the fertile technology hypothesis.

dependent variables: patents per R&D expenditure, and patents per domestic patent applications. The rationale is that the innovative effect of patent rights would be captured in the overall pool of patentable inventions. But since this is unobservable, they posit that this pool is a function of R&D expenditures. Alternatively, this pool could be a function of domestic patent applications. Focusing then on foreign patenting would help capture the filing – or propensity – effect. It is not perfectly clear, though, that innovative effects could not be absorbed in foreign filings; that is, foreign filings can represent both *new* inventions as well as *existing* inventions that have not yet been patented (as long as they are still novel and fall within any grace periods or priority filings). Nonetheless, this study helps to identify important determinants of patenting. As in previous studies, they find geographic distance to have a negative effect on foreign patenting, while average years of schooling, real GDP, imports as a share of GDP, and patent protection all positively stimulate foreign patenting. They also find that anti-trust factors play a role: the tighter anti-trust scrutiny is, the weaker patent rights are, and therefore the lower the incentive to patent.

To summarize, the existing literature suggests a variety of explanations for patenting: that it is driven by technological capacity and R&D, by strategic behavior, by legal regime changes, by the attractions of destinations (market size), and by the ability of destinations to absorb new technologies. In actuality, no single variable may dominate. Not only could these variables play complementary roles, but it is not always clear that they represent distinct factors or developments. The studies are either from the perspective of national patenting behavior (e.g. U.S. or Canadian) or from the world as a whole (e.g. TRIPS-related countries). None of these studies have thus far analyzed regional patenting systems, such as the EPO. Thus existing work has overlooked the role that major institutions might play in explaining the recent growth and distribution of international

patent rights.

2. Theoretical Framework

This section develops a conceptual model of international patenting flows as a basis for the empirical analysis. The following model is adapted from Eaton and Kortum (1996) and Park (2001). First some definitions and notation. Let the *source country* be the country of origin of patent applications (or the country to which patents are granted). Let the *destination country* be the country in which a patent application is filed (or the country which grants a patent right). In this paper, we are primarily interested in the special case where the destination is not a specific country but a regional office (such as the EPO).

The variations in international patenting depend on three kinds of heterogeneity: (1) market heterogeneities (some destinations are more attractive than others); (2) invention heterogeneities (some inventions are more valuable than others); (3) heterogeneity between source countries (some source countries are more inventive than others).

Let the source country be indexed by i and the destination country by j , such that $i = j$ refers to *domestic* patent applications, and $i \neq j$ to *foreign* patent applications. Let P_{ij} denote the number of patent applications from country i to country j . Each source country produces each period a flow of inventions. Let λ_i denote this flow of ideas, some of which may be patentable. Of these, some fraction, f_{ij} , is applied for a patent in country j .⁷ Hence the number of patent applications from

⁷ Of course, this is not to say that they are “patentable” – only that patent applications are filed for them. Whether inventions qualify or meet the standards of patentability (as set out in country j ’s laws) is determined at the patent granting phase.

country i in country j is:

$$(1) \quad P_{ij} = \alpha_i f_{ij}$$

We will model each of the three components on the right hand of this equation. First, we assume inventive output to be produced according to a Cobb-Douglas production function:

$$(2) \quad \alpha_i = \alpha R_i^{\beta_1} S_i^{\beta_2} L_i^{1-\beta_1-\beta_2}, \quad \alpha > 0$$

where R denotes the stock of research and development (R&D) capital, S the supply of scientists and engineers, and L labor. The exponents $(\beta_1, \beta_2, \dots)$ represent *elasticities*: the percentage change in inventive output for a 1% change in input.

It will be shown that those inventions which cross a particular quality threshold will be the ones for which patents are sought in the destination market. To develop this idea, assume that each of the various inventions of the source country can be indexed by its “quality” level, associated say with the inventive step of an invention. The quality level is assumed to be a random variable, Q , drawn from an exponential distribution. Let the cumulative distribution function be:

$$F(q) = \Pr(Q < q) = 1 - e^{-\psi q}$$

which essentially captures the stylized fact that the distribution of invention quality is skewed: a

small percentage of the top inventions account for a large majority of the total value of patent rights.⁸

This can be seen from the fact that $F'(q) > 0$ and $F''(q) < 0$. Thus the median quality is less than the mean quality (hence the distribution of Q is positively skewed – or skewed to the right).

The mean inventive step from such a distribution is $1/\lambda$. An invention of size q is assumed to augment a firm's productivity by a factor of e^q . For example, if A is an index of productivity, the new level would be $A' = e^q A$. Thus, under this formulation, q is the growth rate of productivity (i.e. $q = \dot{A}/A$). Firm productivity could be enhanced either because the invention improves production potential or is a cost-saving innovation. We assume that the productivity increase is reflected in firm profits. A tractable formulation is:

$$(3) \quad \pi(q) = e^q \bar{\pi}, \quad \pi'(q) > 0$$

where π denotes the instantaneous flow of profits and $\bar{\pi}$ some base flow. The derivative of π with respect to q is of course assumed to be positive. However, it is also necessary to adjust the level of a firm's profits due to imitation activities. Because of imitation or infringement, the returns to inventions are not fully appropriable. In each market or destination j , firms face hazards of imitation. Assume that imitation acts like a tax on profits, and denote by h the rate at which profits can actually be appropriated. Thus *net* instantaneous profits are:

$$(3)' \quad \pi(q) = h e^q \bar{\pi}, \quad 0 \leq h < 1$$

⁸ See Putnam (1996) for some empirical evidence and review of the literature.

where $h < 1$ signifies that the returns are imperfectly appropriated and $h = 0$ denotes that they are completely dissipated. However, with patent protection the ability to appropriate increases, depending on the strength of the intellectual property regime. Let θ denote the increase in the rate of appropriation due to patent protection. θ is assumed to be a positive function of the strength of patent rights.⁹ Thus with patent protection, the rate at which returns are appropriated equals $h + \theta$, where $0 \leq h + \theta \leq 1$.¹⁰

The value of a firm equals the presented discounted value of the future stream of profits, and depends on whether or not a firm has a patent. *With a patent*, the value is:

$$(4) V^{PAT} = \int_0^{\infty} (h + \theta) \bar{\pi} e^q e^{-rt} dt = \frac{(h + \theta) e^q \bar{\pi}}{r}$$

where r is the real interest rate. *Without a patent*, the value of a firm is the above expression with θ set to zero:

$$(5) V^{NO PAT} = \int_0^{\infty} h \bar{\pi} e^q e^{-rt} dt = \frac{h e^q \bar{\pi}}{r}$$

Hence the value of patent protection is:

⁹ If the enforcement of patent rights is completely ineffective or if patent protection, for some reason, is not necessary for the appropriation of investment returns, then θ would be zero.

¹⁰ Note that the increase in appropriability could have been modeled multiplicatively (as $h \theta$). However, the results are qualitatively similar but analytically less tractable.

$$(6) \Delta V = V^{PAT} - V^{NOPAT} = \frac{\theta e^q \pi}{r}$$

That is, a patent in market j enables the patentee to ‘purchase’ a reduction in the incidence of imitation, the benefit from which is reflected in an increase in firm value. Thus a firm will seek patent protection if the net benefit of patenting exceeds the cost of filing for protection:

$$(7) \quad V = V^{PAT} - V^{NO PAT} \geq c$$

where c denotes the cost of obtaining a patent (e.g. filing fees, agent fees, and possibly translation fees). The underlying logic is that inventors have means other than patent protection to appropriate the rewards from their innovation (such as lead times, reputation, secrecy). Thus the value of a patent is the incremental return an inventor can get above and beyond that which can be realized by alternative (non-patenting) means.

Equation (7) helps determine which of the source country inventions will be applied for patent protection. A critical threshold quality of inventions can be identified (using (6) and (7)), namely:

$$(8) \quad q^* = \ln\left(\frac{rc}{\pi\theta}\right)$$

The more expensive it is to file a patent (i.e. the higher c is), the higher the quality threshold (indicating that only inventions of higher quality are worth patenting). Furthermore, the stronger the patent regime (i.e. the higher π is), the lower the quality threshold. Thus not surprisingly, patent

rights are more valuable, holding other factors constant, if patent protection is stronger. A higher base flow of profits (π) also contributes to a lower critical threshold quality. Firms that are in general more productive or have more valuable products are likely to have a higher base flow of profits.

The number of patents filed can now be determined. Recall the cumulative distribution function $F(q)$. Given a critical threshold quality q^* , $F(q^*)$ is the fraction of source country inventions that are not patented and $(1 - F(q^*)) = e^{-\psi q^*}$ the fraction that are. Thus, using our notation above, the third term in equation (1) is:

$$(9) \quad f_{ij} = e^{-\psi q^*} = \left(\frac{r c_j}{\pi_i \theta_j} \right)^{-\psi}$$

where subscripts i, j are brought back to clarify that it is the filing cost and the strength of patent rights of the destination that matter; while it is the base flow of profits of the source country firm that matters. Now, putting it all together (i.e. substituting (2), (9), and the assumption that $\eta_{ij} = 1$, into (1)) yields the prediction for patenting flows from i to j :

$$(10) \quad P_{ij} = \alpha R_i^{\beta_1} S_i^{\beta_2} L_i^{1-\beta_1-\beta_2} \left(\frac{r c_j}{\pi_i \theta_j} \right)^{-\psi}$$

Taking natural logs of both sides of (10) yields:

$$(11) \ln\left(\frac{P_i}{L_i}\right) = \gamma_0 + \gamma_1 \ln\left(\frac{R_i}{L_i}\right) + \gamma_2 \ln\left(\frac{S_i}{L_i}\right) + \gamma_\pi \ln(\bar{\pi}_i) + \gamma_\theta \ln(\theta_j) - \gamma_c \ln(c_j) + \mu$$

where μ denotes the error term and γ_0 the constant.¹¹ A few modifications will be made before applying the model to the data. First, if j denotes the EPO region, there exist specific costs (c_j) of filing in the EPO; however, there exists no aggregate measure of the state of intellectual property rights in the EPO. Currently, there are measures of patent strength (to be discussed below) for the individual countries that comprise the EPO. Thus one possibility is to take a weighted average of the values of the index of patent strength of the individual countries. However, the issue is how to construct those weights. Secondly, the levels of patent rights are high among EPO countries, and they do not vary much over time (though there is a slight upward trend in the strength of patent regimes during the sample period). Thus, as a proxy, we will incorporate a time trend to capture variables, such as aggregate EPO patent strength, that evolve over time.¹²

Another modification to (11) is that we need to be more concrete about the base flow of profits term (i.e. $\ln \bar{\pi}_i$). We postulate that this is a function of overall source country productivity and the level of patent rights in the source country.¹³ The rationale is that the more productive a country is the more inventive output it may produce; secondly, the stronger a patent regime is the greater the incentive to innovate may possibly be (referred to in the literature as the “inventive

¹¹ Furthermore, the parameters in (11) are functions of previous parameters; i.e. $\gamma_0 = \ln r$, $\gamma_1 = \alpha$, $\gamma_2 = \beta$, $\gamma_\pi = \theta$, and $\gamma_c = \phi$.

¹² In other words, we will assume that $\theta_j = e^{\beta_{st}}$

¹³ That is, $\bar{\pi}_i = \text{GDPC}^{\beta_3} \text{IPR}^{\beta_4}$, where GDPC and IPR as defined below.

effect” of patent rights). A third modification to (11) is to add time subscripts since data on international patenting are available over time. Finally, given the panel data nature, the error term may consist of important individual (country) effects, which may be fixed or random. The individual effect may capture any specific interaction effect that individual source countries have with the EPO.

Thus the equation to be estimated is:

$$(12) \ln\left(\frac{P_{ijt}}{L_{it}}\right) = \gamma_0 + \gamma_1 \ln\left(\frac{R_{it}}{L_{it}}\right) + \gamma_2 \ln\left(\frac{S_{it}}{L_{it}}\right) \\ + \gamma_3 \ln(GDPC_{it}) + \gamma_4 \ln(IPR_{it}) + \gamma_5 t - \gamma_c \ln(c_{jt}) + \mu_{it}$$

where GDPC denotes real GDP per capita and IPR an index of patent rights – both for the source country.¹⁴ The error term $\mu_{it} = \alpha_i + \xi_{it}$, where α_i is the individual effect component and ξ_{it} the orthogonal and spherical disturbances component. The error term is motivated, as in Eaton and Kortum (1996), by the fact that some profitable inventions fail to be patented while some unprofitable ones are patented. Note that in (12), j refers to the EPO. Thus the dependent variable is the natural log of EPO patent applications from country i divided by the labor force in country i .

Equation (12) constitutes the basic model specification for explaining patenting behavior in the EPO. Other independent variables can be added to equation (12), but in order to avoid cluttering up the basic model, the discussion of these variables is deferred to the empirical section. The context for these variables will be clearer when they are brought up.

¹⁴ The new parameters are: $\gamma_3 = \beta_3 \gamma_\pi$, $\gamma_4 = \beta_4 \gamma_\pi$, and $\gamma_5 = \beta_5 \gamma_0$.

3. Data and Sample Statistics

A panel data set is used to estimate equation (12). The sample consists of 53 source countries over the period 1980-2000. In this sample, there is only one destination, namely the EPO. Given a number of missing observations (due to incomplete data on the independent variables of interest), in practice this sample reduces to about 30 source countries over a 21 year period (providing 630 observations = 21 x 30).

The data used in this paper can be put into five categories: Patenting Data, Macroeconomic Data, Science and Technology Data, Trade Data, and Intellectual Property Rights Data. The following describes the data in more detail:

(A) Patenting Data: Data on EPO patenting (applications and grants) and on patent filing costs are from the European Patent Office.¹⁵ The breakdown of EPO patents by designation (i.e. by individual member states) is derived from the World Intellectual Property Office's (WIPO) *Industrial Property Statistics*. Patent applications are the sum of direct EPO filings (or grant) and indirect filings (via the *Patent Cooperation Treaty (PCT)* process, as administered by the WIPO). This sum helps control for the fact that inventors can switch between these different modes of obtaining patent rights, without affecting the overall demand for a patent in the EPO region. Here, we do not distinguish between first or subsequent filings, as the determinants of this may be somewhat more

¹⁵ I thank Peter Hingley for providing these data.

complex and micro-oriented.¹⁶

Patent filing costs are converted into real terms using the GDP deflator for Germany, since the filing costs are in Deutsche Marks. The deflator's base year is 1995.

(B) Macroeconomic Data: Gross Domestic Product (GDP) are in real 1995 U.S. dollars. GDP, population, and labor force data are from the World Bank *World Development Indicators*.

(C) Science and Technology Data: Data on national, public, and private research and development (R&D) and on the supply of scientists and engineers are from the OECD's *Main Science and Technology Indicators* database.¹⁷ For non-OECD and/or developing countries, we obtained R&D data (including the share of public R&D in national) from UNESCO's *Statistical Yearbook*. For several countries, data are missing. For data that were missing between years, we filled in gaps by a linear interpolation.¹⁸ The reported R&D data are all investment (flow) data. To convert to R&D stocks, we used the *perpetual inventory method* assuming a 10% geometric depreciation rate. The Appendix describes in more detail how the stocks were constructed from flow data.

The supply of scientists and engineers is our measure of scientific human capital. Studies on economic growth have often used education-related variables, such as the secondary school

¹⁶ In the sense that detailed information about inventor propensities or preferences may be required. Part II does provide a first look at distinguishing among different modes of filing.

¹⁷ This can be accessed through www.sourceoecd.org.

¹⁸ For example, if there were a three year gap in R&D flows, we divided the total change in R&D values over that period by three and incremented the value of R&D by that figure for each year that was missing. For example let $\Delta = (x(t+3) - x(t))/3$, where t denotes time. Then $x(t+1) = x(t) + \Delta$, $x(t+2) = x(t+1) + \Delta$.

enrollment rate (see, for example, Mankiw et. al. (1992)). Thus as a more general measure of human capital, we examine both the secondary school enrollment rate and tertiary (post-secondary) school enrollment rate. The enrollment rates are used on the assumption that spending on (or investment in) education is correlated with the student population enrolled. Data on school enrollment rates are obtained from the World Bank *World Development Indicators*.

(D) Trade Data: We employed bilateral trade and foreign direct investment (FDI) positions between countries in order to capture technological interactions between them. Bilateral trade data for 1996-2000 are from the United Nations *Comtrade* database. For earlier years (1980-1995), we used Statistics Canada's *World Trade Analyzer*. Bilateral FDI data are from the OECD's *International Direct Investment Statistics*. We then examine total source country exports to the EPO region and FDI stocks in the EPO region. To obtain total source country and FDI vis-a-vis the EPO as a whole, we simply summed up a source country's exports to or FDI in each of the EPO member states (in real dollar terms (where 1995=100)). Both FDI and exports matter because generally inventors seek patents in regions where they expect to market their goods and services.

(E) Patent Rights: The measure of patent rights is taken from Ginarte and Park (1997) and Park and Wagh (2000). These studies provide an index of the *strength* -- not necessarily *quality* -- of patent rights across countries and over time. The index of patent rights (henceforth IPR) ranges from zero (weakest) to five (strongest). The value of the index is obtained by aggregating five sub-indices: extent of coverage, membership in international treaties, enforcement mechanisms, duration of protection, and provisions against loss of protection (against, for example, compulsory licensing).

These features (coverage, membership, duration, enforcement, and loss of protection) were chosen as a reference point because of their adoption in international standards (for example, the *TRIPs* Agreement of 1994).

The numerical value of each of these sub-indices (which range from zero to one) indicates the fraction of legal features in that sub-index available in the particular country. For example, a value of $\frac{1}{3}$ for *enforcement* indicates that a country has one-third of the possible enforcement mechanisms listed under that sub-index. A value of $\frac{1}{2}$ for *duration* implies that a country grants protection for half the international standard time (of 20 years from the date of application or 17 years from the date of grant). The value for *membership in international treaties* indicates the fraction of available treaties to which the nation is a signatory. The value for *coverage* indicates the fraction of invention classes the country allows as patentable subject matter. Finally, several conditions exist under which authorities can revoke or reduce patent rights. The value for *provisions against loss of protection* indicates the fraction of those conditions which are not exercised in the country.¹⁹ The Appendix summarizes the scoring method for determining the strength of national patent protection.

Next, we turn our attention to some sample statistics. Table 1 shows the sample means (over the period 1980-2000) of the following variables: EPO patent filings, EPO patent grants, R&D as a percentage of GDP, the shares of private and public financing of R&D, the supply of scientists and engineers per 10,000 workers, exports to and FDI positions in the EPO (as a percentage of source

¹⁹ Of course, actual patent protection may deviate from statutory (“laws on the books”) protection. The authors investigate this possibility but find that most firms’ complaints are not so much about the *execution* of laws but about the *statutory differences* between industrialized countries and the *lack of laws* in less developed countries, both of which the index incorporates.

country GDP), and the index of patent rights (for the source country). The countries in the sample are grouped according to their sample average real GDP per capita. The high-income nations are those whose GDP per capita exceeded \$17,000 real (1995) U.S. dollars; the low-income nations are those whose GDP per capita was less than \$3,500 real 1995 U.S. dollars; and the medium-income countries are those whose GDP per capita lies between those two limits. We also indicate in the table whether a country is a member state of the EPO (and the year of accession). For each income group, we provide the within-group mean values, and at the end of the table, we provide the overall mean values of the variables.

In general, the high-income nations do most of the patenting in the EPO and receive most of the EPO patent rights, with the U.S. being the leader in both categories. The high-income nations also have the highest rates of R&D spending (averaging 1.53% of GDP), while the low-income nations have an average R&D to GDP ratio of 0.48%. Generally, the share of private R&D in national R&D is higher among high-income groups. For low-income nations, the public sector has a larger share (and larger role) in national R&D. This might be explained by the fact that the index of patent rights is lowest among the low-income nations (and highest among the high-income nations). Weak patent rights might have led to low rates of private sector innovation, placing the burden on the public sector to conduct and finance national R&D. Or it may be that because the share of private R&D in national R&D is low, the governments have had less incentive to invest in and strengthen their patent systems. Either way it is not common for a country to have both strong levels of patent protection and large shares of government R&D.

The high-income nations have large exports to and stocks of foreign direct investment in EPO countries, signifying that the EPO is an important market for their goods and services. Here,

exports and FDI are the sum of exports and FDI across EPO member states. In general, however, it is the EPO member states themselves that have high ratios of EPO exports and FDI to GDP, reflecting a good degree of intra-European trade and investment activity. For example, Netherlands' FDI position in the EPO is about 67% of Netherlands GDP (the highest in the sample). Belgium's exports to the EPO region are about 36% of Belgian GDP. However, the U.S.'s EPO exports and FDI (while large in absolute magnitudes) are 1.87% and 3.41% respectively of U.S. GDP.

In terms of the ratio of scientists and engineers to workers in the sample, Israel is an outlier. It has 122.47 scientists and engineers per 10,000 workers, the highest in the sample. (The second highest is Japan, with a ratio of 94.5.) Granted that labor market classifications and definitions vary somewhat across countries, the figure is quite large. What is puzzling about that figure is that it is not reflected in the amount of patenting Israel does. But this is only when Israel is compared to the high-income nations. Within the medium-income group, it is in third place (in terms of EPO filings) next to S. Korea and Spain. Another possible outlier is Hong Kong which has only 1.83 scientists and engineers per 10,000 workers. This might be due to Hong Kong's economy being driven more by financial activity (rather than technology).

Among the low-income nations, the most EPO patent filings are done by emerging economies such as India, China, and Russia. But in general, their rates of R&D investments are much lower than the overall sample mean and their patent regimes are not very strong or fully developed. Scientific and engineering resources are limited. Thus they will not likely account for a significant share of EPO patenting activity – at least until some major developments occur.

Table 2 presents some patenting statistics for the leading source countries of the EPO. In both 1985 and 1999, the U.S., Germany, Japan, France, and the U.K. (in this order) are the top five

countries in terms of EPO patent applications and EPO patent grants. In terms of applications, Switzerland and the Netherlands are in sixth and seventh place in 1985, and switch places in 1999. Austria and Canada are in tenth place in 1985 and 1999 respectively. In terms of grants, Switzerland is in sixth place in both years. Overall, there is remarkable stability in ranking (in both the applications and grants category).

Over time, however, all of the leading source countries have tripled or quadrupled their EPO filings (a phenomenon noted for the U.S. by Kortum and Lerner (1998) and others). Thus there too appears to be an “explosion” in patenting in the EPO as well. Most of the leaders in patenting are fellow EPO states. Increased patenting could have been the result of increased inventiveness, lower costs of filing, or of new uses for patenting (such as strategic rivalry). In terms of patent grants, there is not the same kind of explosive growth. This might be the result of lags in processing applications (i.e. increases in patent pendency), due to the increased influx of applications over time; of a more selective search and examination process; or of a rise in minor (or not so high quality) inventions.

Table 3 examines patenting trends from the other perspective – the destination’s. The destinations or designations in EPO filings or grants are sorted in descending order so as to give an idea of which EPO countries were most attractive to patent applicants or patentees. For both EPO applications and grants, the five leading EPO states are Germany, U.K., France, Italy, and the Netherlands. In 1985, out of nearly 40,000 EPO filings, 96.1% designated Germany and 68.4% the Netherlands. In 1999, nearly all member states were designated in EPO filings. The country with the lowest number of designations is Cyprus, yet it is designated in 95.3% of all EPO filings that year. As the data for 1994 show, designating almost all the member states is a fairly recent trend. Even in 1994, many countries were not designated. In fact, fifth place Netherlands was designated

in 72.7% of all filings. Monaco was designated in about half the EPO filings. The pattern for EPO patent grants is different. For grants, a noticeably smaller percentage of countries is designated. For example, in 1999, Cyprus was designated in 2% of EPO grants. In fact, most countries are designated in less than half of all EPO patent grants in 1999. Germany, France, and the U.K., though, are consistently designated in about 85% (or more) of EPO grants in those periods.

The EPO patent applications described thus far have revenue implications for the EPO. Fees from patent applications are a source of income for the EPO. Table 4 provides a very rough view of EPO patent revenues from filings. We simply take the filing fee per application and multiply it by the number of applications (per year). Needless to say, the EPO has other sources of revenue (for example, it shares renewal fee revenues with national offices), so that the table does give a limited view of EPO's finances.

In order to obtain real revenues from filings, we first convert nominal fees into real 1995 values (in Deutsche Marks). Columns 1 and 2 show the nominal and real fees respectively, and columns 3 and 4 show the EPO direct and indirect (via PCT) filings respectively. Note that nominal filing fees were changed in particular months of the year. We rounded off to the nearest year-end (depending on whether the month was closer to December of the current year or December of the previous year). The real fees in column 2 are multiplied by the sum of filings in column 5. The resulting real revenues are in column 6. The figures show that revenues rose steadily from 1980 to 1990, then fluctuated somewhat, and then peaked in 1996, after which they declined. The reason is that, with inflation, real fees have followed a declining trend. Nominal filing fees thus have not kept pace with (German) inflation. Still until 1996, real revenues increased because the increased quantity of filings compensated for the drop in real fees per application. However, after 1996, the

cut in nominal and real fees were rather significant so that the growth in applications did not compensate – hence the decline in the filing fee revenues in the late 1990s.

4. Empirical Results

This section focuses on the estimation of equation (12). We first present the results of the basic model and then discuss the sensitivity of the results to alternative specifications, providing commentary along the way.

Table 5 presents estimates of the basic model represented by equation (12). The dependent variable is EPO patent filings per source country worker. The first column contains the random effects estimates and the second fixed effects estimates. The coefficient estimates do not vary radically but the Hausman specification test in some cases rejects the null of no correlation between the independent variables and the error term.

The results show that countries with greater GDP per capita – that is, the more productive economies – have greater EPO patenting per worker, holding other factors constant. Moreover, source country patent rights also positively and significantly stimulate EPO patenting. Thus, the results are supportive of an “inventive effect” coming from stronger patent protection. Given the log-log specification of the equation, the coefficients represent “elasticities.” The response of EPO patenting to GDP per capita is elastic and to the index of patent rights inelastic. More on the importance of the elasticity of demand for patents (vis-a-vis fees) later.

The source country’s stock of R&D per worker and number of scientists and engineers per 10,000 workers have a positive effect on stimulating the overall flow of ideas in the source country

and consequently the number of EPO patents per worker it files. Real EPO fees have the expected negative effect on EPO patent applications. The coefficient estimate is less than one (in absolute value) suggesting that the “demand for EPO patents” is inelastic. Thus higher fees, holding other factors constant, may actually raise EPO revenues. The demand would be inelastic if firms view patent protection as a necessity in order to conduct business or trade in the EPO region. The fact that other firms are applying for patents may stimulate others to feel that they “must” get them too (in order to have strategic bargaining chips for purposes of, say, future cross-licensing or other negotiations). The demand may also be inelastic if technologies are becoming easier to imitate and the returns harder to appropriate without intellectual property protection. Overall the coefficient estimates are significant at the 1% level (or better) and the model explains about 80% of the variation in the data.

Column 3 examines how sensitive the results are to the use of R&D stocks, rather than flows. Replacing the R&D stock measure with R&D flows (or R&D investment per worker) affects the results very marginally. The similarity might have to do with the relative smoothness of R&D investment spending, which make the flows and stocks fairly proportional over time. R&D may be smooth because of high costs of adjustment (relative to physical capital). Much of R&D consists of wages and salaries paid to the R&D scientists and engineers, whose employment is structured in a way that hiring and removing staff is fairly difficult. Thus from a practical point of view, the flows and stocks of R&D yield similar results. Theoretically, however, R&D stocks capture the notion that what matters to the production of patentable ideas is the existing stock of knowledge (after accounting for depreciation and obsolescence), rather than the flow of knowledge in a particular time period. Certainly, a new ‘flow’ of knowledge may be associated with a particular contemporaneous

technological breakthrough, but that piece of knowledge often builds upon previous knowledge.

Column 4 splits the stock of R&D per worker between private and public R&D per worker.²⁰ Since data on this composition is not available for all the countries in the sample (see Table 1), we lose about a fourth of the observations (making direct comparisons to the results in other columns difficult since the samples are different). Nonetheless, qualitatively, the results are similar, except that the source country's government or public R&D per worker is not an important explanatory factor. Private R&D may be the more important determinant since private agents tend more to market their innovations and to seek proprietary rights, whereas the mission of public R&D is often to contribute knowledge to the public domain. Public R&D also tends to finance basic, generic ideas which are not patentable by nature. However, in previous studies (see Park (2003)), public R&D is found to stimulate private R&D. In other words, public knowledge is important to conducting and producing private sector research results. If this is the case, then summing private and public R&D helps to capture the indirect effect of public R&D on private sector patenting behavior. Nonetheless, the results in column 4 do indicate that public and private R&D should not be treated as substitutes – and that the relative (and more direct) contribution of private R&D to patenting is greater.

In preliminary investigations, we did examine the implications of including a time trend (results not shown) and found that the time trend was a significant explanatory factor, but that it reduced the statistical significance of the scientists and engineers variable. The time trend may capture destination factors not explicitly modeled. Here there is only one destination, namely the EPO, so that the time trend could capture the overall growth of the market in this region or some

²⁰ Note that public and private R&D would not add up to national R&D since non-profit and higher education R&D are left out.

other phenomenon that has changed over time. But the time trend could also be capturing source country factors that also evolve over time. Indeed the variables we look at explicitly, such as the source country's GDP per capita and supply of scientists and engineers, are variables that vary over time. Thus, for this sample, we will include the scientists and engineers variable (rather than the time trend) since it provides information (on human capital) that we can explicitly allude to.

Table 6 examines slightly different dependent variables. For column 1, we take the ratio of EPO patent filings per scientist and engineer in the source country, and in column 2, the ratio of EPO patent filings per stock of R&D in the source country. The results in the table suggest that the results thus far are not sensitive to the choice of these dependent variables. The coefficient estimates are exactly similar to those shown in the second column of Table 5, except for the constant term and one other variable. In column 1 of Table 6, that variable is the natural log of scientists and engineers per 10,000 workers (which has a coefficient of -0.80), and in column 2 it is the natural log of the stock of R&D per worker (which has a coefficient of -0.42). Both coefficient estimates are the previous coefficient estimates (in Table 5, column 2) minus one. This is not surprising since to get these alternative dependent variables, we subtract either $\ln(\text{SE})$ or $\ln(\text{RD})$ to both sides of equation (12) (and move the natural log of workers to the right-hand side). The negative coefficient estimates signify that neither the supply of scientists and engineers nor the stock of R&D have increasing returns effects. For example, the ratio of patent filings to R&D stock (or to the supply of scientists and engineers) is not increasing in the stock of R&D (nor in the supply of scientists and engineers). Thus R&D stocks (or scientists and engineers) have a positive, but diminishing impact on patenting. Overall, the models explain a smaller percentage of the variation in the data than the model where patents per worker is the dependent variable.

Table 7 examines alternative independent variables. Column 1 shows the results of adding two “market size” variables: exports to the EPO region and FDI position in the EPO region (both as a ratio to source country GDP). Both variables measure the extent to which the source country trades and invests in the EPO. They indicate the relative importance of the EPO market to the source country. Patent rights are usually desired if firms have significant marketing interests in say the EPO region. However, these two variables are statistically insignificant explanatory factors. But in column 2, where we dropped the GDP per capita variable, we can partially see why the results turned out they way they did in column 1. Both FDI and exports – as previous studies have shown²¹ – are positive functions of GDP per capita. The latter might capture a country’s productive capacity to export and to manufacture abroad. Once GDP per capita is controlled for, the additional FDI and export variables do not matter. This suggests that GDP per capita may capture the influences of exports and FDI.

But note that, when per capita GDP is dropped, the FDI variable is a statistically significant factor but the exports variable is not. One possibility is that the demand for patent protection is greater if production facilities are being set up abroad. Manufacturing abroad exposes firms to risks of getting their knowledge (say, production methods) appropriated. Hence, foreign patents are sought when firms locate abroad (which explains why FDI matters to patent filings). Exports, on the other hand, may expose firms to reverse engineering, but in the EPO region this does not appear to be a significant risk factor and may be why it does not significantly affect the patenting decision. If this interpretation is correct, it would be consistent with a view that firms are concerned with managing local knowledge spillovers – within the EPO region – arising out of their investments in

²¹ See Maskus (2000) for a survey.

foreign affiliates.

In columns 3 and 4 of Table 7, alternative measures of human capital are introduced. Thus far the scientists and engineers variable may be capturing R&D-related human capital, which is very specialized. Education investments may capture more general measures of human capital. To that end, we try educational enrollment rates which are used in standard empirical growth studies. The secondary school enrollment rate is statistically insignificant (see column 3), while the tertiary enrollment rate (in college or universities) is significant at the 10% level. Thus the more appropriate type of human capital for innovation and patenting activity seems to be a specialized one. At least the tertiary education variable comes close to explaining patenting.

In Table 8, we report on the extent to which patent filings respond to cyclical shocks and lagged effects. At one level, one could argue that patenting is costly so that patenting decreases during recessions and increases during booms. At another level, one could argue that technological investments are typically for the longer run and hence should be less sensitive to short run considerations. Thus it would be of both policy and academic interest to pursue the connection between patenting and business cycles. This paper provides a brief glimpse at the issue. First, in order to obtain a measure of cyclical shocks, we ran an AR(1) regression of the natural log of GDP with a time trend, as follows:

$$\ln(\hat{GDPC}) = -2.61 + 0.95 \ln(GDPC_{-1}) + 0.0015 \text{ Time Trend}$$

$$(0.46) \quad (0.008) \quad (0.0002)$$

$$Adj. R^2 = 0.99, \quad No. of Observ. = 1056$$

where the standard errors are in parentheses.²² Let the measure of cyclical shocks be the residual to this equation: $\hat{u} = \ln(GDPC) - \ln(\hat{GDPC})$, where the symbol $\hat{}$ denotes fitted values.

Column 1 of Table 8 displays the results of regressing the dependent variable on $\hat{u}$ (and a constant term). The coefficient is significantly positive, suggesting that patent filings move with the business cycle. However, once the other RHS variables are controlled for, no independent influence of cyclical shocks on patenting is present (see column 2 of Table 8). Any cyclical movements are likely to be reflected in the movements of RHS variables themselves (such as R&D).

Column 3 of Table 8 reports on the lagged effects of the variables (so as to examine any adjustment effects). Most of the lagged variables are insignificant in explaining patenting (except for the lagged dependent variable and lagged R&D). Thus most of the variables we have been modelling have relatively strong contemporaneous effects.

The positive effect of the lagged dependent variable suggests that there is some momentum in patenting. Nearly two-thirds of last period's patenting accounts for the current period patenting. But since the coefficient (of the lagged dependent variable) does not exceed one, the momentum fades (holding other factors constant). In fact, focusing only on the dependent variable and its lag, the change in $\ln(PAT)$ is negatively related to lagged patenting:

$$\ln(PAT) = -0.33 \ln(PAT_{-1}) + \dots$$

²² This coefficient of lagged GDP per capita is not unity so that the resulting shocks to GDP per capita are not permanent ones.

where $\ln(PAT_{t-1})$ was subtracted from both sides of the model (in column 3, Table 8). The growth rate of patenting per worker is negatively related to the initial level of patenting per worker (holding other factors constant). This is reminiscent (in the growth literature) of some form of economic “convergence.” Here it is convergence in patenting per worker across source countries (conditional on other fundamentals – i.e. other RHS variables – being the same). The significance of this is that it suggests that there are underlying “diminishing returns” at work – operating on the inputs used to produce patentable output. Thus, to the extent that there are global differences or divergences in patenting per worker across countries, they are likely to be due to differences in the endowment (or efficiency) of scientific resources across countries, rather than to some technological “increasing returns” effect operating on those inputs (whereby the high patenting nations produce more patents and the low patenting nations produce less; i.e. “the rich getting richer, and the poor getting poorer”).

In fact, the coefficient estimate of lagged R&D is somewhat consistent with this explanation. The negative (and significant) estimate implies that if R&D strays too far above some trend (or long run) value, the effect on patenting is reduced or diminished. In other words, there may be some stable relationship between patenting and R&D. The divergence between the growth rates of R&D and patenting discussed in the previous literature may be transitory.

Taking the estimates in column 3 of Table 8 somewhat seriously, we can calculate the steady-state estimates (by equating the variables and their lags, e.g. $PAT = PAT_{t-1}$, and so forth). This yields:

$$\ln PAT = -32.7 + 1.59 \ln GDPC + 0.24 \ln RD + 0.07 \ln SE + 1.29 \ln IPR - 0.36 \ln FEE$$

where the variables are as defined in the tables. Compared to the estimates obtained previously

(without the dynamic specification) – namely the results in column 2 of Table 5 – the impact of GDP per capita (GDPC) and the index of patent rights (IPR) are fairly similar. The measured long run effect of R&D is half the measured static effect (i.e. 0.24 vs. 0.58). The measured effect of patenting costs (FEE) is one-third higher than the static effect. The measured effect of scientists and engineers is one-third the measured static effect. Nonetheless, the qualitative impacts are similar.

Part II. Forecasting

1. Overview

As emphasized earlier in the introduction, good forecasts of patent filings (by technology and/or by mode of filing) are useful to the EPO for purposes of allocating internal resources. For the EPO's external relations with other supranational offices, coordination of tasks is enhanced by good forecasts of the breadth of international patent filing activities (whether they involve two or more countries, or blocs of countries).

In this part, we conduct forecasting exercises with a modified version of the basic model analyzed in Part I. We examine these forecasting exercises in three different contexts:

- Overall Filings in the EPO, broken down by *modes of filing*; in particular, applicants can file patents directly at the EPO or indirectly via the PCT. Furthermore, these filings may be 'first filings' or 'subsequent filings'. Part I did not consider the breakdown of filings by *mode*.

- Patent filings broken down by *technological field*. The technological classification adopted here is that of the EPO's *Joint Cluster (JC)* system, which consists of fourteen technological units (e.g. unit 1 is electrical machines and electricity, unit 2 is handling and processing, etc.). The EPO consists of directorates assigned to particular JC's.
- Patent filings comprising patent families. The patent family database is indexed by priority forming filings and provides related subsequent filing activity in the major blocs: EPC (including the EPO), US, Japan, and Others. The database thus enables users to pick out the type of patent families they desire to examine.²³

We consider each of these different contexts (i.e. Aggregate Filings, Sectoral Filings, and Family Filings) in turn. For each of these, the structure of discussion is as follows:

- i. Sample Statistics
- ii. Regression Equations
- iii. Forecast Accuracy (RMSPE)
- iv. Sample Forecasts

²³ For more details on patent family statistics at the EPO, see Hingley and Park (2003). This study draws upon material in this earlier paper.

That is, we first provide some (additional) sample statistics to get an impression of the underlying trends or patterns. We then examine a few different forecasting models; typically, for forecasting purposes, it is useful to control for the dynamics arising from lags of the dependent variable. Thirdly, we look at summary statistics of forecast accuracy; in particular the *root mean squared percentage errors* (RMSPE). Finally, we show some sample forecasts for individual source countries for a particular year.

The general methodology is to estimate equations of the form:

$$(13) \ln y_t = \alpha + \beta \ln x_t + \varepsilon_t, \quad \text{for } t = 1, \dots, T$$

where $y_t = Y_t/L_t$ is patent filings (Y_t) per worker (L_t) at time t , x_t could be R&D per worker or a lagged value of the dependent variable, and ε_t the error term. As in Part I, the model is log-linear and the variables are expressed in per worker units.

To forecast values of y_t for $t > T$ – that is, make “out-of-sample” forecasts – we use the parameter estimates $(\hat{\alpha}, \hat{\beta})$ and known (or realized) values of x_t for $t > T$ to derive these forecasts.

For example, for time $t = T+1$, the fitted equation for (13) predicts the following value for the natural log of y_{T+1} :

$$(14) \ln \hat{y}_{T+1} = \hat{\alpha} + \hat{\beta} \ln x_{T+1}$$

Then to get the predicting patent filings $\hat{Y}_{T+1}$, we take the exponential of the above predicted

dependent variable and multiply it by the labor force at T+1:

$$\text{i.e. } \hat{Y}_{T+1} = \exp(\ln \hat{y}_{T+1}) L_{T+1}$$

The root mean squared percentage error (RMSPE) formula is then:

$$\text{RMSPE} = \sqrt{\left(\frac{Y_{T+1} - \hat{Y}_{T+1}}{Y_{T+1}} \right)^2}$$

To get an anchor for the root mean squared percentage error, note that $\text{RMSPE} = 1$ if the predicted value is either twice that of the actual value (i.e. if $\hat{Y}_{T+1} = 2 Y_{T+1}$) or equal to zero (i.e.

$\hat{Y}_{T+1} = 0$). In other words, it gives us an idea of the percentage deviation.

To illustrate some sample forecasts by source country, we shall show and compare the Y_{T+k} and $\hat{Y}_{T+k}$ for each country, where k denotes the forecast horizon (e.g. k-step ahead).

To keep the forecasting exercise at a “practical” level, we focus on those determinants or explanatory variables (x_t) that are easily obtained. This means employing R&D flows rather than stocks (to obviate the need to construct capital measures, particularly since the empirical results are not too sensitive to the choice of stock vs. flow). It also means focusing on variables in per (general) worker units (rather than per scientific and engineering workforce, as again the results are not too

sensitive to the choice of labor force measure).²⁴ The regression models also fit much better if the lagged values of the dependent variable are included, given the serial correlation or momentum in patent filings over time. As will be seen, we focus primarily on the R&D variable. Other variables such as output are correlated with R&D (since output is a function of R&D, among other factors) and would create multicollinearity problems. Variables that are imperfect proxies for some important factor tend to create greater forecast errors (e.g. patent filing costs).²⁵

Thus, without the patent cost variable, we no longer have a variable that represents the *destination* – namely the EPO. The R&D variable represents a *source* country characteristic. It should be noted, though, that the characteristics or attributes of the EPO vary not across source countries but over time. In other words, the source countries all face (largely) the same *conditions* in the EPO destination (whether it be EPO policy, institutional factors, rules, market size, market conditions, and so forth). Thus most of the variation between the EPO filings of different source countries is likely to be due to source country factors. Nonetheless, developments in the EPO do occur over time that could stimulate or decrease the patenting of source countries (though not necessarily in the same way or to the same extent), and it would therefore be useful to develop proxy measures of EPO-destination characteristics. However, we had earlier discussed some of the difficulties in defining and deriving EPO destination variables (e.g. weighting and aggregating the member country characteristics). If the destination were a single country, this is easy to do. But for

²⁴ Putting variables in per worker units helps avoid non-stationarity problems, since the levels of filings, rather than filings per worker, tend to be integrated of order one – i.e. $I(1)$.

²⁵ For example, filing fees at the EPO may explain Euro-direct filings, but not PCT filings. Filing fees thus help to explain overall filings (as shown in the previous part), but do not help to discriminate among the different modes of filing.

a bloc (such as the EPO) one needs a measure of the market size or other characteristics of the bloc as a whole, and then to weight the underlying individual countries comprising the bloc. In Part I, we tried using different measures of EPO-bloc trade and FDI, but not successfully (recall Table 7).

As a substitute (for some ‘factor’ that varies over time), we employed the time trend. However the time trend was typically insignificant once the lagged dependent variable was included. Also, the time trend is deterministic. It captures the past trend, or fits a trend over past data; using the trend’s estimated coefficient in forecasting is likely to be inappropriate if there are stochastic shifts (due say to regime changes). Certainly patent filings exhibit some persistence over time (rather than follow pure random walks). Some shock that affects patenting in one period is likely to carry over and affect patenting in the next period. However, using a trend to capture that (and then fix its coefficient value for forecasting) would neglect the potential role that additional (unanticipated) shocks can play in perturbing filings from past trend. We therefore used the lagged terms of the dependent variable to proxy for time shifts in EPO destination conditions. Another variable that varies over time and is specific to the EPO would be the filing fees at the EPO. Incorporating this variable, as mentioned above, tended to make the models produce larger forecast errors, particularly in predicting the composition of filings by mode of filing. Thus developing destination variables for the EPO is a work in progress.

Lastly, we discuss the possibility that R&D has lagged effects on patenting. In preliminary analyses, we found that the results were not qualitatively different if we used the first, second, or third lags of R&D flows. This may be due to a couple of factors. First, R&D is correlated with its past values, reflecting the fact that the R&D behind an innovation is not a one-shot investment but part of a cumulative effort, which is why the stock of R&D was important. Secondly, while a given

period's R&D may yield a patentable innovation with a lag, this is not to say that current R&D activity cannot influence current *patenting activity*. Firms may wish to file for patents before further refining their research projects or devoting more resources to them. The priority right gives added security and incentive to continue their R&D, if only to acquire proprietary rights to early versions of their innovations. Thus, R&D activity may stimulate current patent filing activity to the extent that firms take anticipatory action and seek priority rights to forthcoming innovations.

2. Aggregating Filings (Mode of Filings)

In this section, we examine patent filings at the EPO (across all technological fields). We also consider four different ways in which an EPO patent may be filed; for instance, a particular filing may be a first or a subsequent filing, and the filing may be a direct EPO filing or one where the EPO is designated in a PCT application. Consider the following notation:

EF Euro-direct First Filings

ES Euro-direct Subsequent Filings

PF Euro via PCT First Filings

PS Euro via PCT Subsequent Filings

Total = EF + ES + PF + PS

In Part I, we did not make a distinction as to the mode of filings, and have thus been focusing on “Total” filings.

i. Sample Statistics

Table 9 shows some sample statistics on total filings as well as the composition of those filings: namely EF, ES, PF, and PS. The table also shows the percentage share of Euro-direct filings that are first (or priority) filings – i.e. $\left(\frac{EF}{EF+ES} \right) \times 100$. To conserve space and to highlight the key

stylized facts, only three years are shown: 1985, 1995, and 2000. The countries are grouped by bloc: EPC Contracting States, Japan, U.S., and Other.²⁶ Note that because the data set goes up to 2000, we treat recently joined members of the EPC such as Hungary and Romania as Other Bloc countries.

For EPC Contracting States, a significant increase in Euro-direct first filings has occurred, not only in levels but as a share of all modes of filing. While a slight decline has occurred in the share of EP filings that are direct subsequent filings (i.e. ES), there has been a greater increase in the share of subsequent filings to the EPO via the PCT (i.e. PS). For example, in the U.K. in 1985, only 16.3% of all EP filings were subsequent filings via the PCT. By 2000, that share was 70.5%. Likewise in Germany in 1985, that share was 8.1%; by 2000, it was 48.3%.

For Japan and the U.S., total EP filings have more than tripled over the 15 year period shown. Yet, compared to the EPC Contracting States, the percentage of EF's are quite small; in other words, not a large share of U.S. and Japanese filings at the EPO are direct first filings. Moreover, this share of EF's by the U.S. and Japan has declined over time. In its place, there has been a tremendous increase in Euro filings via the PCT (particularly subsequent filings). In other words, PS is the most popular mode of filing for Japanese and U.S. patent applicants.

²⁶ Note that the table does not provide an exhaustive list of EPC members. A select sample of the main patenting economies are shown.

Among the Other Bloc countries, there is quite a bit of variance in modes of filing. Some countries (Argentina, Bulgaria, and Indonesia) have a high share of priority filings directly at the EPO but the actual volume of filings is relatively small. During the early 1980s, countries like India and Poland have had a relatively high share of EF's (i.e. Euro-direct first filings) in their EP filings; however, by 2000, these countries have switched to using the PCT system to obtain patent protection in the EPO. The rest of the Other bloc countries tend mostly to file subsequent patent applications to the EPO (whether directly or via the PCT).

Overall, Table 9 shows that the majority of EPO patent applications are subsequent filings. Most of those are filed directly at the EPO rather than through the PCT. Of all the Euro-direct filings, only 17.8% are first (priority) filings. However, the trend in Euro-direct first filings is increasing, particularly among the EPC Contracting States and the U.S.

ii. Regression Models

Estimates of the forecasting equations are shown in Table 10A - E. For each variable (i.e. Total, EF, ES, PF, and PS), we look at different model representations: AR1 (autoregressive model of order 1), AR3 (autoregressive model of order 3), RE1 (regression model with a lagged dependent variable and R&D), and RE3 (regression model with the first three lagged values of the dependent variable and R&D). In other words, RE1 is AR1 augmented with R&D, and RE3 is AR3 augmented with R&D. Other variables were included in preliminary analyses, such as GDP per capita, but were found to be correlated with R&D and/or contributed marginally to the forecasting exercises.

Thus the RE3 model is:

$$(15) \ln\left(\frac{P_{it}}{L_{it}}\right) = \eta_0 + \eta_1 \ln\left(\frac{P_{it-1}}{L_{it-1}}\right) + \eta_2 \ln\left(\frac{P_{it-2}}{L_{it-2}}\right) + \eta_3 \ln\left(\frac{P_{it-3}}{L_{it-3}}\right) + \eta_4 \ln\left(\frac{R_{it}}{L_{it}}\right) + \mu_{it}$$

where P denotes patent filings at the EPO, L labor, R research and development expenditures (in real 1995 U.S. dollars), and μ_{it} the error term. Subscript i indexes the source country and subscript t indexes time. The AR3 model is where η_4 is set to zero. The RE1 model is where $\eta_2 = \eta_3 = 0$ and the AR1 model is where $\eta_2 = \eta_3 = \eta_4 = 0$.

The motivation for these different models is to highlight the role of R&D in predicting patent filings. A comparison between AR1 and RE1, or between AR3 and RE3, determines whether R&D has any predictive power over and above the autoregressive model. A large fraction of the variation in the data can be captured by autoregressive terms (i.e. by the lags of the dependent variable) without any additional variables like R&D.²⁷

Table 10A presents the estimated equations for forecasting total filings at the EPO. The models were estimated over the period 1980 - 1996, so that out of sample forecasts for 1997 - 2000 can be made. Column 1 presents the results of the AR1 model. Judging by the coefficient estimate of the own lag of 0.973, there appears to be quite a bit of persistence in patent filings at the EPO. The coefficient on the first lag is not as high (i.e. is just under 0.7) once the second and third lagged dependent variables are included (see column 2). The estimated coefficient on the first lag is also lowered if, instead of adding two more lags, we include the R&D variable. The patent-elasticity of

²⁷ Formally, we could determine the appropriate lag length, via quasi-likelihood ratio (QLR) tests and so forth, for each model specification (and each variable). However, for manageability, we examine either 1 or 3 lags of the dependent variable. For practical purposes, this tends to suffice.

R&D is measured to be 0.106 (meaning that a 1% increase in a source country's R&D leads, on average, to a 0.106% increase in its EPO patent filings). On the other hand, this measured elasticity falls to 0.073 once all three lags of the dependent variable are included (as in the AR3). The R&D variable, however, is still strongly significant statistically. The various models explain about 98% of the variation in EPO filings.

As for EF filings (i.e. Euro-direct first filings), the R&D variable is also an important predictor and the lags of the dependent variable are also statistically significant (although the level of significance of the third lag diminishes somewhat). About 91-93% of the variations in EF are captured by the models. As for ES filings (i.e. Euro-direct subsequent filings), R&D is statistically significant at the 1% level if the first lag of the dependent variable is included, but is significant at the 5% level if all three lags of the dependent variable are included. About 96-97% of the variations in ES are accounted for.

As for the Euro via PCT filings, R&D is statistically significant at the 1% level when the first lag of the dependent variable is included, but only at the 10% level if further lags of the dependent variable are included. Thus R&D does not explain PF (i.e. Euro via PCT first filings) relatively well. A lot more "noise" appears to be present in the variations in PF. As for PS (i.e. Euro via PCT subsequent filings), R&D and only the first lag of the dependent variable appear to have predictive power.

Thus, to sum up, R&D matters for EP filings in the aggregate (that is, across technological fields), but matters moderately for first filings arriving at the EPO via the PCT. R&D typically has a patent-elasticity of less than 0.2 (and more than 0.06).

iii. Forecast Accuracy

For each of the forecasting equations discussed above, we can examine their forecasting performance. As mentioned earlier, each of the above equations was estimated over the period 1980-1996. The estimated – or fitted – equation can thus be used to generate out-of-sample forecasts of filings in 1997 - 2000. In Table 11, we show the root mean squared percentage error (RMSPE) associated with each equation's performance in predicting patent filings (whether it be total, EF, ES, PF, or PS) in year 1997, 1998, 1999, and 2000. Thus for each year, each of the models generates a forecast for each source country.²⁸ By comparing these forecasts with the *actual* source country filings, the RMSPE computes a summary measure of the forecast errors across the source countries. (The next table will show the underlying sample forecasts for individual countries, but first we examine the RMSPE's).

In general, the AR3 model produces lower forecast errors than the AR1, and RE3 produces lower forecast errors than RE1, particularly for the later years. If the more parsimonious specifications do better, they do so usually for the 1-step ahead forecasts (i.e. for predicting 1997's values). Moreover, in general, adding R&D helps to improve forecast accuracy (but again there are some exceptions). It is ambiguous, though, as to whether forecast accuracy could be improved by adding more lagged dependent variables to the AR1 model or by adding R&D. In other words, it is generally difficult to compare between AR3 and RE1.

²⁸ That is, predict patent filings given the actual values of the right hand side variables (RHS), such as R&D.

iv. Sample Forecasts

Table 12 provides some sample forecasts for the underlying sample of source countries. In this table, we select forecasts for 1998 as an example. This step could easily be replicated or reproduced for other years. A comparison between actual and predicted values is made for each of the five modes of filing: Total, EF, ES, PF, and PS. To conserve space, the predictions from just two models are reported: the AR3 and RE3. The bottom of the table reports again the RMSPE associated with a model for that year (which matches the information given in Table 11).

For Total filings, the AR3 and RE3 both under-predict the overall (i.e. all country) filings, but not by much. The shortfall is about 7% of the actual filings. For the U.S., the RE3 model's forecast is 97% of actual; for Japan, it is about 95%, and Germany 86%. For overall EF (Euro-direct first filings), the AR3 and RE3 models are off by about 1800 filings (which are about a quarter of actual filings). The models under-predict EF for the U.S., Germany, France, Switzerland, and the Netherlands, but over-predicts it for the U.K. and Japan. For ES (Euro-direct first filings), the models perform somewhat better. The RE3 model, for example, under-predicts overall filings by about 2400 (or 5.75% of actual). By individual country, the RE3 over-predicts for the U.S., under-predicts for Germany, and is nearly exact for Japan, U.K., and Korea.

For Euro via PCT filings, the forecast of overall (across countries) PF is about 82% of actual filings. Quantitatively, there are not as many first filings in the EPO coming via the PCT system. For subsequent EPO filings arriving via the PCT, the models do reasonably well. The overall prediction is about 99% of actual PS filings. The RE3 model slightly over-predicts for the U.S. and U.K. as well as for some of the smaller economies, under-predicts for France, Germany, and Japan,

and is nearly on target for Canada and the Netherlands.

3. Sectoral Filings (Joint Clusters)

We now examine the EP filings by technological field. The patent applications at the EPO are put into one of fourteen technological divisions, called Joint Clusters (JC):

- JC1 Electrical Machines and Electricity
- JC2 Handling and Processing
- JC3 Industrial Chemistry
- JC4 Measuring and Optics
- JC5 Computers
- JC6 Human Necessities
- JC7 Organic Chemistry
- JC8 Audio Visual
- JC9 Civil Engineering and Thermodynamics
- JC10 Electronics
- JC11 Polymers
- JC12 Biotechnology
- JC13 Telecommunications
- JC14 Vehicles and General Technology

For this paper, we do not analyze the filings in each of these JC technological fields. Rather we examine functionally related groups of JC's. A judgement call has to be made as to which JC classes should be grouped together. An alternative way might have been to examine the statistical properties (such as whether certain series of JC filings move together over time), but this does not in and of itself imply any functional comparability.²⁹ Thus this paper has put the fourteen JC fields into the following five groups (G1 to G5):

G1.	Group 1	Electronics-Related	Consists of JC 1, 8, and 10
G2.	Group 2	Chemicals-Related	Consists of JC 3, 7, and 12
G3.	Group 3	Manufacturing-Related	Consists of JC 2, 6, and 11
G4.	Group 4	Physics-Related	Consists of JC 4, 9, and 14
G5.	Group 5	Computer-Related	Consists of JC 5 and 13

i. Sample Statistics

Table 13 shows the percentage distribution of filings by Joint Cluster (1 - 14) and by the larger meta-categories G1 - G5. We compare two years: 1985 and 1998, and we provide sample statistics for the five leading patenting source countries (U.S., Japan, Germany, U.K., and France) as well as for countries as a whole. The table shows the technological composition of filings for four variables: EF (Euro-direct first filings), ES (Euro-direct subsequent filings), PF (Euro via PCT first

²⁹ For example, two fields of invention Y and Z may both pertain to mechanical engineering, but inventions in field Z may occur erratically over time. This does not have any bearing on whether Y and Z should be grouped together as a class on functional grounds.

filings), and PS (Euro via PCT subsequent filings). There are thus 8 panels in the table (2 years per variable).

The following are the main highlights: in 1985, industrial and organic chemistry dominated U.S. direct first and subsequent filings at the EPO. By 1998, computers and telecommunications dominated U.S. EF's and electronics and human necessities dominated U.S. ES's. In 1985 and 1998, handling and processing dominated U.S. PF's (as did chemicals in 1998). In 1985, chemicals and human necessities tended to dominate US subsequent Euro via PCT filings; but by 1998, biotechnology emerged as an important source of PS filings from the U.S. For Germany, patent filings tended largely to be in the fields of civil engineering and handling and processing (in both 1985 and 1998). This finding does not seem to vary by much among the different modes of filing. Germany does exhibit emerging strength in telecommunications in first filings (both Euro-direct and Euro via PCT). For Japan, electronics emerges as an important field for EPO patenting, particularly as subsequent filings; for the U.K. and France, patenting is relatively active in telecommunications as well as in civil engineering and general technology.

For countries as a whole, among EF filings, we observe a transition from industrial chemistry and handling and processing in 1985 to organic chemistry, telecommunications, human necessities, and civil engineering in 1998. Among ES filings, patent applications in handling and processing account for the largest share. Among Euro via PCT filings, civil engineering used to dominate in 1985, but by 1998, filings are more evenly spread across technological fields.

Next, we comment on the R&D data by sector. In analyzing patent filings by JC, it would be ideal to be able to match (and regress) those filings against the R&D conducted in those technological fields. Unfortunately, R&D data are not disaggregated along the lines of any particular

patent classification system, be it Joint Clusters, U.S. patent classification, or International Patent Classification (IPC). R&D data are typically broken down by industries (according to standard industrial economic classifications). Clearly there is overlap between the patent and industrial classifications: each JC group, for instance, could apply to two or more industries, and vice versa.

On the other hand, it is not clear (a priori) that the influence on a technological field should be restricted to the R&D of a certain group of industries. Due to the potential for R&D knowledge spillovers, across sectors, it is possible that the R&D conducted in one technological field may have potential benefits and applicability in other fields.³⁰ It is therefore difficult to preclude the R&D of any particular sector from having influence. For this reason, a broader definition of R&D may be useful in explaining JC filings. Of course, some sectors have more influence than others in influencing the innovative activity in any particular technological field. In other words, there should be a probability distribution defined over the various sectors that can have influence on the R&D of each field (with some sectors having greater weight than others).³¹ The degree of spillovers between sectors or between lines of research should depend, among other factors, on how closely related the sectors or fields are (for instance, a greater likelihood of knowledge spillovers may occur between Aircraft and other transportation than between Aircraft and Chemicals).

Methods exist for producing a concordance between patent classifications and industrial classification, and/or for deriving the probability distributions discussed above. The methodologies

³⁰ For example, the semi-conductor industry enjoyed positive externalities from the quartz watch industry. Various consumer goods have also been developed based on R&D conducted in the defense industry.

³¹ A problem that needs to be confronted is that this probability distribution may change over time, or be influenced by policy regime shifts.

vary in approach and are subject to various limitations. No clear consensus on the “ideal” approach exists to date. As a compromise this study uses (a) business enterprise research and development (BERD) for all industrial sectors (in order to take into account the possibility of far reaching inter-sectoral R&D spillovers) and (b) ad-hoc matching of industrial sectors to JC groups.

For the second approach, a judgement has to be made as to which types of R&D may be most relevant to each JC Group. The following ad-hoc matches were made (after various trial and error at explaining the group filings):

<u>Group 1</u> (Electrical Machines and Electronics)	⇐	R&D in Electronics and Electrical Machines And Utilities (Electricity, Gas, and Water)
<u>Group 2</u> (Chemicals and Biotechnology)	⇐	R&D in Food, Petroleum, Chemicals, Rubber and Plastics.
<u>Group 3</u> (Handling & Processing, Human Necessities, and Polymers)	⇐	R&D in Agriculture, Textiles, Food, Wood, Petroleum, Chemicals, and Rubber/Plastics.
<u>Group 4</u> (Measuring & Optics, Civil Engineering, Thermodynamics, Vehicles & General Technology)	⇐	R&D in Manufacturing (at large) ³²
<u>Group 5</u> (Computers and Telecommunications)	⇐	R&D in Nonmetal Products, Electronics, Electrical Machines, Services, and Office Equipment and Machinery

Table 14 provides some sample statistics of R&D broken down by industrial sectors.³³ Again

³² It was difficult to find a narrower range of R&D to explain patent filings in Group 4. Manufacturing R&D is BERD minus agricultural, utilities, construction, and services R&D.

³³ The source of data is the OECD’s *Basic Science and Technology Statistics*, Table 9 (Revisions 2 and 3), accessed via www.sourceoecd.org.

two years of data are examined (1985 and 1998) as well as the same selection of countries as in Table 13. In 1985, total R&D expenditures – across the sample of countries – was \$232.6 billion U.S. dollars (in real 1995 terms). In 1998, this figure increased to \$345.3 billion in real terms. The U.S. accounted for nearly half the world's R&D in both years. France, Germany, Japan, and the U.K. together accounted for about 38% of total R&D in 1985 and about 36% in 1998.

For all countries in 1985, the leading R&D sectors were Electrical Machines and Electronics, Transportation, Office and Computing, and Chemicals. In 1998, the leading sectors were Services, Electrical Machines and Electronics, Transportation, and Chemicals. The service sector reflects the increased importance of finance and business services (including software services). R&D in services also complements the development of machines and equipment (for transportation, communications, wholesale and retail trade, real estate, and financial intermediation). The U.S. especially leads in services-related R&D. For Japan, electronics R&D is a major part of national R&D, and for Germany and France, it is transportation R&D, and for the U.K., chemicals R&D.

ii. Regression Models

The forecasting equations are estimated separately for each JC Group. These results are in Tables 15 - 19 (where Table 15 pertains to Group 1, Table 16 to Table 2, and so forth). For each of the five groups, we consider total filings as well as filings broken down by mode of filing (EF, ES, PF, and PS). Part A of each table pertains to total filings, Part B to EF filings, Part C to ES filings, and so forth.

For each of the variables to be forecasted – i.e. total filing, EF, ES, PF, and PS – we compare

the AR3 model to two other models incorporating R&D: namely RE1 and RE3. Recall that RE1 is AR1 augmented with the R&D per worker variable, and RE3 is AR3 augmented with R&D. The R&D variable in both the RE1 and RE3 represents business enterprise research and development (BERD). The BERD is our general, sector-wide measure of R&D. We then replace BERD with a measure of R&D that is tailored for a particular JC group. (The previous sub-section discussed the construction of this second measure of R&D.) We will refer to this more narrowly measured R&D variable as a *sector-specific* one. We will also refer to *RE1s* as the RE1 model where the sector-specific R&D variable replaces the more general BERD. Likewise *RE3s* is the RE3 model with the sector-specific R&D variable replacing the BERD variable. Thus, for each of the variables to be forecasted, five models are estimated (AR3, RE1, RE3, RE1s, and RE3s). The estimation results of each model correspond to columns 1 - 5 respectively.

The models are estimated from 1980 - 1996, so that out-of-sample forecasts can be generated for 1997 - 2000. Instead of going through each of the estimated models in detail, we will provide the main highlights. For JC Group 1, as shown in Table 15 A - E, business enterprise R&D is a statistically significant determinant of total filings, Euro-direct subsequent filings, and Euro via PCT subsequent filings (though the significance level falls to the 10% level for Euro via PCT subsequent filings when the first three lags of the dependent variable are controlled for). Business enterprise R&D is not important for first filings (directly or indirectly) at the EPO. The more narrow, sector-specific R&D is statistically significant at the 1% level but only if just the first lag of the dependent variable is included. The patent elasticity of R&D (where statistically significant) appears to be no more than 0.4%.

For JC Group 2, as shown in Table 16 A - E, BERD and sector-specific R&D are also

statistically significant for total filings (with a patent-elasticity ranging from 0.072% to 0.348%). By mode of filing, R&D in general is mildly significant for subsequent filings, ES and PS, while sector-specific R&D is mildly significant for PS only.

For the other JC groups (3, 4, and 5), business enterprise R&D is consistently important for explaining total filings. Sector-specific R&D can strongly explain total filings only if just the first lag of the dependent variable is included. Thus the narrowly defined measures of R&D spending of specific sectors tend to proxy for some of the dynamics captured by the autoregressive terms. The more general measure of R&D – namely BERD – appears to better capture the extent of the inter-sectoral spillovers of knowledge that influence innovative activity than do the narrower measures of R&D.

By mode of filing, BERD is strongly significant in explaining the Euro-direct subsequent filings of Groups 3, 4, and 5. Sector-specific R&D is statistically significant at explaining the Euro-direct first filings (EF's) and the Euro via PCT first and subsequent filings (i.e. PF's and PS's) of Group 3.

In summary, it is relatively more difficult to use measures of R&D to explain the variations in the data by mode of filing. R&D is likely to better explain the *volume* of innovative activity and diffusion of technology in particular technological fields; however, it does not strongly aid in explaining the choice of route or of the timing of patent applications. Other factors are needed to help distinguish the advantages or disadvantages of each mode of filing (such as cost, procedural factors, or institutional factors). To the extent that R&D can explain some modes of filing, it is the variations in subsequent filings that it best explains. Most of the EP filings are subsequent filings. Thus this is consistent with the idea that R&D helps to explain the volume of patenting activity more

than the composition by mode of filing.

iii. Forecast Accuracy

Table 20 A - E reports on the root mean squared percentage errors associated with each of the JC regressions discussed above. Each of the estimated models was used to make forecasts for 1997 - 2000 inclusive. Note that panels A - E of Table 20 refer to JC Groups 1 to 5 respectively. Again we describe the main highlights.

In general, the performance between the AR3 and RE3 models are comparable, particularly where R&D is statistically insignificant; after all if R&D is important, it would contribute to lowering the RMSPE's. The models where sector-specific R&D measures are used (i.e. RE1s and RE3s) typically do not perform as well as models where general business enterprise R&D is employed, particularly for forecasts by mode of filing.

For group 1, the RE3 model generally performs best for predicting total filings, and RE3s also performs comparably well. For group 2, the RE3 model performs better than the AR3 for total filings as well as for subsequent filings. The RE3s model (using sector-specific R&D) particularly does not do well at predicting Euro-direct first filings (as the $\text{RMSPE} > 1$). For group 3, the use of sector-specific R&D provides good predictions for 1997 and 1998, relative to the use of BERD, but yields mixed predictive success for later years (or longer horizons). For first filings (EF or PF), the AR3 and RE3 provide comparable predictions. For group 4, the models using general R&D perform better than the models using sector-specific R&D, suggesting that for this group the BERD contributes more to predictive accuracy than manufacturing R&D. For predicting subsequent filings

(ES and PS), the RE3 performs better than the AR3. Finally, for group 5, the models using sector-specific R&D (i.e. RE1s and RE3s) produce large forecast percentage errors. The smaller sample sizes may be a factor (due to limited data on R&D in the computer and telecommunication sectors). The RE3 model (using general R&D) performs better than the AR3 at predicting total filings and subsequent filings.

iv. Sample Forecasts

Next we show some forecasts for individual source countries. The above table gave only the overall summary statistic of the forecasts (in terms of the RMSPE). In Table 21 A - E, we show the individual country forecasts behind the summary statistics for 1998. To conserve space, we show only the results from the RE3 model using business enterprise R&D. We compare the actual values and predicted values by JC group for each of the five variables of interest: total filings, EF, ES, PF, and PS. The table reproduces the value of the RMSPE at the bottom of the table.

For total filings, the model does quite well for JC Groups 1 - 4. For instance, in terms of overall country filings, the predicted sum of group 1 filings is 90% of the actual sum (i.e. aggregating across countries). The predicted cross-country sum of group 2 filings is 94.3% of the actual sum; for group 3 it is 95.8%, and for group 4, it is 91.3%. But for group 5, the predicted overall sum of filings is 74.1% of actual. Quantitatively, there are not as many filings in Group 5 as there are in each of the other groups. Perhaps a greater degree of uncertainty or unpredictability is a characteristic of innovation in computers and telecommunications such that actual patenting activity deviates substantially from what trends in R&D and past patenting behavior would suggest.

As for forecasts of EF, the volume of filings is small, but the model's forecasts are fairly tight. Again, forecasts for group 5 filings are poorest. The overall predicted sum of filings is 58% of the actual. For forecasts of ES, the predictions are quite close to actual for all JC groups. For example, for group 1, overall forecasts (across countries) are 96.8% of actual; for group 2, they are 99.4% of actual, and even for group 5, they are 94.5% of actual.

Forecasts of Euro via PCT first filings are relatively not as good. For group 1, for instance, overall forecasts are 68.4% of actual; for group 3, 79.2% of actual; and for group 5, 64.1% of actual. Forecasts of Euro via PCT subsequent filings are better. For instance, for group 2, overall forecasts are 96.4% of actual; for group 4, 92.2% of actual; and for group 5, 77.6% of actual.

These results support the earlier contention that predictive accuracy is greater for subsequent filings. Nevertheless, the forecasts are on the whole relatively close to actual. This is attributable to the inclusion of the autoregressive terms. Patent filings tend to exhibit a fair degree of momentum or persistence over time. Moreover, R&D is an important influence on innovation and decisions to patent and to file internationally.

4. Patent Family Filings

In this section we shift our attention from patent applications to patent families. International patent applications and other documents relating to the *same invention* would comprise a patent family. The patent family data are indexed by the priority number of the first filing, with information on subsequent filings for that invention in four blocs (EPC contracting states (including the EPO),

U.S., Japan, and Other).³⁴ The database can be filtered to select different types of patent families (for example, Trilateral patent families, which are patent families involving patent filing activity in each of the trilateral blocs: U.S., Japan, and EPC).

Patent family data depend on the appearance of patent publications that can *index* a patent family (link subsequent filings and priority filings). Due to this dependence, there is a *timeliness problem* because of patent publication lags. Consequently, there may be some under-reporting of subsequent filing activities connected with an earlier priority forming filing. This particularly affects the more recent years, such as 1999 and on. The timeliness problem particularly affected data for the U.S. where, until 2000, patents were published only upon grant. Thus the figures for the U.S. could only be updated after several years of delay.

i. Sample Statistics

The following tables (22 – 26) describe some sample statistics derived from the EPO's patent family database. The database provides patent family counts by priority country or priority bloc as well as by bloc of destination (for subsequent filing) and by the associated route of filing (e.g. via national offices, PCT, direct EPO, or EPO via the PCT). The database provides information annually and by field of technology (e.g. International Patent Classification Code). The data list four possible blocs that can comprise a patent family -- namely the EPC, Japan, U.S., and Other. For purposes of deriving these sample statistics, several indicator (or 'dummy') variables were

³⁴ At the EPO, the data file containing information on patent families is called *PRI*, which in turn is derived from the database of published patent documents (called *DOCDB*), which also is maintained at the EPO.

constructed. For example, for each priority country (per year and per IPC code), there are fifteen possible permutations for a family: (1) EPC, (2) Japan, (3) US, (4) Other, (5) EPC and Japan, (6) EPC and the US, (7) EPC and Other, (8) US and Japan, (9) US and Other, (10) Japan and Other, (11) EPC, Japan, and the US, (12) EPC, Japan, and Other, (13) EPC, US, and Other, (14) US, Japan, and Other, and (15) and all four (EPC, Japan, US, and Other). Of course, combinations (11) and (15) comprise trilateral patent families. Thus, for each combination, an indicator variable was created such that it equals one if this combination describes the data point (that is, characterizes a patent family per priority country, per year, per IPC code, and per given route of filing) and zero otherwise. The combinations are mutually exclusive and thus exhaust the whole data set. In other words, the sum of the counts of all the combinations equals the total counts in the data set (or percentage shares of each combination add up to 100%). These indicator variables make it easy to identify the types and quantity of secondary filings associated with each priority country.

Table 22 provides a breakdown of international patent families that are based on priorities filed in 1979, 1984, 1989, 1994, and 1999. (The format follows that of the Statistical Annex of the *Trilateral Statistical Report*. The main difference is that the EPC and Other Country Blocs for which priority is claimed are disaggregated by country.) The 1999 figures (in panel 1 of Table 22) are provisional, so that there will be an underestimation of patent family formation for this particular year (as will be pointed out below). In Table 22, the subsequent filing destinations are available only by blocs (EPC, US, Japan, and Other). The first column provides the quantity of first filings associated with the priority country and the remaining columns show different types of subsequent filings as a percentage of first filings. Note that these percentages (in each row) need not sum to 100% since the various bloc combinations shown are not mutually exclusive.

In 1999 (as in all other years), Japan leads with the most first filings (in excess of 350,000), the vast majority of which are domestic filings. The very large number of such filings has been attributed to the practice in Japan of filing domestic patent applications with single claims. One observer has suggested scaling Japanese domestic patent applications down by 5 to make them comparable to patents elsewhere in the world; however, this is not attempted here.³⁵ The U.S. has the second most priority filings, with a total of 153,350, which is slightly more than the Other Country Bloc. The EPC as a whole has 130,999 first filings, almost half of which come from Germany. The second most first filings within the EPC are from the U.K., followed by France. About 12,000 direct first filings at the EPO take place in 1999, which are just under 10% of all first filings within the EPC bloc. Among the Other Country Bloc, most first filings occur in China, followed by Korea, Russia, and Australia.

In 1999, more than 40% of U.S. first filings formed patent families with at least one other bloc (EPC, Japan, or Other), 30.1% formed patent families with at least one other trilateral bloc (Japan, EPC, or both), 32.2% formed patent families with at least a non-trilateral country or bloc, 28.7% formed patent families with at least the EPC, 9.2% with at least Japan, and just under 8% formed a trilateral patent family. For Japan, the percentages of patent family formation are generally smaller due to the large number of its first filings (i.e. in the denominator). For the EPC as a whole, the rates of patent family formation are generally between those of the U.S. and Japan. Switzerland leads with the highest rate of trilateral patent family formation, followed by France, then Germany. The medium to lower income EPC states, such as Greece, Ireland, Monaco, Portugal, and Spain produce a negligible number of trilateral patent families. Among the Other Bloc countries, the

³⁵ See Y. Okada (1992).

largest number of trilateral patent family counts comes from Korea, and very small or zero trilateral patent families come from Brazil, Bulgaria, China, Hungary, Mexico, Philippines, Poland, Romania, and Russia.³⁶ The EPC obtains relatively most secondary filings from Korea, Australia, Canada, New Zealand, Norway, South Africa, and Israel. Again, it should be reminded that these observations for year 1999 should be treated as provisional until further subsequent filing data are obtained.

In fact, the 1994 figures (see the second panel of Table 22) show that the rates of patent family formation are noticeably higher. For example, 13.4% of EPC first filings formed trilateral patent families (vs. 3.9% in 1999) while 14.2% of U.S. priority filings formed trilateral patent families (vs. 7.8% in 1999). For Japan, the rate of trilateral patent family formation is slightly different (5.2% in 1994 vs. 3.8% in 1999). But the ranking and distribution of sources of subsequent filings is fairly stable; for example, Germany is the leading source of patent families involving the EPC, followed by the U.K. and France. Among the Other bloc countries, Korea remains the leading source of international subsequent filings (i.e. EPC, US, Japan, and Other Blocs as a whole), followed by Australia, Canada, Norway, and South Africa; overall, however, the rates of patent family formation are smallest among the other country bloc.

The pattern of first and secondary filings is largely similar in 1989 for the EPC and Japan as in 1994. The rate of patent family formation for the U.S. is generally higher in 1989 (for example, 20.6% of U.S. priority filings formed trilateral patent families vs. 14.2% in 1994). Among other countries, a noticeable fact is that Korea's patent priority counts in 1989 are about a third of that in

³⁶ Note that for the Other Bloc, the second and third columns coincide as a result of the fact that where a patent family is formed with at least one other bloc, that bloc is one of the trilateral blocs.

either 1994 or 1999, since around that time Korea had reformed and strengthened its patent regime and the effects on innovation in Korea appeared (later) with a lag.³⁷

In 1984 and 1979, the volume of first and secondary filings is smaller than in later years, but the rates of family formation are fairly similar to those in 1989. The U.S., however, has a greater proportion of international patent families in 1984 than in the later years.

As an overall summary (of Table 22), there has been a rising trend in priority filings (with the exception of a slight cyclical downturn for the Other Country bloc in 1994). The level and proportion of Japanese and EPC priority filings that form patent families in at least one other bloc has increased over the period 1984-1994, as has the level and proportion of first filings that form trilateral patent families. For the U.S., however, the proportion of priority filings that form trilateral patent families has been falling since the mid-1980s. This decline may not necessarily reflect a decrease in the propensity to patent abroad, but reflect other factors. One possibility (held by critics of the U.S. patent system) is that the U.S. experienced an increase in marginal patent applications due to a relaxation of novelty and nonobviousness standards. This would explain the large increase in first filings and the little impact this had on subsequent international filings (since those patents are not valuable enough to warrant filing abroad). But another possibility is that the cost of patenting has increased for U.S. patentees (particularly the translation costs in the EPC, Japan, and Other bloc,

³⁷ See Park (2001) for a global comparative analysis of patent regime changes. Korea had significantly increased the strength of its patent system between 1990 and 1995, as its domestic firms developed increased innovative potential and international presence. As a comparison, see the Gadgaw and Richards (1988) study which shows how Korea (prior to 1990) had one of the weakest patent regimes (similar to that of India, China, and Brazil). Korea's experience supports Evenson's (1990) observation that intellectual property laws and policies are often endogenous (or respond to) domestic R&D and innovation activity, as the latter gives policymakers something of interest to protect. Otherwise, authorities may favor local imitators.

and possibly the costs of agent representation), and thus lowering the rate of subsequent filings. Yet another possibility is that since patentability standards vary across countries, U.S. applicants may file for protection for certain inventions (such as software, biotechnological innovations, internet-related innovations, and so forth) at home but not elsewhere where those inventions are not considered patentable subject material.

Table 23 presents a breakdown of international patent families by IPC code for priority years 1984 and 1994. In a typical year, comparing across IPC codes, it can be observed that patenting activity is relatively greatest (in terms of the volume of filings) in Performing Operations (IPC code B), followed by Physics (code G), Electricity (code H), Human Necessities (code A), and Chemistry and Metallurgy (code C). Mechanical engineering (code F) is of medium intensity in Japan and the EPC, but of relatively low intensity in the U.S. Patenting activity is least intense in Textiles and Paper (code D) and Fixed Construction (code E). In terms of the rates of international patent family formation, the 1999 figures are underestimates due to data reporting lags. In the years prior to 1999, more normal rates of international patent family formation can be observed. In terms of forming trilateral patent families, there is quite a high volume in Chemistry and Metallurgy, particularly among EPC states, where 23.8% of priority filings in this field are subsequently filed in the U.S. and Japan. There is also quite a high volume of trilateral patent families among Physics and Electricity inventions. The rate of trilateral patent family formation is also fairly high in Textiles and Paper, but the number of counts is relatively modest. Thus the distribution of international patent families by technological field is not even, with a slight bias among trilateral patent families toward the hard sciences fields (e.g. Chemistry, Physics, and Electricity).

Overall, the rate of international patent family formation by fields had increased by 1994. This reflects a greater increase in priority filings (i.e. the denominator) than in subsequent filings (i.e. numerator). However, the rates of international patent family formation by the Other Bloc countries as a whole have generally increased throughout the sample period.

Table 24 shows the growth rates of priority filings by IPC field in selected countries over the period 1984 to 1999, as well as 1984 to 1994 (given the provisional nature of 1999 figures). The purpose is to analyze which particular technological fields drove the increase in (priority) patenting. This too was not even across IPC fields. For example, for the U.S., significant increases in patenting can be observed in the broad fields of Physics, Electricity, and Human Necessities (related partly to developments in pharmaceutical innovations). Priority filings in Chemistry and Metallurgy also experienced robust growth. Growth in such filings was modest in Mechanical Engineering. For Japan, Electricity, Chemistry, Human Necessities, and Fixed Constructions drove the growth in priority filings, while Textiles and Paper and Performing Operations contributed weakly. For the EPC, Physics, Electricity, and Chemistry and Metallurgy are behind the growth in priority filings, with each of the leading three countries (Germany, U.K., and France) contributing strongly. There is, on the other hand, weak growth in Mechanical Engineering filings in the EPC, with the U.K. experiencing negative growth. As for the Other Country Bloc, Korea has had strong growth in all fields, more than doubling or tripling filings in the fields of Physics, Electricity, Chemistry and Metallurgy, and Mechanical Engineering. Australia also had strong growth in the hard sciences fields. Canada, on the other hand, had weak (in many cases, negative) growth in filings in many

fields. This prompted the Canadian government to undertake a review of the Canadian innovation system and innovation policies.³⁸

Table 25 focuses attention on priority and subsequent filings in the EPO. The data combine filings directly at the EPO as well as indirectly via the PCT. The table shows both the volume of subsequent filings and those filings as a percentage of the previous year's first filings. The percentages provide some idea of the transfer rate or transfer coefficient (i.e. fraction of priority filings that are subsequently filed). The lagged one year priority filings are used as a base (or denominator) since, under the Paris Convention, patent applicants have up to one year to exercise their priority rights in subsequent filings. The table presents figures for selected years, namely every five years from 1980 to 2000, and for each priority bloc (EPC, Japan, U.S., and Other).

The information on transfer rates from world wide first filings to subsequent EPO filings in Table 25 shows that the EPC contracting states are the source of most EPO subsequent filings, followed by the U.S., Japan, and Others. The general rise in transfer rates over time (particularly after 1985) reflects a "centralization effect" in which patent applicants discovered the option of using the EPO system that had started in 1978.

Over this period, the EPC itself is the source of most EPO subsequent filings, followed by the U.S., Japan, and Other. The other country bloc has the smallest share of subsequent EPO filings. Among EPC states, Germany is the leading source of EPO filings (with over 10,000 filings per year since 1990), with a transfer rate exceeding 20%; that is, a little more than one-fifth of Germany's

³⁸ Industry Canada, the government ministry in Canada which oversees the *Canadian Intellectual Property Office* (CIPO), had commissioned a series of research and other investigative studies on intellectual property policy. See, for example, Putnam (2003) for studies from a May 2001 conference organized by Industry Canada and for other references.

priority filings are filed subsequently at the EPO. Next are the U.K., France, Italy, and Switzerland. These countries' transfer rates are higher. Switzerland in 1995, for example, had over half of its first filings subsequently finding their way to the EPO. France in 1995 had 43% of its priority filings subsequently filed at the EPO. On the other hand, countries like Monaco conduct a negligible amount of secondary filings at the EPO (namely 1 or 2 filings per year). Greece, Ireland, Portugal, and Turkey file less than 100 subsequent patent applications at the EPO.

Overall the mean number of EPO filings is 1401 (with a coefficient of variation³⁹ of nearly 200%, reflecting the wide variation among the EPC states). The average transfer rate among EPC states is 22.6%. By comparison, the mean annual subsequent EPO filings from the U.S. and Japan exceed 10,000 with much less variation over time (that is, the coefficient of variation is about half). The average transfer rate for the U.S. is 19.9% (meaning that about one-fifth of U.S. priority filings find their way to the EPO within a year). For Japan, the average transfer rate is under 5% (due to its large number of domestic priority filings with single claims and which are not destined for foreign filings).

Among other countries, Australia has been the leading source of EPO filings, but Korea has been catching up and may well have exceeded Australia. The transfer rate of Korean priority filings to the EPO is only 2%, however, while that of Australia exceeds 10% per year. Canada and Israel, as mentioned before, are also important sources of subsequent EPO filings, though the quantity of filings is comparable to that of medium-income EPC states, such as Austria, Denmark, and Finland. Overall, the other country bloc on average transfers 6.9% of first filings to the EPO.

³⁹ That is, the standard deviation as a percentage of the mean.

Table 26 provides some sample statistics on the various ways in which patent applicants have filed for patents in the EPO. For example, different routes can be involved (e.g. national, direct EPO, PCT, or EPO via the PCT). Some applicants file only in the EPC area or form families with 2 or more blocs (such as the U.S., Japan, or Other). Thus the table shows patent family counts for various combinations of destination blocs and routes, for two particular years (i.e. 1984 and 1994) and for the period as a whole (1979-1999). Note that, in the table, reference to EPO and PCT indicates their involvement *somewhere* in the family.

The key observations on filings at the EPO are as follows: Most families that have patent activity only in EPC contracting states involve first filings at the national offices to establish priority rights. From a volume of such filings that exceeds 90 000, only a small proportion involve first filings at the EPO, but this proportion has increased in recent years. The PCT involvement is rather low when an applicant files only in the EPC contracting states. But the use of the PCT has been expanding considerably over the period from 1985 to 2000.

For patent families involving only the EPC and Japan, the routes of choice are still filing at the national office and EPO directly. The share of PCT involvement is small. The same pattern applies for patent families involving only the U.S. and EPC together.

For trilateral patent families, there is a more significant use of the direct-EPO route. Almost twice as many on average use the direct EPO filing versus national filings (i.e. 9908 vs. 5487). For patent families involving all four blocs, direct EPO filings are nearly four times as frequent as national filings (i.e. 12929 vs. 3716). The involvement of the PCT (either PCT or EPO via the PCT) is more significant when applicants choose to file in all four blocs. Involvement of the PCT is also popular for trilateral patent families as well as patent families involving the EPC, US, and Other; the

PCT route is modestly used when applying to the EPC, Japan, and Other. In other words, the U.S. destination appears to be important when taking the PCT route.

ii. Regression Models

We next consider the forecasting of patent families. We focus first on predicting first (priority) filings, and then on predicting numbers of families containing subsequent filings in the EPC, Other Bloc, Other Trilateral, or Trilateral. As explained earlier, the patent families database shows subsequent filings in one of four blocs: EPC (including the EPO), U.S., Japan, and Other.⁴⁰ The first three blocs are referred to as the Trilateral.

Conceptually, the priority filings reflect the overall level of new inventions that are patented in a particular period; the subsequent filings measure the “transfer” of those patentable inventions abroad. Thus priority filings are a measure of inventiveness⁴¹ and subsequent filings a measure of international technology diffusion. Here, first filings refer to priority filings of the source country.⁴²

R&D is modelled as determining both the first *and* subsequent filings. That is, R&D affects the innovative potential of a source country and the propensity to make subsequent filings. One reason that research and development can stimulate the “transfer” of technologies abroad is that the R&D expenditures may reflect the investment effort level (and possibly thereby the quality level)

⁴⁰ The destinations are not further disaggregated by country; for example, countries in the Other Bloc are not explicitly identified or separated out.

⁴¹ Subject to the qualification that not all inventions are patented or are patentable.

⁴² Here source country refers to the country of origin of the first filings (from which priority claims are made by all other patents). It does not refer necessarily to the country of residence of the patent owner or inventor.

of a source nation's patentable inventions. Thus the greater a source country's resources devoted to R&D per worker, the greater the number of innovations that might be worthy of patenting subsequently in other markets. Support for this view comes from previous studies on patent valuation. While one way to screen for valuable patents has been to observe renewable behavior over time, an alternative way is to observe the spatial dimension: to see which patents are applied for in more destinations (markets). In other words, larger patent families tend to be more valuable (holding other factors constant).⁴³ Thus, on theoretical grounds, R&D could stimulate both first and subsequent filings. The greater the R&D content, for example, in a nation's supply of patentable inventions, the greater is the proportion of priority filings that will be likely to be used as a basis for subsequent filings in other markets.

In what follows, we estimate models of subsequent filing behaviour for the sample years 1981 - 1994, corresponding to the priority filing years 1980 and 1993. The estimated models are then used to generate out-of-sample forecasts for families containing subsequent filings in 1995 - 2000. We then compare these forecasts with actual data. It will be seen that the models generally over predict for 1999 and 2000, which is consistent with the fact that there is a timeliness problem in reporting families incorporating subsequent filing activity for these years. It is for this reason that we expanded the out-of-sample forecast interval and decreased the in-sample estimation period.

We first examine the behavior of priority filings. First filings are a function of lagged first filings and R&D (in natural log and per worker units). First filings in turn are a determinant of

⁴³ For studies that infer patent value from patent renewal behaviour, see Pakes (1986), Schankerman et al. (1986), and Lanjouw, Pakes, and Putnam (1998). For studies that infer patent value from patent family size, see Harhoff et. al. (2003), Lanjouw and Schankerman (1999), and Putnam (1996). Harhoff et. al. (p. 1343), for example, argue that "patents representing large international patent families are especially valuable."

subsequent filings (since the latter can be traced back to the former). Table 27 reports on the estimation of the forecasting equation for first filings. As column 1 indicates, a high degree of dynamic persistence is exhibited in first filings. But the impact of the lagged dependent variable is reduced once other variables are controlled for; its coefficient for instance drops to the 0.719 - 0.756 range. The patent elasticity of R&D is 0.187 – that is, a 1% increase in R&D stimulates a 0.187% increase in priority filings.

In place of R&D, we also investigated the model using GDP per capita. The results are qualitatively and quantitatively similar; the patent elasticity of GDP per capita is 0.183. But GDP per capita tends to be correlated with R&D per worker since output per worker (or per capita) is a function of inputs like R&D. Thus it is not surprising that the two variables yield similar results. However, when both GDP per capita and R&D are entered jointly, the GDP variable is statistically insignificant (and of the wrong sign). Overall, these simple models capture 93% of the variation in international priority filings.

We next turn to the behavioral equations for subsequent filings. The results are in Table 28 A - D. Panel A focuses on counts of families with subsequent EPO filings; panel B on counts of families with subsequent ‘Other Bloc’ filings; panel C on counts of families with subsequent ‘Other Trilateral’ filings; and panel D on counts of families with subsequent ‘Trilateral’ filings. In these tables, we essentially compare two models: AR3 versus a model that uses the lagged dependent variable, lagged first filings, and R&D (referred to as the RE1 model).⁴⁴ Thus in effect we are comparing whether priority filings and R&D produce better forecasts than extending the AR1 model to an AR3. Note that first filings are lagged one period because, in accordance with the *Paris*

⁴⁴ That is, the random effects regression model with the first lagged dependent variable.

Convention, applicants have up to one year to file subsequent patent applications and to refer to the priority date associated with the first filing.

In Table 28 A, we find that a 1% increase in priority filings leads to a 0.062% growth in families with subsequent filings at the EPO, whereas a 1% increase in R&D stimulates a 0.25% increase in those filings. Table 28 B - C repeat the exercise for families with other kinds of subsequent filings (involving the Other Bloc, Other Trilateral, and Trilateral). Qualitatively, there is a similar pattern. The previous period's patenting activity or intensity influences a current period's activity. R&D also has a statistically significant influence on the technology "transfer" rates. However, lagged priority filings weakly determine the number of families with subsequent Trilateral filings and insignificantly determines those with subsequent Other Trilateral and Other Bloc filings. This finding implies that the mere size or stock of patentable inventions (as measured by the flow of priority filings) does not influence international technology diffusion (holding other factors constant). It is, in other words, not necessarily the case that the more inventions there are the more international patent family formation. Some countries have a relatively large number of priority filings, yet have a comparatively smaller propensity to patent abroad (e.g. Japan), while others have a relatively small number of priority filings, yet have a comparatively high propensity to patent internationally (such as Canada). This result might suggest that patentees have separate decisions concerning their priority filings and subsequent filings.

Yet in column 4 of Tables 28 B - C, we find that if we drop the lagged dependent variable, the lagged first filings variable is statistically significant at the 1% level. Thus this variable appears to be picking up the effects of lagged dynamics, or proxies for lagged patenting effects. Note, however, that the adjusted R-squared values are lower. A smaller percentage of the variations in the

data are captured without the inclusion of the lagged dependent variable. Furthermore it results in much higher root mean squared errors (not reported).

In an earlier study, Hingley and Nicolas (2003) find that transfer ratios – i.e. ratio of subsequent filings to first filings – tend to be stable per applicant country (over time). To reconcile this finding with our finding that first (priority) filings only weakly determine the subsequent family filings associated with those first filings, it should be noted that the analysis here is across source countries (as well as time). The transfer rate may still be stable over time for each country, yet across countries first and subsequent filings need not be strongly correlated.

iii. Forecast Accuracy

Table 29 shows the root mean squared percentage errors (RMSPE) associated with each patent family model estimated thus far. Each entry in Table 29 has a corresponding forecast equation to refer to (or “column” to refer to) in Table 28 (specifically the first three columns only, where applicable).⁴⁵ Again, since the estimated equations described in Table 28 were estimated up to year 1994, and we then use the actual (realized) values of the independent variables for 1995 - 2000 to make predictions of the dependent variable.

The forecast performance between AR3 and RE1 is fairly similar for families containing subsequent EPO filings. The AR3 performs relatively better for the earlier years (1995 - 1997) but the momentum captured in the AR3 does not extend well into a longer time horizon. For the number

⁴⁵ In other words, we do not report the RMSPE associated with the model without the lagged dependent variable (i.e. column 4 results).

of families containing Other Bloc, Other Trilateral, and Trilateral subsequent filings, the AR3 performs generally better than the RE1 model (which incorporates lagged priority filings and R&D per worker). However, the RE1 model generally performs better than the simple AR1 model. Nonetheless the differences in forecast errors between AR3 and RE1 are of small magnitude. Note the relatively large forecast errors for predicting Trilateral family filings for 1999 and 2000. This reflects the “timeliness” problem of obtaining actual, or realized, Trilateral patent family data for those years.

iv. Sample Forecasts

Table 30 provides some sample forecasts by source country for 1998. The first part of the table shows the predictions for the priority filings. The simple autoregressive model is outperformed by the RE1 model (i.e. AR1 augmented with R&D). Both models, however, under-predict the total filings (of all countries listed in the table) by about 200,000. This is due mostly to the prediction error for Japan. As indicated earlier, Japan’s system or practice of admitting single claims per patent application has resulted traditionally in a large number of first filings. The models have difficulty capturing this idiosyncrasy. Japan’s priority filings pose an outlier for these models. The models also under-predict the first filings of the U.S. and Germany.

For predicting the number of families containing subsequent EPO family filings, the RE1 model does generally better overall. The predictions are quite close for the U.S., Japan, and Germany. This is attributable to the high adjusted R-squares of the forecasting equations (indicating that a large fraction of the variation in the data are captured by the model). The RE1 predicts a total,

country-wide, forecast of 102,150 families with subsequent EPO filings. This is just 158 filings shy of the actual (or an error rate of just 0.154%).

For predicting the formation of other patent families, the models do reasonably well at predicting the subsequent filings. Though the RMSPE is slightly smaller for the AR3 (i.e. in terms of average individual country forecasts), the RE1 model produces a total, country-wide, forecast that is slightly closer to the actual. For instance, the RE1 model predicts 146,670 families containing subsequent filings in an ‘Other Bloc’ country. This is 1151 filings less than the actual (which is again an error rate of less than 1%). The RE1 model over-predicts actual families with subsequent filings in an ‘Other Trilateral’ country by 598 filings. The actual is 123,250. Lastly, the RE1 model over-predicts families with subsequent filings involving the Trilateral. The error rate is larger (namely 13.5%).

While the total, country-wide, forecasts of families containing subsequent Trilateral filings are relatively not as good, the forecasts for individual countries vary in accuracy. For example, for the U.S., the RE1 model produces good forecasts of families with subsequent Trilateral filings. But generally the model over-predicts these kinds of filings for most countries. The errors appear to be systematic and suggests that forecasts of 1998 may also suffer from the “timeliness” problem. Clearly the model has fit well historically over the truncated sample period (up to 1994), yet the RMSPE for 1998 exceeds 0.8 (suggesting that the predictions are 80% greater than, or almost double, the actual). This suggests either that the model has omitted important variables with predictive content, or that there is a severe lag in the reporting of actual families with subsequent Trilateral filings. It would be useful to re-do the forecasts at a later point in time in order to determine the more likely source of the forecast errors.

Conclusion

Patenting activity is intense within the EPO. The U.S., Japan, and the EPO account for the bulk of world patenting activities. This is in large part due to their relatively high incomes (which provides for larger markets), their greater productivity (which makes their inventors more prolific producers of knowledge capital), their greater R&D and science and engineering resources, and their stronger intellectual property systems. But the increased patenting in the EPO is also attributable to the institutional system itself. With the benefits of single filing and centralized procedures, the economic cost of patenting in EPO member countries has been reduced. Moreover, membership in the EPO has particularly helped the smaller member economies to obtain increased technology inflows. This is largely due to the low marginal cost of designating additional EPO states (beyond the top three to five states).

By 1999, nearly all the EPO states have been designated in EPO patent applications. Despite reducing fees and improving procedures for filing patents, the EPO receives very little patents from developing countries or emerging markets. This is due in good measure to factors internal to those nations (their policies and environment). However, to the extent that they depend on access to foreign markets in Europe, the U.S., and Japan for their development, their lack of involvement in international patenting activities becomes an important issue.

To highlight the main findings in this study, we found EPO filings per source country worker to be significantly explained by source country GDP per capita, R&D stock per worker (particularly

private R&D), scientists and engineers per 10,000 workers, patent protection levels, and EPO filing fees and EPO membership. We found the results to be fairly insensitive to changes in the specification of the basic model, such as the use of R&D flows instead of stocks, of alternative measures of human capital, the inclusion of FDI and export variables, and the introduction of cyclical shocks. The results are also insensitive to whether the dependent variable is patents per scientist and engineer or patents per R&D stock. An alternative model would relate cross-country patenting to trade and FDI. This alternative model explains a smaller percentage of the variations in the data. Moreover, there are likely to be simultaneity problems if trade and FDI are associated with patenting (as a means of gaining entry into a market).

In the forecasting exercises, the main highlights are:

- R&D better explains total filings rather than the different modes of filing (EF, ES, PF, and PS). It does not explain well why agents choose to file at the EPO first or subsequently or why they would (or would not) opt to seek protection in the EPO via the PCT system.
- A general, economy-wide measure of R&D (such as business enterprise R&D) predicts sectoral filings better than sector-specific R&D because it likely captures inter-sectoral knowledge spillovers more broadly, though this finding might be due to the imprecise concordance between industrial R&D and patents by technological fields.
- Models with lags of the dependent variable (usually the first three lags) plus R&D produce quite good forecasts. Typically 90% or more of the actual filings is predicted. Note though that these forecasts are generated using actual R&D values realized during the forecasting period. In practice, for predicting filings beyond the sample period (e.g. 2005 and beyond),

we will need to generate forecasts of R&D as well (and thus develop models of R&D behavior).

- Generally, predictions of Euro-direct (or Euro via PCT) *subsequent* filings are better (in the sense of lower RMSPE) than predictions of Euro-direct (or Euro via PCT) *first* filings.
- Forecasts by technological (Joint Cluster) groups (i.e. meta-Groups 1 - 5) were also quite close to actual, except for Group 5 (computers and telecommunications). As with forecasting aggregate filings (across technological fields), the models forecast better for total filings than for filings by different modes of filing. Also, at the sectoral level, subsequent filings are easier to predict than first filings.
- For predicting counts of patent families, R&D is useful for predicting both the priority filings that index patent families as well as the families containing subsequent filings in the EPO, Other Bloc, Other Trilateral, and Trilateral.
- These priority filings in turn were useful at predicting numbers of families containing subsequent filings in the EPO but weakly useful at predicting families containing the other kinds of subsequent filings. This is because, across countries, there is not a tight connection between domestic filings and international filings (since some countries have large domestic filings and relatively less international, and vice versa). Of course, the transfer rate (i.e. ratio of subsequent filings to first filings) may be stable over time for each country, but across countries there does not seem to be a monotonic relationship between the transfer rate and the level of first filings.
- Due to the timeliness problem, forecasts of patent families were less accurate for 1999 and 2000 (and even for 1998 in the case of Trilateral patent families). For periods where the

timeliness problem is not a concern, the forecasts are generally quite good (i.e. close to 99% of actual).

In conclusion, there are several possible extensions to this study. First, more research is needed to better understand and explain patent *granting* behavior. This requires more detailed knowledge about the patenting authorities – their objectives and constraints. The existing literature has predominantly focused on the behavior of patent applicants. Secondly, it would be of interest to explore the *feedback effects*, if any, from the EPO system to R&D activities in various source countries. Inventors or firms should perceive the EPO region to represent a large (or larger) market. This surely should have impacted on their incentives to do R&D (including inventors in developing countries). The larger market should justify a larger investment in innovation or greater risk-taking. Thirdly, it would also be useful to derive *destination variables* for the EPO. Thus far the models of patenting behavior in this study have largely focused on source country characteristics. We discussed in the text the challenges posed in constructing destination variables.

Finally, this paper has examined patent families and sectoral (Joint Cluster) filings separately. As there have been few, if any, studies on patent filings by patent family or by technological fields, the simpler task has been to consider them separately. Future research should therefore seek to analyze and forecast patent family filings by *Joint Clusters* (or Groups).

Appendix

Derivation of R&D Capital Stocks:

The perpetual inventory method can be used to derive the stock of R&D capital, R , from data on gross R&D flows, I . At period t , the stock of R&D capital equals the stock left over from the previous period (net of depreciation) *plus* new R&D investments made during period t :

$$(A1) R(t) = I(t) + (1 - d) R(t-1),$$

where d is the depreciation rate. It is assumed that $d = 0.1$. The *initial* stock of capital can be derived by rolling the above identity backwards in time:

$$\begin{aligned} R(0) &= I(0) + (1 - d) R(-1) \\ &= I(0) + (1 - d) [I(-1) + (1 - d) R(-2)] \\ &= I(0) [1 + (1 - d)(I(-1)/I(0)) + (1 - d)^2(I(-2)/I(0)) + \dots] \end{aligned}$$

Let $g(t)$ be the growth rate of investment (i.e. $(1 + g(t)) = I(t)/I(t-1)$), and assume that $\bar{g}$ is the historical *average* growth rate of investment. (This growth rate can be approximated using the growth rate of investment during the sample period, under the assumption that the growth rate during the sample period is representative of the immediate past). Thus:

$$(A2) R(0) = \frac{I(0)(1 + \bar{g})}{(d + \bar{g})}$$

assuming $(\bar{g} + d) > 0$.⁴⁶ Using this initial value, the rest of the series $R(1)$, $R(2)$, ..., $R(t)$ can be generated using equation (A1). Dividing by the number of workers in the corresponding period - i.e. $L(1)$, $L(2)$, ..., $L(t)$ - gives the *R&D capital per worker* in each period.

The quantitative results in this paper are not very sensitive to the assumption of a 10% geometric depreciation rate. We tried 5% and 15% rates and found the coefficients of R&D capital per worker to differ on average by a low order of magnitude (i.e. second decimal point).

⁴⁶ Where $\bar{g} < 0$, we set $\bar{g} = 0$ to avoid abnormally large initial stocks.

Construction of the Index of Patent Rights

The following is an outline of how the strength of national patent regimes is scored. A country receives points for having various legal features (as indicated in the last two columns). The total points will range from zero to five.

(1) Membership in International Treaties	Signatory	Not Signatory
-- Paris Convention and Revisions	1/3	0
-- Patent Cooperation Treaty	1/3	0
-- Protection of New Varieties (UPOV)	1/3	0
(2) Coverage	Available	Not Available
-- Patentability of pharmaceuticals	1/7	0
-- Patentability of chemicals	1/7	0
-- Patentability of food	1/7	0
-- Patentability of plant and animal varieties	1/7	0
-- Patentability of surgical products	1/7	0
-- Patentability of microorganisms	1/7	0
-- Patentability of utility models	1/7	0
(3) Restrictions on Patent Rights	Does Not Exist	Exists
-- "Working" Requirements	1/3	0
-- Compulsory Licensing	1/3	0
-- Revocation of Patents	1/3	0
(4) Enforcement	Available	Not Available
-- Preliminary Injunctions	1/3	0
-- Contributory Infringement	1/3	0
-- Burden-of-Proof Reversal	1/3	0
(5) Duration of Protection	Full	Partial
	1	$0 < f < 1$

-- where f equals the duration of protection as a fraction of the full (potential) duration. FULL duration is either 20 years from the date of application or 17 years from the date of grant (for grant-based patent systems).

Source: Ginarte and Park (1997).

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... Table 1 cont'd/

Medium Income cont'd	EPO Member Entry Year	EPO Patent Filings	EPO Patent Grants	R&D as % of GDP	% R&D Privately Financed	% R&D Publicly Financed	Sci & Eng per 10,000 Workers	Exports to EPO as % of GDP	FDI Stock in EPO as % of GDP	Index of Patent Rights
SAfrica		120	31	0.86	63.78	36.22	15.23	6.29		3.64
Slovakia		14	2	1.02			34.35	42.18		3.36
Spain	1986	351	89	0.73	46.21	30.86	22.93	7.92	1.18	3.66
Uruguay		3	3					2.88		2.46
Venez		5	2	0.36			5.60	4.19		1.94
Low Income:										
Bulgaria		15	5	1.25			75.49	11.02		2.90
China		146	7	0.65			6.44	2.55		2.02
Colombia		4		0.12	28.10	71.90	0.94	3.28		1.77
Costa Rica		3		0.21	8.20	91.80	14.10	6.97		1.75
Egypt		11		0.58	10.00	90.00	12.63	4.59		2.06
India		57	10	0.71			3.33	2.15		1.64
Indonesia		5		0.20			3.74	3.32		0.82
Malaysia		6	3	0.22	46.90	53.10	4.55	9.90		2.74
Mexico		31	5	0.38			5.54	1.65	0.01	2.01
Panama		11	6	0.01			6.53	3.33		2.88
Philippines		3		0.19	37.61	62.39	3.38	3.52		2.67
Poland		37	6	0.89			29.87	11.53	0.84	3.07
Romania		16		0.76			41.35	10.64		2.71
Russia		221	16	0.79			88.95			3.28
Sri Lanka		3		0.19	18.31	81.69	4.48	6.95		3.14
Thailand		6	3	0.21	25.60	74.40	1.80	5.27		2.00
Turkey		23	2	0.84	32.07	30.53	5.35	5.48	0.20	1.95

Notes:

Countries are classified as high income if their sample average GDP per capita exceeds \$17,000 U.S. (real 1995) dollars; low income if their sample average GDP per capita is below \$3,500 real U.S.; and medium if their incomes are in-between.

Patenting Data are from the European Patent Office (EPO), Index of Patent Rights from Ginarte and Park (1997) and Park and Wagh (2002), and Research and Development (R&D), Exports, Foreign Direct Investment (FDI), and Scientists and Engineers data are from the OECD.

Table 2. Leading Source Countries**(A) Patent Applications**

<u>Top Ten</u> <u>In 1985:</u>	<u>EPO</u> <u>Filings</u>	<u>EPO</u> <u>Member</u>	<u>Top Ten</u> <u>in 1999:</u>	<u>EPO</u> <u>Filings</u>	<u>EPO</u> <u>Member</u>
USA	10990	No	USA	40419	No
Germany	8317	Yes	Germany	21864	Yes
Japan	6369	No	Japan	17316	No
France	3200	Yes	France	7410	Yes
UK	2812	Yes	UK	5700	Yes
Switzerl.	1762	Yes	Netherlands	4537	Yes
Netherlands	1230	Yes	Switzerl.	3910	Yes
Italy	1160	Yes	Italy	3434	Yes
Sweden	934	Yes	Sweden	2997	Yes
Austria	482	Yes	Canada	1794	No

(B) Patent Grants

<u>Top Ten</u> <u>In 1985:</u>	<u>EPO</u> <u>Grants</u>	<u>EPO</u> <u>Member</u>	<u>Top Ten</u> <u>in 1999:</u>	<u>EPO</u> <u>Grants</u>	<u>EPO</u> <u>Member</u>
USA	6529	No	USA	8985	No
Germany	5825	Yes	Germany	7036	Yes
Japan	5230	No	Japan	7023	No
France	2350	Yes	France	2912	Yes
UK	1491	Yes	UK	1805	Yes
Switzerl.	1282	Yes	Switzerl.	1252	Yes
Netherlands	905	Yes	Italy	1101	Yes
Italy	665	Yes	Netherlands	1072	Yes
Sweden	567	Yes	Sweden	661	Yes
Austria	312	Yes	Finland	347	Yes

Note:

EPO Patent Filings (and Grants) are the sum of direct filings (and grants) and Euro-PCT International Filings (and Grants).

Table 3. Leading Destination Countries**(A) Designations in EPO Filings**

<u>Order in 1985</u>	<u>EPO Filings</u>	<u>% of Total</u>	<u>Order in 1994</u>	<u>EPO Filings</u>	<u>% of Total</u>	<u>Order in 1999</u>	<u>EPO Filings</u>	<u>% of Total</u>
EPO total	39618	100.0%	EPO total	73008	100.0%	EPO total	121749	100.0%
Germany	38056	96.1%	Germany	72144	98.8%	Germany	118964	97.7%
UK	37044	93.5%	UK	70152	96.1%	UK	118849	97.6%
France	36375	91.8%	France	69494	95.2%	France	118834	97.6%
Italy	28753	72.6%	Italy	58901	80.7%	Italy	118062	97.0%
Netherlands	27116	68.4%	Netherlands	53078	72.7%	Netherlands	117718	96.7%
Sweden	23432	59.1%	Spain	50975	69.8%	Spain	117546	96.5%
Switzerland	22923	57.9%	Switzerland	48653	66.6%	Sweden	117484	96.5%
Belgium	22848	57.7%	Belgium	48592	66.6%	Switzerland	117416	96.4%
Austria	19336	48.8%	Sweden	48215	66.0%	Belgium	117398	96.4%
Luxembourg	13911	35.1%	Austria	46215	63.3%	Austria	117280	96.3%
Cyprus	n/a		Denmark	43036	58.9%	Denmark	117175	96.2%
Denmark	n/a		Portugal	40998	56.2%	Luxembourg	117040	96.1%
Finland	n/a		Luxembourg	40560	55.6%	Greece	117037	96.1%
Greece	n/a		Greece	40415	55.4%	Portugal	116955	96.1%
Ireland	n/a		Ireland	40100	54.9%	Ireland	116944	96.1%
Monaco	n/a		Monaco	37059	50.8%	Monaco	116772	95.9%
Portugal	n/a		Cyprus	n/a		Finland	116355	95.6%
Spain	n/a		Finland	n/a		Cyprus	116036	95.3%

(B) Designations in EPO Grants

<u>Order in 1985</u>	<u>EPO Grants</u>	<u>% of Total</u>	<u>Order in 1994</u>	<u>EPO Grants</u>	<u>% of Total</u>	<u>Order in 1999</u>	<u>EPO Grants</u>	<u>% of Total</u>
EPO total	14708	100.0%	EPO total	40891	100.0%	EPO total	34355	100.0%
Germany	13828	94.0%	Germany	40668	99.5%	Germany	34271	99.8%
UK	13570	92.3%	UK	39076	95.6%	UK	32514	94.6%
France	12435	84.5%	France	36450	89.1%	France	30887	89.9%
Italy	9685	65.8%	Italy	28012	68.5%	Italy	24499	71.3%
Netherl	9186	62.5%	Netherl	22860	55.9%	Netherl	17636	51.3%
Belgium	7929	53.9%	Sweden	20663	50.5%	Spain	17065	49.7%
Switzer	7899	53.7%	Switzer	20077	49.1%	Sweden	14926	43.4%
Sweden	7309	49.7%	Belgium	19520	47.7%	Belgium	14707	42.8%
Austria	6350	43.2%	Spain	18001	44.0%	Switzer	14302	41.6%
Luxem	3277	22.3%	Austria	17259	42.2%	Austria	13189	38.4%
Cyprus	n/a		Luxem	10748	26.3%	Denmark	9565	27.8%
Denmark	n/a		Greece	9852	24.1%	Greece	8399	24.4%
Finland	n/a		Denmark	7426	18.2%	Portugal	7860	22.9%
Greece	n/a		Portugal	1395	3.4%	Luxem	6975	20.3%
Ireland	n/a		Monaco	778	1.9%	Ireland	6916	20.1%
Monaco	n/a		Ireland	484	1.2%	Monaco	4240	12.3%
Portugal	n/a		Cyprus	n/a		Finland	2067	6.0%
Spain	n/a		Finland	n/a		Cyprus	732	2.1%

Notes: n/a indicates not available since the country was not yet an EPO member state; percentages refer to the percent of total EPO filings

Table 4. EPO Fees and Revenue Estimates

Year	(1) Nominal Fees (in DM)	(2) Real 1995 Fees (in DM)	(3) EPO Direct Filings	(4) EPO Filings via PCT	(5) Filings Sum = (3) + (4)	(6) Revenues from Filing = (2) x (5)
1980	450	702.6	17495	2520	20015	14061893
1981	450	674.4	22421	3063	25484	17187224
1982	520	746.2	25318	3640	28958	21608172
1983	520	722.7	28132	4014	32146	23232088
1984	520	708.1	33094	4413	37507	26558523
1985	560	747.2	33748	6242	39990	29880322
1986	560	724.2	36783	7314	44097	31933394
1987	560	710.8	39961	8380	48341	34360529
1988	560	700.2	44755	11136	55891	39132134
1989	560	683.7	49282	13307	62589	42789903
1990	560	662.7	52361	18563	70924	46999490
1991	600	683.3	45648	21890	67538	46150128
1992	600	650.5	46053	24252	70305	45734469
1993	600	627.5	42709	28078	70787	44421951
1994	600	612.2	41374	32877	74251	45455698
1995	600	600.0	40651	38608	79259	47555781
1996	600	593.9	41078	46330	87408	51913443
1997	250	245.8	45308	55082	100390	24679382
1998	250	243.1	48540	64798	113338	27556167
1999	250	242.0	50103	73763	123866	29975067
2000	250	243.0	54618	90569	145187	35275362

Notes:

DM refers to Deutsche Marks. Real fees were derived using the GDP deflator for Germany
(Source: World Bank Development Indicators 2001)

Table 5. EPO Filings – Basic Specification

Dependent Variable:	ln (PAT)			
Estimation Method:	<u>RE</u>	<u>FE</u>	<u>FE</u>	<u>FE</u>
C	-26.44*** (0.94)	-31.6*** (1.62)	-30.4*** (1.67)	-23.9*** (1.78)
ln (GDPC)	1.34*** (0.14)	1.71*** (0.20)	1.69*** (0.20)	0.97*** (0.20)
ln (RD)	0.34*** (0.093)	0.58*** (0.12)		
ln(RD _p)				0.48*** (0.09)
ln (RD _g)				-0.08 (0.103)
ln (RD _f)			0.61*** (0.11)	
ln(SE)	0.32*** (0.078)	0.20*** (0.082)	0.22*** (0.08)	0.15*** (0.071)
ln(IPR)	1.34*** (0.18)	1.13*** (0.184)	1.06*** (0.18)	2.11*** (0.21)
ln(FEES)	-0.34*** (0.044)	-0.24*** (0.048)	-0.24*** (0.048)	-0.25*** (0.047)
Adj. R-squared	0.84	0.82	0.81	0.92
No. of Observ.	630	630	630	483
² (5)	49.53			

Notes: PAT denotes the number of EPO patent applications per source country worker, C the constant term, GDPC gross domestic product per capita, RD the stock of research and development (R&D) capital per worker, RD_p privately financed R&D stock per worker, RD_g publicly financed R&D stock per worker, RD_f the flow of R&D investment per worker, SE number of scientists and engineers per 10,000 workers, IPR index of patent rights, and FEE real filing fees (in 1995 Deutsche Marks (DM)). RE denotes random effects estimation and FE fixed effects estimation. * indicates statistical significance between the 5% and 10% levels of significance; ** indicates between the 1% and 5% levels; and *** at the 1% level of significance or better.

² is the chi-square statistic for testing the null of no correlation between the independent variables and the error term. Standard errors are in parentheses.

Table 6. EPO Filings – Alternative Dependent Variables

	(1)	(2)
	ln (Patent Filings per S&E worker)	ln (Patent Filings per R&D Stock)
C	-22.4*** (1.62)	-31.6*** (1.62)
ln (GDPC)	1.17*** (0.20)	1.17*** (0.20)
ln (RD)	0.58*** (0.12)	-0.42*** (0.12)
ln(SE)	-0.80*** (0.082)	0.20*** (0.082)
ln(IPR)	1.13*** (0.184)	1.13*** (0.184)
ln(FEES)	-0.24*** (0.048)	-0.24*** (0.048)
Adj. R-squared	0.65	0.34
No. of Observ.	630	630

Notes: S&E denotes science and engineering. For other variable definitions, see notes to Table 5. Estimation is by *Fixed Effects*. * indicates statistical significance at the 10% level; ** 5% level; and *** 1% level. Standard errors are in parentheses.

Table 7. EPO Filings – Alternative Independent Variables

Dependent Variable:	ln (PAT)			
	(1)	(2)	(3)	(4)
C	-37.6*** (2.68)	-20.0*** (1.83)	-31.6*** (1.64)	-30.9*** (1.68)
ln (GDPC)	2.22*** (0.26)		1.60*** (0.21)	1.55*** (0.21)
ln (EXP)	0.04 (0.09)	0.02 (0.11)		
ln(FDI)	0.004 (0.018)	0.05*** (0.019)		
ln (RD)	0.51*** (0.17)	1.20*** (0.16)	0.58** (0.12)	0.52*** (0.13)
ln(SE)	0.23 (0.015)	0.35*** (0.17)	0.24*** (0.09)	0.24*** (0.09)
ln(ENROLS)			0.13 (0.14)	
ln (ENROLT)				0.21* (0.11)
ln(IPR)	1.57*** (0.28)	1.26*** (0.31)	1.12*** (0.21)	0.89*** (0.19)
ln(FEES)	-0.08 (0.05)	-0.21*** (0.05)	-0.20*** (0.05)	-0.16*** (0.05)
Adj. R-squared	0.88	0.83	0.81	0.81
No. of Observ.	403	403	512	514

Notes: PAT denotes Patent Filings per worker, EXP exports to the EPO region per GDP, FDI stock of foreign direct investment in the EPO region per GDP, ENROLT enrollment rate in tertiary (higher) education, and ENROLS enrollment rate in secondary education. For other variable definitions, see notes to Table 5. Estimation is by *Fixed Effects*. * indicates statistical significance at the 10% level; ** 5% level; and *** 1% level. Standard errors are in parentheses.

Table 8. EPO Filings – Role of Cyclical Disturbances and Lags

Dependent Variable:	ln (PAT)		
	(1)	(2)	(3)
C	-10.7*** (0.031)	-31.7*** (1.66)	-10.9*** (1.95)
$\hat{u}$	1.83*** (0.84)	-0.08 (0.56)	
ln (GDPC)		1.72*** (0.21)	1.31*** (0.47)
ln (RD)		0.59*** (0.12)	2.39*** (0.83)
ln(SE)		0.19*** (0.083)	0.099 (0.132)
ln(IPR)		1.10*** (0.19)	0.75 (0.53)
ln(FEES)		-0.24*** (0.049)	-0.16*** (0.05)
ln(PAT(-1))			0.67*** (0.034)
ln (GDPC(-1))			-0.58 (0.45)
ln (RD(-1))			-2.31*** (0.78)
ln(SE(-1))			-0.077 (0.131)
ln(IPR(-1))			-0.32 (0.52)
ln(FEES(-1))			0.04 (0.05)
Adj. R-squared	0.84	0.82	0.96
No. of Observ.	716	629	535

Notes: (-1) indicates lagged one period. $\hat{u}$ is the residual from a regression of log GDP per capita on its one-period lag and a time trend (see text for estimates). For other variable definitions, see notes to Table 5.

Table 9: Breakdown of EP filings by route (EPO vs. PCT) and type of filing (First or Subsequent)**A. EPC Bloc**

Source Country	Year	Total EP Filings	Percentage Breakdown:				% of Euro-Direct that are First Filings
			EF (%)	ES (%)	PF (%)	PS (%)	
Austria	1985	482	6.2	80.1	1.0	12.7	7.2
Austria	1995	683	5.1	49.8	1.8	43.3	9.3
Austria	2000	994	9.8	44.2	0.2	45.9	18.1
Belgium	1985	363	20.7	72.5	0.6	6.3	22.2
Belgium	1995	851	36.4	34.9	1.4	27.3	51.1
Belgium	2000	1299	37.3	20.9	0.7	41.2	64.1
Cyprus	1985	1	0.0	0.0	0.0	100.0	
Cyprus	1995	8	25.0	50.0	0.0	25.0	33.3
Cyprus	2000	25	20.0	4.0	12.0	64.0	83.3
Denmark	1985	198	1.0	45.5	3.0	50.5	2.2
Denmark	1995	605	2.3	13.1	3.3	81.3	15.1
Denmark	2000	978	10.6	13.4	2.7	73.3	44.3
Finland	1985	166	4.2	50.0	3.6	42.2	7.8
Finland	1995	899	0.9	24.4	4.1	70.6	3.5
Finland	2000	1918	1.6	21.4	3.7	73.3	7.0
France	1985	3200	3.3	88.8	0.5	7.3	3.6
France	1995	5377	3.5	64.3	1.6	30.5	5.2
France	2000	8168	7.1	45.5	1.0	46.4	13.5
Germany	1985	8317	5.3	86.3	0.4	8.1	5.8
Germany	1995	13650	4.7	59.9	0.7	34.7	7.2
Germany	2000	24493	9.0	42.1	0.6	48.3	17.6
Greece	1985	9	22.2	77.8	0.0	0.0	22.2
Greece	1995	32	6.3	37.5	6.3	50.0	14.3
Greece	2000	55	5.5	20.0	3.6	70.9	21.4
Ireland	1985	34	8.8	85.3	0.0	5.9	9.4
Ireland	1995	118	3.4	22.0	5.9	68.6	13.3
Ireland	2000	325	15.4	19.4	2.2	63.1	44.2
Italy	1985	1160	5.3	86.1	0.5	8.0	5.8
Italy	1995	2291	13.4	63.1	1.7	21.9	17.5
Italy	2000	3819	18.1	48.6	2.0	31.3	27.2
Netherlands	1985	1230	3.6	91.9	0.1	4.5	3.7
Netherlands	1995	3077	32.0	25.3	1.1	41.6	55.8
Netherlands	2000	5331	34.4	14.0	1.0	50.6	71.0
Portugal	1985	5	0.0	100.0	0.0	0.0	0.0
Portugal	1995	33	42.4	30.3	6.1	21.2	58.3
Portugal	2000	36	25.0	22.2	2.8	50.0	52.9
Spain	1985	84	8.3	88.1	0.0	3.6	8.6
Spain	1995	392	5.6	50.8	4.8	38.8	10.0
Spain	2000	789	9.8	30.3	7.5	52.5	24.4
Sweden	1985	934	2.8	48.7	2.7	45.8	5.4
Sweden	1995	1753	0.6	13.7	3.0	82.7	4.0
Sweden	2000	3457	6.9	7.7	1.9	83.5	47.0
Switzerl.	1985	1762	6.8	79.9	0.7	12.6	7.9
Switzerl.	1995	2514	15.2	51.4	2.2	31.2	22.8
Switzerl.	2000	4248	31.7	26.0	3.4	38.9	54.9
Turkey	1985	2	0.0	100.0	0.0	0.0	0.0
Turkey	1995	1	0.0	100.0	0.0	0.0	0.0
Turkey	2000	63	1.6	1.6	22.2	74.6	50.0
UK	1985	2812	2.1	81.2	0.4	16.3	2.6
UK	1995	4237	5.1	28.4	1.3	65.2	15.2
UK	2000	6290	7.6	21.0	0.9	70.5	26.5

B. Other Trilateral

Source Country	Year	Total EP Filings	Percentage Breakdown:				% of Euro- Direct that are First Filings
			EF (%)	ES (%)	PF (%)	PS (%)	
Japan	1985	6369	3.4	87.5	1.2	7.9	3.7
Japan	1995	11178	2.2	76.5	2.7	18.7	2.7
Japan	2000	20428	1.2	60.1	3.2	35.5	2.0
USA	1985	10990	4.5	74.2	2.1	19.2	5.8
USA	1995	26116	1.9	35.4	3.3	59.4	5.1
USA	2000	46866	2.6	20.3	1.6	75.4	11.4

C. Other Bloc

Source Country	Year	Total EP Filings	Percentage Breakdown:				% of Euro- Direct that are First Filings
			EF (%)	ES (%)	PF (%)	PS (%)	
Argen	1985	3	0.0	100.0	0.0	0.0	0.0
Argen	1995	12	16.7	66.7	8.3	8.3	20.0
Argen	2000	31	9.7	64.5	3.2	22.6	13.0
Australia	1985	402	4.0	25.1	4.7	66.2	13.7
Australia	1995	873	0.9	3.9	5.4	89.8	19.0
Australia	2000	1532	0.1	2.6	4.2	93.1	2.4
Brazil	1985	16	0.0	31.3	18.8	50.0	0.0
Brazil	1995	66	3.0	15.2	9.1	72.7	16.7
Brazil	2000	176	1.7	11.9	5.1	81.3	12.5
Bulgaria	1985	24	29.2	66.7	0.0	4.2	30.4
Bulgaria	1995	11	0.0	18.2	18.2	63.6	0.0
Bulgaria	2000	29	0.0	0.0	6.9	93.1	0.0
Canada	1985	354	11.3	83.3	0.6	4.8	11.9
Canada	1995	946	1.3	17.2	7.3	74.2	6.9
Canada	2000	2223	1.0	22.4	2.5	74.0	4.2
Chile	1985	3	0.0	100.0	0.0	0.0	0.0
Chile	1995	1	0.0	100.0	0.0	0.0	0.0
Chile	2000	4	0.0	75.0	0.0	25.0	0.0
China	1985	3	33.3	33.3	0.0	33.3	50.0
China	1995	94	0.0	2.1	14.9	83.0	0.0
China	2000	708	3.8	8.1	5.5	82.6	32.1
Colombia	1985						
Colombia	1995	5	0.0	40.0	0.0	60.0	0.0
Colombia	2000	5	0.0	80.0	0.0	20.0	0.0
Costa Rica	1985						
Costa Rica	1995						
Costa Rica	2000	2	0.0	0.0	0.0	100.0	
Egypt	1985						
Egypt	1995	1	100.0	0.0	0.0	0.0	100.0
Egypt	2000	1	0.0	100.0	0.0	0.0	0.0
Hungary	1985	120	9.2	36.7	2.5	51.7	20.0
Hungary	1995	71	0.0	18.3	18.3	63.4	0.0
Hungary	2000	124	0.8	2.4	4.8	91.9	25.0
Iceland	1985	2	0.0	100.0	0.0	0.0	0.0
Iceland	1995	4	0.0	0.0	50.0	50.0	
Iceland	2000	22	0.0	4.5	0.0	95.5	0.0
India	1985	8	25.0	75.0	0.0	0.0	25.0
India	1995	15	86.7	13.3	0.0	0.0	86.7
India	2000	167	9.6	6.6	28.1	55.7	59.3
Indonesia	1985	1	0.0	100.0	0.0	0.0	0.0
Indonesia	1995	1	100.0	0.0	0.0	0.0	100.0
Indonesia	2000	11	18.2	9.1	45.5	27.3	66.7

C. Other Bloc cont'd

Source Country	Year	Total EP Filings	Percentage Breakdown:				% of Euro-Direct that are First Filings	
			EF (%)	ES (%)	PF (%)	PS (%)		
Israel	1985	94	8.5	88.3	0.0	3.2	8.8	
Israel	1995	364	3.8	33.5	3.6	59.1	10.3	
Israel	2000	1060	1.0	11.7	4.3	82.9	8.1	
Korea	1985	20	5.0	0.0	10.0	85.0	100.0	
Korea	1995	446	6.3	57.2	4.0	32.5	9.9	
Korea	2000	1830	2.0	28.0	4.3	65.7	6.7	
Malaysia	1985	4	0.0	100.0	0.0	0.0	0.0	
Malaysia	1995	6	16.7	66.7	16.7	0.0	20.0	
Malaysia	2000	12	16.7	50.0	0.0	33.3	25.0	
Mexico	1985	2	0.0	100.0	0.0	0.0	0.0	
Mexico	1995	12	8.3	8.3	8.3	75.0	50.0	
Mexico	2000	65	0.0	7.7	13.8	78.5	0.0	
NewZealand	1985	32	6.3	90.6	0.0	3.1	6.5	
NewZealand	1995	138	0.7	4.3	2.2	92.8	14.3	
NewZealand	2000	261	0.0	5.0	3.4	91.6	0.0	
Norway	1985	121	3.3	40.5	7.4	48.8	7.5	
Norway	1995	300	0.3	12.0	3.7	84.0	2.7	
Norway	2000	539	2.6	6.1	3.0	88.3	29.8	
Panama	1985	8	0.0	62.5	0.0	37.5	0.0	
Panama	1995	11	0.0	90.9	0.0	9.1	0.0	
Panama	2000	9	0.0	33.3	0.0	66.7	0.0	
Philippines	1985	1	0.0	100.0	0.0	0.0	0.0	
Philippines	1995	1	0.0	0.0	100.0	0.0	0.0	
Philippines	2000	1	0.0	100.0	0.0	0.0	0.0	
Poland	1985	15	13.3	86.7	0.0	0.0	13.3	
Poland	1995	87	81.6	2.3	1.1	14.9	97.3	
Poland	2000	95	0.0	3.2	9.5	87.4	0.0	
Romania	1985	2	0.0	0.0	50.0	50.0		
Romania	1995	14	0.0	7.1	21.4	71.4	0.0	
Romania	2000	21	0.0	0.0	4.8	95.2		
Russia	1985	4	0.0	0.0	75.0	25.0		
Russia	1995	198	0.0	1.5	28.8	69.7	0.0	
Russia	2000	413	0.0	0.7	13.8	85.5	0.0	
SAfrica	1985	71	9.9	90.1	0.0	0.0	9.9	
SAfrica	1995	64	3.1	42.2	7.8	46.9	6.9	
SAfrica	2000	396	0.5	4.0	4.5	90.9	11.1	
Singapore	1985	5	0.0	80.0	0.0	20.0	0.0	
Singapore	1995	33	3.0	18.2	6.1	72.7	14.3	
Singapore	2000	196	1.5	18.4	32.1	48.0	7.7	
Slovakia	1985							
Slovakia	1995	8	0.0	0.0	12.5	87.5		
Slovakia	2000	29	0.0	6.9	10.3	82.8	0.0	
Thailand	1985							
Thailand	1995	4	0.0	50.0	0.0	50.0	0.0	
Thailand	2000	19	36.8	26.3	5.3	31.6	58.3	
Venez	1985	2	0.0	100.0	0.0	0.0	0.0	
Venez	1995	2	50.0	50.0	0.0	0.0	50.0	
Venez	2000	9	0.0	88.9	0.0	11.1	0.0	

Percentage Breakdown:				% of Euro-Direct that are First Fil.	
EF (%)	ES (%)	PF (%)	PS (%)		

All Countries, Sample Average:

5.7	55.1	2.1	37.1	17.8
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Table 10a. Filings by Route: Total

Dependent Variable: ln (TOTAL/Labor)

	(1)	(2)	(3)	(4)
Constant	-0.178*** (0.069)	-0.109 (0.071)	-1.806*** (0.333)	-1.002*** (0.250)
Lag 1	0.973*** (0.007)	0.694*** (0.052)	0.878*** (0.019)	0.653*** (0.056)
Lag 2		0.123** (0.065)		0.021 (0.069)
Lag 3		0.159*** (0.072)		0.256*** (0.049)
ln RD			0.106*** (0.027)	0.073*** (0.020)
No. of Obs.	473	371	427	331
Adj. R-sq	0.98	0.98	0.98	0.98
Method of Estimation	OLS	OLS	GLS	GLS

Notes: The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. TOTAL refers to the sum of EF (Euro-Direct First Filings), ES (Euro-Direct Subsequent Filings), PF (Euro via PCT First Filings), and PS (Euro via PCT Subsequent Filings). RD refers to research and development expenditures per worker (in real 1995 US dollars).

Table 10b. Filings by Route: EF

Dependent Variable: ln (EF/Labor)

	(1)	(2)	(3)	(4)
Constant	-0.493*** (0.182)	-0.193 (0.186)	-1.416*** (0.381)	-1.142*** (0.434)
Lag 1	0.958*** (0.014)	0.523*** (0.053)	0.918*** (0.020)	0.536*** (0.056)
Lag 2		0.318*** (0.059)		0.296*** (0.061)
Lag 3		0.137*** (0.072)		0.106** (0.056)
ln RD			0.071*** (0.025)	0.071*** (0.028)
No. of Obs.	471	362	424	330
Adj. R-sq	0.91	0.93	0.92	0.93
Method of Estimation	OLS	OLS	GLS	GLS

Notes: The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. EF denotes Euro-Direct First Filings and RD research and development expenditures per worker (in real 1995 US dollars).

Table 10c. Filings by Route: ES

Dependent Variable: ln (ES/Labor)

	(1)	(2)	(3)	(4)
Constant	-0.203** (0.089)	-0.107 (0.090)	-0.839*** (0.238)	-0.418*** (0.207)
Lag 1	0.981*** (0.007)	0.688*** (0.041)	0.952*** (0.013)	0.656*** (0.043)
Lag 2		0.195*** (0.049)		0.214*** (0.051)
Lag 3		0.105*** (0.041)		0.108*** (0.043)
ln RD			0.055*** (0.019)	0.032** (0.016)
No. of Obs.	652	511	590	460
Adj. R-sq	0.96	0.97	0.96	0.97
Method of Estimation	OLS	OLS	GLS	GLS

Notes: The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. ES denotes Euro-Direct Subsequent Filings and RD research and development expenditures per worker (in real 1995 US dollars).

Table 10d. Filings by Route: PF

Dependent Variable: ln (PF/Labor)

	(1)	(2)	(3)	(4)
Constant	-1.179*** (0.276)	-1.213*** (0.365)	-2.953*** (0.526)	-2.191*** (0.586)
Lag 1	0.901*** (0.020)	0.576*** (0.057)	0.829*** (0.027)	0.548*** (0.058)
Lag 2		0.100* (0.063)		0.106* (0.062)
Lag 3		0.216*** (0.053)		0.198*** (0.053)
ln RD			0.130*** (0.035)	0.07* (0.04)
No. of Obs.	342	249	330	247
Adj. R-sq	0.85	0.81	0.85	0.81
Method of Estimation	OLS	OLS	GLS	GLS

Notes: The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. PF denotes Euro via PCT First Filings and RD research and development expenditures per worker (in real 1995 US dollars).

Table 10e. Filings by Route: PS

Dependent Variable: ln (PS/Labor)

	(1)	(2)	(3)	(4)
Constant	-0.449*** (0.127)	-0.064*** (0.116)	-1.682*** (0.267)	-0.802*** (0.298)
Lag 1	0.944*** (0.011)	0.843*** (0.043)	0.891*** (0.014)	0.863*** (0.043)
Lag 2		0.126*** (0.049)		0.038 (0.049)
Lag 3		0.005 (0.035)		0.039 (0.034)
ln RD			0.107*** (0.020)	0.063*** (0.023)
No. of Obs.	505	393	459	353
Adj. R-sq	0.94	0.95	0.95	0.97
Method of Estimation	OLS	OLS	GLS	GLS

Notes: The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. PS denotes Euro via PCT Subsequent Filings and RD research and development expenditures per worker (in real 1995 US dollars).

Table 11. Summary of Forecast Errors: Aggregate EP Filings

			<i>Root Mean Square Percentage Error</i>			
Model	Variable	N	1997	1998	1999	2000
AR1	Total	473	0.189	0.174	0.183	0.194
AR3	Total	371	0.153	0.149	0.125	0.190
RE1	Total	427	0.201	0.205	0.202	0.237
RE3	Total	331	0.139	0.145	0.126	0.189
AR1	EF	471	0.353	0.617	0.468	0.454
AR3	EF	362	0.378	0.545	0.359	0.422
RE1	EF	424	0.351	0.612	0.455	0.489
RE3	EF	330	0.379	0.539	0.341	0.455
AR1	ES	652	0.309	0.332	0.576	0.411
AR3	ES	511	0.298	0.319	0.502	0.278
RE1	ES	590	0.313	0.332	0.546	0.441
RE3	ES	460	0.307	0.315	0.483	0.273
AR1	PF	342	0.409	0.424	0.760	0.873
AR3	PF	249	0.473	0.424	0.418	0.759
RE1	PF	330	0.365	0.352	0.645	0.749
RE3	PF	247	0.409	0.341	0.383	0.719
AR1	PS	505	0.350	0.386	0.202	0.350
AR3	PS	393	0.293	0.289	0.175	0.275
RE1	PS	459	0.291	0.327	0.174	0.288
RE3	PS	353	0.245	0.242	0.170	0.225

Notes: The forecast errors correspond to the estimated models in Table 10 A - E. Each entry is the RMSPE (root mean square percentage error) associated with each model. AR1 denotes the autoregressive model of order 1, AR3 autoregressive model of order 3, RE1 regression model of R&D and one lagged dependent variable, and RE3 regression model of R&D and three lags of the dependent variable. AR1, AR3, RE1, and RE3 correspond to the regression results shown in columns 1 - 4 respectively in Table 10 A- E. N denotes the number of observations, EF Euro-direct first filings, ES Euro-direct subsequent filings, PF Euro via PCT first filings, and PS Euro via PCT subsequent filings. Total is the sum (=EF+ES+PF+PS).

Table 12. Sample Forecasts of EP Filings: Actual vs. Predicted Values for 1998

TOTAL Filings:				EF Filings:			
	<u>Actual</u>	<u>AR3</u>	<u>RE3</u>		<u>Actual</u>	<u>AR3</u>	<u>RE3</u>
Australia	1034	997	1017	Australia	3	5	5
Austria	938	787	789	Austria	67	48	48
Belgium	1260	1079	1073	Belgium	433	354	330
Brazil	113	111	117	Brazil	1		
Canada	1712	1420	1446	Canada	18	18	20
China	247	189	179	China	2		
Cyprus	10	13	11	Cyprus	1	5	4
Denmark	787	771	754	Denmark	56	18	19
Finland	1413	1123	1106	Finland	10	7	8
France	7115	6671	6653	France	559	320	329
Germany	19860	17557	16971	Germany	1171	827	818
Greece	68	53	54	Greece	8	3	3
Hungary	94	104	95	Hungary	0		
Ireland	229	200	197	Ireland	23	9	9
Israel	804	592	596	Israel	15	10	10
Italy	3281	3034	2929	Italy	603	432	407
Japan	16169	15395	15479	Japan	237	267	284
Korea	829	711	744	Korea	21	23	24
Malaysia	8	8	9	Malaysia	1		
Mexico	70	41	39	Mexico	2		
Netherland	4046	3889	3695	Netherland	1298	1114	1004
NewZealar	190	189	183	NewZealar	2		
Norway	482	449	437	Norway	4	5	6
Poland	49	41	49	Poland	1		
Portugal	25	31	34	Portugal	5	5	5
Russia	326	351	327	Russia	2		
SAfrica	138	120	115	SAfrica	2	1	1
Singapore	119	79	81	Singapore	1		
Spain	701	600	599	Spain	47	35	35
Sweden	2869	2432	2380	Sweden	69	31	34
Switzerl.	3606	3328	3152	Switzerl.	1147	707	659
UK	5041	5449	5370	UK	368	488	489
USA	36860	34918	35548	USA	904	628	676
Sum	110493	102733	102227	Sum	7081	5361	5230
RMSPE		0.149	0.145	RMSPE		0.545	0.539

Notes: AR3 refers to the autoregressive model of order 3

RE3 refers to the regression model with 3 lags of the dependent variable

RMSPE denotes root mean square percentage error

Table 12 cont'd ../

ES Filings:				PF Filings:			
	<u>Actual</u>	<u>AR3</u>	<u>RE3</u>		<u>Actual</u>	<u>AR3</u>	<u>RE3</u>
Australia	36	29	30	Australia	62	48	46
Austria	470	338	339	Austria	14	11	11
Belgium	379	320	324	Belgium	19	16	16
Brazil	18	13	13	Brazil	23	16	16
Canada	354	193	196	Canada	74	77	76
China	13	7	7	China	85	59	50
Cyprus	2	2	2	Cyprus	4		
Denmark	114	98	99	Denmark	18	20	19
Finland	323	269	273	Finland	82	39	37
France	3518	3379	3410	France	123	118	118
Germany	9739	8489	8493	Germany	184	162	164
Greece	23	10	10	Greece	0	4	4
Hungary	4	4	4	Hungary	14	16	13
Ireland	50	41	41	Ireland	9	7	7
Israel	75	80	83	Israel	72	39	38
Italy	1756	1585	1569	Italy	44	44	43
Japan	10581	9885	10011	Japan	620	458	459
Korea	354	340	346	Korea	68	41	42
Malaysia	4	3	3	Malaysia	0		
Mexico	10	2	3	Mexico	5		
Netherland	804	814	820	Netherland	51	51	50
NewZealar	7	11	11	NewZealar	8	4	4
Norway	39	41	42	Norway	20	12	11
Poland	4	1	1	Poland	5		
Portugal	13	8	8	Portugal	2	2	2
Russia	1	3	3	Russia	93	113	94
SAfrica	36	31	30	SAfrica	2	8	7
Singapore	11	10	10	Singapore	49	9	9
Spain	278	201	200	Spain	61	37	36
Sweden	267	246	254	Sweden	56	64	63
Switzerl.	1117	1147	1156	Switzerl.	112	76	73
UK	1254	1212	1223	UK	80	81	81
USA	9161	9183	9453	USA	1228	1104	1109
<i>Sum</i>	<i>40815</i>	<i>37998</i>	<i>38466</i>	<i>Sum</i>	<i>3287</i>	<i>2735</i>	<i>2699</i>
<i>RMSPE</i>		<i>0.319</i>	<i>0.315</i>	<i>RMSPE</i>		<i>0.424</i>	<i>0.341</i>

Table 12 cont'd ../

	PS Filings:		
	<u>Actual</u>	<u>AR3</u>	<u>RE3</u>
Australia	933	973	961
Austria	387	379	378
Belgium	429	359	366
Brazil	71	94	95
Canada	1266	1255	1260
China	147	146	134
Cyprus	3	3	2
Denmark	599	676	656
Finland	998	860	853
France	2915	2746	2774
Germany	8766	8245	8146
Greece	37	42	42
Hungary	76	94	85
Ireland	147	155	152
Israel	642	505	507
Italy	878	873	870
Japan	4731	4433	4600
Korea	386	306	317
Malaysia	3		
Mexico	53	42	39
Netherland	1893	1942	1886
NewZealar	173	188	180
Norway	419	434	422
Poland	39	29	29
Portugal	5	13	13
Russia	230	275	255
SAfrica	98	92	86
Singapore	58	67	71
Spain	315	343	337
Sweden	2477	2311	2273
Switzerl.	1230	1340	1308
UK	3339	3679	3612
USA	25567	25828	26150
<i>Sum</i>	<i>59310</i>	<i>58726</i>	<i>58861</i>
<i>RMSPE</i>		<i>0.289</i>	<i>0.242</i>

Table 13. Sample Statistics -- EPO Filings Disaggregated by Joint Clusters and Filing Mode, 1985 and 1998

		G1	G1	G1	G2	G2	G2	G3	G3	G3	G4	G4	G4	G5	G5
	1985 total_ef	ef1%	ef8%	ef10%	ef3%	ef7%	ef12%	ef2%	ef6%	ef11%	ef4%	ef9%	ef14%	ef5%	ef13%
All	1795	5.63	0.77	6.24	15.31	7.89	1.24	15.22	11.20	6.80	3.33	16.48	4.83	4.20	0.87
Fra	106	5.66	0.94	0.94	10.38	9.43	3.77	16.98	15.09	3.77	3.77	18.87	9.43	0.94	0.00
Ger	431	10.67	3.48	7.66	7.66	4.87	1.39	16.94	12.06	1.62	8.82	12.99	9.74	1.39	0.70
Japan	215	9.77	1.40	3.26	17.67	13.49	1.86	19.07	8.37	8.84	2.79	3.26	8.84	0.93	0.47
UK	59	11.86	0.00	3.39	5.08	11.86	1.69	15.25	0.00	0.00	5.08	23.73	11.86	5.08	5.08
USA	500	8.20	3.40	5.00	20.40	11.20	2.00	10.40	8.60	6.00	4.80	7.80	6.80	2.40	3.00
	1998 total_ef	ef1%	ef8%	ef10%	ef3%	ef7%	ef12%	ef2%	ef6%	ef11%	ef4%	ef9%	ef14%	ef5%	ef13%
All	7209	7.75	5.97	3.79	6.24	12.06	5.50	9.58	13.01	3.02	7.30	11.48	4.88	3.08	6.34
Fra	494	10.12	6.28	5.06	4.45	8.30	10.73	6.48	7.29	2.83	3.24	9.11	3.24	4.25	18.62
Ger	1167	8.65	3.26	6.17	7.37	9.51	9.51	10.80	7.54	2.23	5.40	9.25	6.51	3.60	10.20
Japan	235	12.77	4.26	5.53	11.91	5.96	1.70	11.91	7.66	7.66	3.83	8.94	9.36	4.26	4.26
UK	368	3.26	3.80	2.17	3.26	14.67	12.50	4.35	2.99	5.71	5.98	4.08	4.62	7.61	25.00
USA	909	7.92	3.63	5.94	5.94	8.80	1.32	8.25	7.48	8.25	5.28	4.51	5.17	13.09	14.41
	1985 total_es	es1%	es8%	es10%	es3%	es7%	es12%	es2%	es6%	es11%	es4%	es9%	es14%	es5%	es13%
All	31797	8.74	0.89	2.89	9.67	15.92	2.79	15.29	10.26	4.18	5.97	12.81	8.16	1.60	0.82
Fra	2840	11.09	2.25	5.99	11.09	9.15	2.96	11.27	8.77	3.24	7.32	11.06	11.73	1.90	2.18
Ger	7163	9.06	1.34	4.01	10.48	14.37	2.44	13.68	9.24	7.69	5.92	9.65	9.27	0.81	2.04
Japan	5574	12.22	5.79	10.73	9.83	11.16	4.92	6.76	4.45	10.42	7.48	5.08	5.53	3.70	1.94
UK	2282	7.19	3.07	4.34	8.94	13.67	2.28	12.88	7.89	6.00	7.84	8.90	13.19	1.93	1.88
USA	8144	10.22	2.80	4.87	12.60	13.81	5.56	8.20	7.44	10.46	7.02	5.57	5.80	4.24	1.40
	1998 total_es	es1%	es8%	es10%	es3%	es7%	es12%	es2%	es6%	es11%	es4%	es9%	es14%	es5%	es13%
All	41109	8.50	2.11	3.13	9.33	12.33	2.96	14.35	12.90	2.82	5.16	8.71	13.26	1.31	3.13
Fra	3509	12.91	2.65	6.47	8.81	6.01	1.40	9.52	10.40	2.99	5.24	10.60	12.77	2.68	7.55
Ger	9715	9.68	1.47	5.26	7.30	5.51	1.77	16.05	8.68	4.82	5.05	13.40	16.51	1.69	2.82
Japan	10588	13.53	9.64	11.28	7.61	4.11	1.81	6.45	7.15	6.44	7.98	6.15	7.15	5.02	5.67
UK	1253	7.26	3.51	4.31	6.30	5.75	5.19	13.09	12.77	3.11	5.99	13.73	11.25	4.47	3.27
USA	9139	11.02	5.36	5.79	6.98	5.60	4.39	7.45	10.20	8.75	7.18	6.17	7.54	7.12	6.44

Table 13 cont'd ../

		G1	G1	G1	G2	G2	G2	G3	G3	G3	G4	G4	G4	G5	G5
	1985 total_pf	pf1%	pf8%	pf10%	pf3%	pf7%	pf12%	pf2%	pf6%	pf11%	pf4%	pf9%	pf14%	pf5%	pf13%
All	469	10.79	1.87	1.38	4.00	4.52	0.94	12.94	16.00	2.38	5.28	28.99	9.44	1.07	0.40
Fra	16	12.50	0.00	0.00	6.25	12.50	0.00	12.50	6.25	0.00	0.00	31.25	6.25	6.25	6.25
Ger	30	3.33	0.00	3.33	6.67	0.00	0.00	33.33	6.67	3.33	10.00	20.00	13.33	0.00	0.00
Japan	78	7.69	5.13	6.41	10.26	15.38	6.41	20.51	5.13	5.13	8.97	5.13	1.28	2.56	0.00
UK	11	0.00	9.09	0.00	9.09	0.00	0.00	9.09	18.18	0.00	0.00	45.45	9.09	0.00	0.00
USA	227	7.49	2.64	1.76	11.01	6.17	3.08	14.54	14.98	7.93	6.61	11.45	7.93	1.76	2.64
	1998 total_pf	pf1%	pf8%	pf10%	pf3%	pf7%	pf12%	pf2%	pf6%	pf11%	pf4%	pf9%	pf14%	pf5%	pf13%
All	3335	5.74	5.65	5.32	6.91	8.59	4.00	11.84	15.28	3.09	4.03	14.39	7.02	2.75	5.39
Fra	122	4.10	9.02	5.74	6.56	5.74	4.10	13.11	17.21	3.28	2.46	9.84	11.48	4.92	2.46
Ger	182	9.89	1.65	5.49	6.59	10.44	6.59	12.64	9.34	3.30	3.85	6.59	9.34	3.30	10.99
Japan	623	14.13	4.65	7.38	11.24	7.06	2.25	10.59	5.94	4.01	7.54	9.31	4.49	5.94	5.46
UK	83	4.82	1.20	6.02	7.23	10.84	7.23	13.25	13.25	3.61	4.82	14.46	7.23	2.41	3.61
USA	1225	4.65	2.69	3.92	7.84	15.35	5.80	12.90	13.71	8.33	4.65	5.39	6.04	4.90	3.84
	1985 total_ps	ps1%	ps8%	ps10%	ps3%	ps7%	ps12%	ps2%	ps6%	ps11%	ps4%	ps9%	ps14%	ps5%	ps13%
All	5505	3.61	1.18	2.88	15.34	10.81	4.71	8.63	11.37	4.59	5.83	18.35	10.05	2.20	0.45
Fra	235	6.81	1.70	5.96	14.47	4.68	6.81	7.66	17.45	2.55	5.11	11.06	11.49	3.83	0.43
Ger	672	5.95	2.83	3.13	9.52	6.10	2.38	17.41	12.65	2.08	8.04	13.39	13.99	1.49	1.04
Japan	500	12.00	4.80	16.40	8.80	9.20	3.80	16.80	5.00	4.60	5.80	3.40	4.40	3.20	1.80
UK	458	6.99	1.31	6.33	7.64	5.24	11.14	11.14	12.45	3.71	8.52	10.26	11.35	2.40	1.53
USA	2111	12.98	3.27	4.88	9.24	10.33	6.35	9.00	10.33	7.20	7.91	5.78	4.78	4.03	3.93
	1998 total_ps	ps1%	ps8%	ps10%	ps3%	ps7%	ps12%	ps2%	ps6%	ps11%	ps4%	ps9%	ps14%	ps5%	ps13%
All	59704	4.72	2.71	2.72	8.10	19.61	5.85	12.59	13.32	4.62	4.89	9.30	6.27	2.14	3.16
Fra	2927	5.71	3.31	4.34	9.70	16.30	8.06	9.36	9.63	5.09	5.53	6.29	8.88	3.66	4.13
Ger	8780	9.32	2.04	4.65	8.78	12.87	4.50	9.72	6.88	6.09	5.91	10.11	12.03	2.30	4.81
Japan	4732	10.80	6.23	4.73	10.19	19.19	7.38	6.93	5.43	9.91	5.75	4.88	3.38	2.49	2.70
UK	3321	4.85	3.16	3.70	7.65	18.25	10.09	9.58	9.70	4.49	7.41	6.56	6.74	3.52	4.31
USA	25579	6.48	3.45	4.56	7.75	15.42	10.81	6.99	11.76	6.42	5.37	4.24	4.72	5.83	6.20

Notation:

Total_ef, total_es, total_pf, and total_ps denote total counts of Euro-direct first filings (ef), Euro-direct subsequent filings (es), Euro via PCT direct filings (pf), and Euro via PCT subsequent filings (ps) respectively. The rest of the entries in the table are percentage shares.

Joint Clusters 1 - 14 are put into five groups: G1, G2, . . . , G5 as defined in the text.

Table 14. Sample Statistics -- R&D Expenditures Disaggregated by Sector, 1985 and 1998

1985																
Business																
Cntry	Enter. R&D	Agri%	Food%	Text%	Wood%	Petro%	Chem%	Rubber%	Nonmet%	Metal%	Elec%	Office%	Trans%	Util%	Const%	Serv%
All	232671.9	0.98	3.63	1.17	1.55	1.29	13.56	1.77	1.63	6.25	19.61	14.38	17.24	2.21	0.76	11.69
Fra	12637.5	1.01	1.28	0.59	0.44	2.85	16.63	2.39	1.12	2.91	24.64	10.06	29.61	2.46	0.66	2.49
Ger	24160.6	0.06	1.01	0.47	0.83	0.68	20.75	1.59	1.14	4.59	25.48	15.13	21.99	1.13	0.39	1.94
Japan	39203.9	0.08	2.45	1.05	1.06	1.15	15.76	2.45	2.93	7.47	32.63	9.84	15.75	1.07	1.87	3.30
U.K.	12358.0	N/A	2.40	0.42	0.47	1.07	18.39	0.75	0.83	1.86	29.80	12.74	23.26	2.58	0.42	5.21
U.S.A.	112257.5	N/A	1.40	N/A	0.17	2.01	8.97	N/A	0.97	1.48	17.13	20.45	35.81	0.21	0.21	7.97

1998																
Business																
Cntry	Enter. R&D	Agri%	Food%	Text%	Wood%	Petro%	Chem%	Rubber%	Nonmet%	Metal%	Elec%	Office%	Trans%	Util%	Const%	Serv%
All	345310.8	1.44	4.12	1.17	2.04	1.46	15.98	2.40	1.24	4.25	19.65	N/A	19.54	1.20	1.05	21.76
Fra	17261.4	1.45	1.87	0.59	0.92	1.50	19.07	2.77	1.24	2.69	16.53	N/A	24.09	2.71	0.97	8.92
Ger	28626.3	0.24	0.67	0.66	0.71	0.20	18.39	1.81	0.99	2.45	14.57	N/A	36.17	0.33	0.30	7.11
Japan	63881.9	0.11	2.43	0.71	1.18	0.47	15.10	2.41	2.01	4.21	34.38	N/A	15.11	0.95	1.64	6.24
U.K.	14154.7	1.01	2.39	0.33	0.48	2.31	28.88	0.65	0.55	1.55	11.79	N/A	20.33	1.38	0.39	16.46
U.S.A.	160723.7	N/A	0.88	0.34	0.17	1.07	12.96	N/A	0.43	1.40	2.84	N/A	19.52	0.13	0.28	28.83

Notes:

NA -- not available or incomplete.

Business Enterprise R&D is in real 1995 US dollars. All Countries refer to the 58 countries in the sample.

Agri refers to Agriculture, Food to Food and Kindred Products, Text to Textiles, Wood to Wood-based products (including printing and publishing), Petro to Petroleum-related, Chem to Chemicals (including Drugs), Rubber to Rubber-based Products, Nonmet to Nonmetallic products, Metal to Metal-related, Elec to Electrical Machines and Electronics, Office to Office Equipment and Machinery, Trans to Transportation, Util to Utilities, Const to Construction, and Serv to Services. Manufacturing R&D consists of the sum of R&D in the Food to Transportation sectors (inclusive).

Table 15a. Sectoral Regressions (Total Filings)Dependent Variable: $\ln(\text{TOTAL}/\text{Labor})$ Technology Group 1 (Electrical Machines & Electricity, Electronics, Audio, Video & Media)

	(1)	(2)	(3)	(4)	(5)
Constant	-0.567*** (0.223)	0.275 (0.246)	-0.054 (0.235)	-0.078 (0.253)	-0.069 (0.28)
Lag 1	0.786*** (0.076)	0.814*** (0.031)	0.586*** (0.074)	0.864*** (0.030)	0.566*** (0.089)
Lag 2	0.112 (0.088)		0.141* (0.079)		0.164* (0.092)
Lag 3	0.041 (0.072)		0.146** (0.067)		0.164** (0.079)
Business- Enter. R&D		0.299*** (0.053)	0.163*** (0.052)		
Specific- sector R&D				0.151*** (0.062)	0.099 (0.065)
No. of Obs.	177	217	175	157	124
Adj. R-sq	0.91	0.93	0.94	0.91	0.91
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. TOTAL refers to the sum of EF (Euro-Direct First Filings), ES (Euro-Direct Subsequent Filings), PF (Euro via PCT First Filings), and PS (Euro via PCT Subsequent Filings). Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Electronic and Utilities R&D per worker.

Table 15b. Sectoral Regressions (Euro-Direct First Filings)

Dependent Variable: ln (EF/Labor)

Technology Group 1 (Electrical Machines & Electricity, Electronics, Audio, Video & Media)

	(1)	(2)	(3)	(4)	(5)
Constant	-0.017*** (0.505)	-0.614 (0.676)	0.020 (0.725)	-0.337 (0.919)	-0.466 (0.845)
Lag 1	0.645*** (0.081)	0.907*** (0.036)	0.650*** (0.081)	0.807*** (0.051)	0.717*** (0.093)
Lag 2	0.264*** (0.091)		0.249*** (0.091)		0.208** (0.104)
Lag 3	0.079 (0.083)		0.102 (0.084)		0.078 (0.093)
Business- Enter. R&D		0.075 (0.084)	-0.022 (0.089)		
Specific- sector R&D				0.283* (0.165)	-0.127 (0.144)
No. of Obs.	140	172	139	128	101
Adj. R-sq	0.85	0.80	0.86	0.80	0.85
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. EF refers to Euro-Direct First Filings. Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Electronic and Utilities R&D per worker.

Table 15c. Sectoral Regressions (Euro-Direct Subsequent Filings)

Dependent Variable: ln (ES/Labor)

Technology Group 1 (Electrical Machines & Electricity, Electronics, Audio, Video & Media)

	(1)	(2)	(3)	(4)	(5)
Constant	-0.678*** (0.270)	0.528 (0.352)	0.531 (0.395)	0.181 (0.348)	-0.285 (0.405)
Lag 1	0.663*** (0.068)	0.836*** (0.029)	0.599*** (0.068)	0.786*** (0.037)	0.537*** (0.081)
Lag 2	0.261*** (0.079)		0.241*** (0.074)		0.246** (0.087)
Lag 3	0.015 (0.066)		0.019 (0.063)		0.018 (0.075)
Business- Enter. R&D		0.325*** (0.069)	0.289*** (0.081)		
Specific- sector R&D				0.338*** (0.091)	0.227** (0.106)
No. of Obs.	219	251	213	187	156
Adj. R-sq	0.88	0.90	0.90	0.87	0.85
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. ES refers to Euro-Direct Subsequent Filings. Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Electronic and Utilities R&D per worker.

Table 15d. Sectoral Regressions (Euro via PCT First Filings)Dependent Variable: $\ln(\text{PF/Labor})$ Technology Group 1 (Electrical Machines & Electricity, Electronics, Audio, Video & Media)

	(1)	(2)	(3)	(4)	(5)
Constant	-4.184*** (1.423)	-4.729*** (1.204)	-3.886*** (1.589)	-1.836 (1.056)	-0.355 (1.479)
Lag 1	0.373*** (0.107)	0.593*** (0.070)	0.361*** (0.108)	0.819*** (0.066)	0.501*** (0.131)
Lag 2	0.132 (0.112)		0.133 (0.112)		0.188 (0.145)
Lag 3	0.192* (0.101)		0.190* (0.102)		0.180 (0.140)
Business- Enter. R&D		0.152 (0.100)	0.067 (0.113)		
Specific- sector R&D				0.002 (0.159)	0.167 (0.225)
No. of Obs.	89	137	89	102	66
Adj. R-sq	0.36	0.41	0.38	0.66	0.64
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. PF refers to Euro via PCT First Filings. Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Electronic and Utilities R&D per worker.

Table 15e. Sectoral Regressions (Euro via PCT Subsequent Filings)

Dependent Variable: ln (PS/Labor)

Technology Group 1 (Electrical Machines & Electricity, Electronics, Audio, Video & Media)

	(1)	(2)	(3)	(4)	(5)
Constant	-1.127*** (0.386)	-0.301 (0.380)	-0.569 (0.457)	-0.155 (0.419)	0.329 (0.537)
Lag 1	0.664*** (0.073)	0.880*** (0.028)	0.618*** (0.073)	0.897*** (0.030)	0.717*** (0.092)
Lag 2	0.085 (0.083)		0.116 (0.081)		0.119 (0.108)
Lag 3	0.138** (0.066)		0.134** (0.066)		0.099 (0.083)
Business- Enter. R&D		0.138*** (0.052)	0.106* (0.059)		
Specific- sector R&D				0.101 (0.075)	0.115 (0.086)
No. of Obs.	192	233	187	167	133
Adj. R-sq	0.80	0.86	0.82	0.86	0.82
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. PS refers to Euro via PCT Subsequent Filings. Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Electronic and Utilities R&D per worker.

Table 16a. Sectoral Regressions (Total Filings)Dependent Variable: $\ln(\text{TOTAL}/\text{Labor})$ Technology Group 2 (Industrial Chemistry, Organic Chemistry, and Biotechnology)

	(1)	(2)	(3)	(4)	(5)
Constant	-0.415*** (0.143)	-0.410 (0.366)	-0.331** (0.149)	-0.324 (0.239)	-0.325*** (0.124)
Lag 1	0.825*** (0.072)	0.699*** (0.034)	0.779*** (0.072)	0.750*** (0.037)	0.798*** (0.080)
Lag 2	0.027 (0.086)		0.064 (0.084)		0.035 (0.094)
Lag 3	0.099 (0.063)		0.064 (0.062)		0.044 (0.067)
Business- Enter. R&D		0.348*** (0.063)	0.072*** (0.025)		
Specific- sector R&D				0.224*** (0.054)	0.078*** (0.026)
No. of Obs.	178	219	176	174	140
Adj. R-sq	0.96	0.93	0.96	0.93	0.96
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. TOTAL refers to the sum of EF (Euro-Direct First Filings), ES (Euro-Direct Subsequent Filings), PF (Euro via PCT First Filings), and PS (Euro via PCT Subsequent Filings). Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Chemicals, Rubber, Petroleum, and Food R&D per worker.

Table 16b. Sectoral Regressions (Euro-Direct First Filings)Dependent Variable: $\ln(\text{EF/Labor})$ Technology Group 2 (Industrial Chemistry, Organic Chemistry, and Biotechnology)

	(1)	(2)	(3)	(4)	(5)
Constant	0.011 (0.411)	-3.51*** (1.302)	-0.356 (0.551)	-0.959 (0.760)	0.292 (0.384)
Lag 1	0.650*** (0.074)	0.674*** (0.053)	0.666*** (0.075)	0.670*** (0.064)	0.615*** (0.086)
Lag 2	0.149* (0.091)		0.139 (0.094)		0.264*** (0.099)
Lag 3	0.195*** (0.077)		0.199*** (0.079)		0.109 (0.087)
Business- Enter. R&D		0.096 (0.166)	-0.068 (0.073)		
Specific- sector R&D				0.376*** (0.155)	0.043 (0.079)
No. of Obs.	169	204	163	169	139
Adj. R-sq	0.86	0.76	0.85	0.78	0.89
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. EF refers to Euro-Direct First Filings. Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Chemicals, Rubber, Petroleum, and Food R&D per worker.

Table 16c. Sectoral Regressions (Euro-Direct Subsequent Filings)

Dependent Variable: ln (ES/Labor)

Technology Group 2 (Industrial Chemistry, Organic Chemistry, and Biotechnology)

	(1)	(2)	(3)	(4)	(5)
Constant	-0.398** (0.182)	-0.128 (0.208)	-0.119 (0.238)	-0.322 (0.170)	-0.377** (0.199)
Lag 1	0.853*** (0.063)	0.943*** (0.019)	0.901*** (0.063)	0.968*** (0.026)	0.935*** (0.074)
Lag 2	0.019 (0.084)		-0.063 (0.085)		-0.062 (0.106)
Lag 3	0.095 (0.065)		0.129** (0.103)		0.103 (0.081)
Business- Enter. R&D		0.071** (0.035)	0.038 (0.042)		
Specific- sector R&D				-0.009 (0.041)	-0.029 (0.050)
No. of Obs.	225	257	216	202	174
Adj. R-sq	0.93	0.94	0.94	0.94	0.93
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. ES refers to Euro-Direct Subsequent Filings. Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Chemicals, Rubber, Petroleum, and Food R&D per worker.

Table 16d. Sectoral Regressions (Euro via PCT First Filings)

Dependent Variable: ln (PF/Labor)

Technology Group 2 (Industrial Chemistry, Organic Chemistry, and Biotechnology)

	(1)	(2)	(3)	(4)	(5)
Constant	-1.487** (0.787)	-3.147*** (0.873)	-1.524* (0.945)	-2.18*** (0.878)	-0.676 (0.957)
Lag 1	0.362*** (0.086)	0.743*** (0.054)	0.362*** (0.087)	0.754*** (0.062)	0.364*** (0.099)
Lag 2	0.274*** (0.081)		0.274*** (0.081)		0.321*** (0.094)
Lag 3	0.245*** (0.080)		0.246*** (0.080)		0.239*** (0.094)
Business- Enter. R&D		0.058 (0.083)	-0.007 (0.094)		
Specific- sector R&D				0.094 (0.111)	-0.010 (0.130)
No. of Obs.	122	169	122	131	96
Adj. R-sq	0.68	0.55	0.68	0.56	0.70
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. PF refers to Euro via PCT First Filings. Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Chemicals, Rubber, Petroleum, and Food R&D per worker.

Table 16e. Sectoral Regressions (Euro via PCT Subsequent Filings)

Dependent Variable: ln (PS/Labor)

Technology Group 2 (Industrial Chemistry, Organic Chemistry, and Biotechnology)

	(1)	(2)	(3)	(4)	(5)
Constant	-0.484** (0.224)	-0.449 (0.511)	-0.298 (0.329)	-0.249 (0.280)	-0.676 (0.957)
Lag 1	0.613*** (0.070)	0.870*** (0.023)	0.572*** (0.072)	0.891*** (0.021)	0.364*** (0.099)
Lag 2	0.185** (0.079)		0.195*** (0.079)		0.321*** (0.094)
Lag 3	0.133** (0.061)		0.147*** (0.063)		0.239*** (0.094)
Business- Enter. R&D		0.112* (0.071)	0.049 (0.042)		
Specific- sector R&D				0.079* (0.044)	-0.010 (0.130)
No. of Obs.	209	247	203	187	159
Adj. R-sq	0.92	0.91	0.91	0.92	0.91
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. PS refers to Euro via PCT Subsequent Filings. Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Chemicals, Rubber, Petroleum, and Food R&D per worker.

Table 17a. Sectoral Regressions (Total Filings)Dependent Variable: $\ln(\text{TOTAL}/\text{Labor})$ Technology Group 3 (Handling & Processing, Human Necessities, and Polymers)

	(1)	(2)	(3)	(4)	(5)
Constant	-0.217 (0.150)	-1.149*** (0.333)	-0.006 (0.165)	-0.329 (0.241)	-0.256** (0.124)
Lag 1	0.929*** (0.081)	0.727*** (0.033)	0.856*** (0.083)	0.736*** (0.033)	0.802*** (0.089)
Lag 2	-0.154 (0.106)		-0.059 (0.109)		0.338*** (0.113)
Lag 3	0.197*** (0.071)		0.153** (0.074)		-0.216*** (0.085)
Business- Enter. R&D		0.208*** (0.057)	0.058*** (0.024)		
Specific- sector R&D				0.215*** (0.054)	0.041* (0.024)
No. of Obs.	178	219	176	154	124
Adj. R-sq	0.95	0.95	0.96	0.96	0.97
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. TOTAL refers to the sum of EF (Euro-Direct First Filings), ES (Euro-Direct Subsequent Filings), PF (Euro via PCT First Filings), and PS (Euro via PCT Subsequent Filings). Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Agricultural, Food, Wood, Petroleum, Rubber, and Textile R&D per worker.

Table 17b. Sectoral Regressions (Euro-Direct First Filings)

Dependent Variable: ln (EF/Labor)

Technology Group 3 (Handling & Processing, Human Necessities, and Polymers)

	(1)	(2)	(3)	(4)	(5)
Constant	-0.051 (0.423)	-1.091*** (0.472)	-0.498 (0.505)	-0.634 (0.410)	0.207 (0.363)
Lag 1	0.490*** (0.073)	0.887*** (0.033)	0.479*** (0.074)	0.793*** (0.045)	0.609*** (0.092)
Lag 2	0.309*** (0.083)		0.321 (0.084)		0.241*** (0.102)
Lag 3	0.190*** (0.075)		0.198*** (0.077)		0.103 (0.087)
Business- Enter. R&D		0.043 (0.054)	-0.075 (0.059)		
Specific- sector R&D				0.223*** (0.071)	0.085 (0.060)
No. of Obs.	176	214	171	157	130
Adj. R-sq	0.84	0.80	0.84	0.88	0.89
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. EF refers to Euro-Direct First Filings. Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Agricultural, Food, Wood, Petroleum, Rubber, and Textile R&D per worker.

Table 17c. Sectoral Regressions (Euro-Direct Subsequent Filings)

Dependent Variable: ln (ES/Labor)

Technology Group 3 (Handling & Processing, Human Necessities, and Polymers)

	(1)	(2)	(3)	(4)	(5)
Constant	-0.581 (0.182)	-0.422 (0.329)	0.008 (0.245)	-0.427*** (0.158)	-0.357** (0.177)
Lag 1	0.808*** (0.064)	0.917*** (0.024)	0.841*** (0.065)	0.931*** (0.025)	0.899*** (0.077)
Lag 2	0.193*** (0.081)		0.147* (0.083)		0.219** (0.096)
Lag 3	-0.054 (0.064)		-0.059 (0.062)		-0.159** (0.074)
Business- Enter. R&D		0.061 (0.050)	0.105*** (0.038)		
Specific- sector R&D				0.019 (0.035)	-0.005 (0.038)
No. of Obs.	225	256	216	182	158
Adj. R-sq	0.93	0.95	0.95	0.95	0.95
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. ES refers to Euro-Direct Subsequent Filings. Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Agricultural, Food, Wood, Petroleum, Rubber, and Textile R&D per worker.

Table 17d. Sectoral Regressions (Euro via PCT First Filings)

Dependent Variable: ln (PF/Labor)

Technology Group 3 (Handling & Processing, Human Necessities, and Polymers)

	(1)	(2)	(3)	(4)	(5)
Constant	-2.105** (0.644)	-3.68*** (0.713)	-1.826* (0.734)	-2.159*** (0.724)	-1.531 (0.729)
Lag 1	0.339*** (0.067)	0.669*** (0.044)	0.337*** (0.067)	0.705*** (0.053)	0.418*** (0.079)
Lag 2	0.263*** (0.066)		0.259*** (0.066)		0.165*** (0.080)
Lag 3	0.231*** (0.060)		0.227*** (0.060)		0.230*** (0.070)
Business- Enter. R&D		0.104 (0.072)	0.056 (0.071)		
Specific- sector R&D				0.189** (0.087)	0.073 (0.085)
No. of Obs.	169	217	169	145	115
Adj. R-sq	0.65	0.54	0.66	0.63	0.71
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. PF refers to Euro via PCT First Filings. Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Agricultural, Food, Wood, Petroleum, Rubber, and Textile R&D per worker.

Table 17e. Sectoral Regressions (Euro via PCT Subsequent Filings)

Dependent Variable: ln (PS/Labor)

Technology Group 3 (Handling & Processing, Human Necessities, and Polymers)

	(1)	(2)	(3)	(4)	(5)
Constant	-0.562 (0.188)	-0.673 (0.469)	-0.289 (0.288)	-0.162 (0.486)	-0.535* (0.307)
Lag 1	0.624*** (0.064)	0.844*** (0.024)	0.626*** (0.065)	0.811*** (0.033)	0.551*** (0.077)
Lag 2	0.185** (0.069)		0.185*** (0.069)		0.229*** (0.081)
Lag 3	0.119** (0.051)		0.110*** (0.051)		0.124** (0.059)
Business- Enter. R&D		0.129** (0.064)	0.050 (0.037)		
Specific- sector R&D				0.208*** (0.081)	0.011 (0.045)
No. of Obs.	216	253	212	174	148
Adj. R-sq	0.93	0.92	0.94	0.90	0.92
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. PS refers to Euro via PCT Subsequent Filings. Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Agricultural, Food, Wood, Petroleum, Rubber, and Textile R&D per worker.

Table 18a. Sectoral Regressions (Total Filings)Dependent Variable: $\ln(\text{TOTAL}/\text{Labor})$ Technology Group 4 (Measuring & Optics, Civil Engineering & Thermodynamics, Vehicles & General Technology)

	(1)	(2)	(3)	(4)	(5)
Constant	-0.549*** (0.144)	-0.589** (0.262)	-0.293 (0.152)	-0.487*** (0.173)	-0.295** (0.124)
Lag 1	0.855*** (0.069)	0.794*** (0.028)	0.827*** (0.071)	0.848*** (0.024)	0.834*** (0.069)
Lag 2	0.072 (0.091)		0.035 (0.087)		0.252*** (0.083)
Lag 3	0.012*** (0.066)		0.065 (0.062)		-0.138** (0.064)
Business- Enter. R&D		0.197*** (0.043)	0.051*** (0.022)		
Specific- sector R&D				0.124*** (0.038)	0.014 (0.018)
No. of Obs.	178	219	176	219	176
Adj. R-sq	0.96	0.96	0.96	0.97	0.97
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. TOTAL refers to the sum of EF (Euro-Direct First Filings), ES (Euro-Direct Subsequent Filings), PF (Euro via PCT First Filings), and PS (Euro via PCT Subsequent Filings). Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Manufacturing R&D per worker.

Table 18b. Sectoral Regressions (Euro-Direct First Filings)

Dependent Variable: ln (EF/Labor)

Technology Group 4 (. . . Technology Group 4 (Measuring & Optics, Civil Engineering & Thermodynamics, Vehicles & General Technology)

	(1)	(2)	(3)	(4)	(5)
Constant	-0.704 (0.415)	-2.125*** (0.659)	-0.746 (0.515)	-1.855*** (0.530)	-0.998** (0.449)
Lag 1	0.649*** (0.076)	0.787*** (0.042)	0.634*** (0.077)	0.799*** (0.042)	0.631*** (0.077)
Lag 2	0.266*** (0.085)		0.282*** (0.086)		0.291*** (0.086)
Lag 3	0.028 (0.072)		0.016 (0.073)		-0.003 (0.073)
Business- Enter. R&D		0.089 (0.076)	0.013 (0.059)		
Specific- sector R&D				0.079 (0.075)	-0.021 (0.061)
No. of Obs.	175	209	170	209	170
Adj. R-sq	0.84	0.80	0.83	0.80	0.83
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. EF refers to Euro-Direct First Filings. Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Manufacturing R&D per worker.

Table 18c. Sectoral Regressions (Euro-Direct Subsequent Filings)

Dependent Variable: ln (ES/Labor)

Technology Group 4 (Measuring & Optics, Civil Engineering & Thermodynamics, Vehicles & General Technology)

	(1)	(2)	(3)	(4)	(5)
Constant	-0.372*** (0.146)	-0.223 (0.232)	-0.076 (0.186)	-1.855*** (0.53)	-0.380*** (0.151)
Lag 1	0.995*** (0.060)	0.918*** (0.019)	0.976*** (0.062)	0.799*** (0.042)	0.954*** (0.063)
Lag 2	-0.088*** (0.081)		-0.060 (0.081)		0.032 (0.079)
Lag 3	0.058 (0.058)		0.036 (0.058)		-0.041 (0.058)
Business- Enter. R&D		0.088** (0.037)	0.061*** (0.031)		
Specific- sector R&D				0.079 (0.075)	0.019 (0.029)
No. of Obs.	223	257	214	257	214
Adj. R-sq	0.95	0.96	0.96	0.96	0.96
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. ES refers to Euro-Direct Subsequent Filings. Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Manufacturing R&D per worker.

Table 18d. Sectoral Regressions (Euro via PCT First Filings)

Dependent Variable: ln (PF/Labor)

Technology Group 4 (Measuring & Optics, Civil Engineering & Thermodynamics, Vehicles & General Technology)

	(1)	(2)	(3)	(4)	(5)
Constant	-1.113* (0.602)	-2.301*** (0.659)	-1.021 (0.689)	-1.945*** (0.565)	-0.801 (0.609)
Lag 1	0.519*** (0.072)	0.788*** (0.042)	0.518*** (0.073)	0.818*** (0.038)	0.541*** (0.072)
Lag 2	0.122 (0.077)		0.121 (0.077)		0.102 (0.078)
Lag 3	0.268*** (0.068)		0.268*** (0.068)		0.281*** (0.068)
Business- Enter. R&D		0.076 (0.064)	0.019 (0.068)		
Specific- sector R&D				0.037 (0.064)	-0.004 (0.069)
No. of Obs.	160	204	160	204	160
Adj. R-sq	0.73	0.66	0.74	0.71	0.77
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. PF refers to Euro via PCT First Filings. Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Manufacturing R&D per worker.

Table 18e. Sectoral Regressions (Euro via PCT Subsequent Filings)

Dependent Variable: ln (PS/Labor)

Technology Group 4 (Measuring & Optics, Civil Engineering & Thermodynamics, Vehicles & General Technology)

	(1)	(2)	(3)	(4)	(5)
Constant	-0.562*** (0.185)	-0.675 (0.507)	-0.415 (0.282)	-0.534 (0.362)	-0.128 (0.230)
Lag 1	0.716*** (0.058)	0.839*** (0.027)	0.707*** (0.059)	0.863*** (0.026)	0.747*** (0.006)
Lag 2	0.053 (0.064)		0.060 (0.066)		0.082 (0.065)
Lag 3	0.164*** (0.054)		0.163*** (0.055)		0.116** (0.054)
Business- Enter. R&D		0.139** (0.069)	0.024 (0.035)		
Specific- sector R&D				0.109* (0.062)	0.030 (0.035)
No. of Obs.	212	252	209	252	209
Adj. R-sq	0.94	0.92	0.94	0.93	0.95
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. PS refers to Euro via PCT Subsequent Filings. Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Manufacturing R&D per worker.

Table 19a. Sectoral Regressions (Total Filings)Dependent Variable: $\ln(\text{TOTAL}/\text{Labor})$ Technology Group 5 (Computers and Telecommunications)

	(1)	(2)	(3)	(4)	(5)
Constant	-0.830** (0.362)	-0.136 (0.361)	-0.269 (0.387)	0.353 (0.721)	0.360 (0.775)
Lag 1	0.519*** (0.074)	0.826*** (0.035)	0.506*** (0.075)	0.852*** (0.045)	0.537*** (0.103)
Lag 2	0.289*** (0.082)		0.283*** (0.080)		0.322*** (0.107)
Lag 3	0.107 (0.081)		0.076 (0.077)		0.023 (0.111)
Business- Enter. R&D		0.250*** (0.067)	0.160*** (0.074)		
Specific- sector R&D				0.248** (0.111)	0.198* (0.117)
No. of Obs.	175	212	173	131	103
Adj. R-sq	0.84	0.86	0.86	0.83	0.83
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. TOTAL refers to the sum of EF (Euro-Direct First Filings), ES (Euro-Direct Subsequent Filings), PF (Euro via PCT First Filings), and PS (Euro via PCT Subsequent Filings). Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Electronics, Nonmetals, Services, and Office Equipment R&D per worker.

Table 19b. Sectoral Regressions (Euro-Direct First Filings)Dependent Variable: $\ln(\text{EF/Labor})$ Technology Group 5 (Computers and Telecommunications)

	(1)	(2)	(3)	(4)	(5)
Constant	-1.048 (0.946)	-2.386* (1.385)	-3.383* (1.798)	-1.282 (1.834)	-4.392 (4.990)
Lag 1	0.502*** (0.108)	0.855*** (0.054)	0.471*** (0.109)	0.813*** (0.079)	0.305* (0.162)
Lag 2	0.339*** (0.119)		0.329*** (0.118)		0.261* (0.158)
Lag 3	0.066 (0.113)		0.085 (0.112)		0.104 (0.149)
Business- Enter. R&D		-0.069 (0.158)	-0.294 (0.193)		
Specific- sector R&D				0.162 (0.204)	-0.019 (0.483)
No. of Obs.	86	117	86	68	45
Adj. R-sq	0.69	0.69	0.72	0.63	0.51
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. EF refers to Euro-Direct First Filings. Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Electronics, Nonmetals, Services, and Office Equipment R&D per worker.

Table 19c. Sectoral Regressions (Euro-Direct Subsequent Filings)

Dependent Variable: ln (ES/Labor)

Technology Group 5 (Computers and Telecommunications)

	(1)	(2)	(3)	(4)	(5)
Constant	-0.489 (0.327)	-0.216 (0.548)	0.250 (0.415)	-0.465 (0.685)	0.399 (0.673)
Lag 1	0.695*** (0.071)	0.746*** (0.039)	0.622*** (0.072)	0.823*** (0.048)	0.634*** (0.092)
Lag 2	0.179** (0.080)		0.219*** (0.079)		0.303*** (0.104)
Lag 3	0.080 (0.066)		0.027 (0.066)		-0.035 (0.088)
Business- Enter. R&D		0.404*** (0.100)	0.249*** (0.085)		
Specific- sector R&D				0.210** (0.110)	0.188* (0.114)
No. of Obs.	173	215	170	140	110
Adj. R-sq	0.89	0.87	0.90	0.84	0.86
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. ES refers to Euro-Direct Subsequent Filings. Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Electronics, Nonmetals, Services, and Office Equipment R&D per worker.

Table 19d. Sectoral Regressions (Euro via PCT First Filings)

Dependent Variable: ln (PF/Labor)

Technology Group 5 (Computers and Telecommunications)

	(1)	(2)	(3)	(4)	(5)
Constant	-1.767 (2.347)	-3.029* (1.645)	0.529 (2.834)	-4.629** (2.217)	-0.773 (3.212)
Lag 1	0.482*** (0.154)	0.732*** (0.078)	0.446*** (0.154)	0.842*** (0.098)	0.672*** (0.215)
Lag 2	0.124 (0.163)		0.138 (0.162)		0.069 (0.219)
Lag 3	0.263*** (0.154)		0.267*** (0.152)		0.269*** (0.193)
Business- Enter. R&D		0.139 (0.166)	0.370 (0.263)		
Specific- sector R&D				-0.303 (0.299)	-0.154 (0.422)
No. of Obs.	45	83	45	49	27
Adj. R-sq	0.44	0.53	0.50	0.64	0.67
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. PF refers to Euro via PCT First Filings. Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Electronics, Nonmetals, Services, and Office Equipment R&D per worker.

Table 19e. Sectoral Regressions (Euro via PCT Subsequent Filings)

Dependent Variable: ln (PS/Labor)

Technology Group 5 (Computers and Telecommunications)

	(1)	(2)	(3)	(4)	(5)
Constant	-0.309 (0.507)	-0.731 (0.607)	-0.027 (0.667)	0.244 (1.077)	0.895 (1.078)
Lag 1	0.553*** (0.074)	0.871*** (0.0337)	0.549*** (0.075)	0.848*** (0.045)	0.498*** (0.098)
Lag 2	0.307*** (0.082)		0.304*** (0.082)		0.382*** (0.099)
Lag 3	0.092 (0.073)		0.093 (0.073)		0.086 (0.095)
Business- Enter. R&D		0.107 (0.073)	0.051 (0.078)		
Specific- sector R&D				0.251* (0.142)	0.123 (0.144)
No. of Obs.	167	208	167	129	103
Adj. R-sq	0.79	0.79	0.80	0.78	0.82
Method of Estimation	OLS	GLS	GLS	GLS	GLS

Notes: Column 4 and 5, the dependent variable is divided by the number of workers in that same specific sectors. The equations are estimated over the period 1980-1996. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. PS refers to Euro via PCT Subsequent Filings. Business Enter. R&D is the natural log of Business Enterprise Research and Development (B.E.R.D.) Expenditures per worker, and specific-sector R&D is the natural log of Electronics, Nonmetals, Services, and Office Equipment R&D per worker.

Table 20. Summary of Forecast Errors: Sectoral Filings (by Joint Cluster)**A. Joint Cluster Group 1 (Electrical Machines, Electricity, Electronics, Audio, Video & Media)**

Model	Variable	N	1997	1998	1999	2000
AR3	Total	177	0.205	0.155	0.111	0.248
RE1	Total	217	0.268	0.160	0.151	0.205
RE3	Total	175	0.198	0.166	0.113	0.199
RE1s	Total	157	0.305	0.159	0.165	0.189
RE3s	Total	124	0.141	0.161	0.133	0.172
AR3	EF	140	0.220	0.152	0.463	0.327
RE1	EF	172	0.259	0.342	0.457	0.329
RE3	EF	139	0.217	0.151	0.459	0.272
RE1s	EF	128	0.297	0.474	0.676	0.427
RE3s	EF	101	0.181	0.146	0.444	0.350
AR3	ES	219	0.358	0.296	0.231	0.361
RE1	ES	251	0.413	0.242	0.353	0.361
RE3	ES	213	0.395	0.255	0.289	0.344
RE1s	ES	187	0.361	0.236	0.427	0.213
RE3s	ES	156	0.329	0.225	0.316	0.224
AR3	PF	89	0.477	0.489	0.503	0.368
RE1	PF	137	0.453	0.508	0.534	0.434
RE3	PF	89	0.439	0.472	0.505	0.345
RE1s	PF	102	0.510	0.428	0.678	0.334
RE3s	PF	66	0.349	0.313	0.722	0.251
AR3	PS	192	0.260	0.266	0.198	0.264
RE1	PS	233	0.204	0.177	0.259	0.229
RE3	PS	187	0.226	0.199	0.167	0.194
RE1s	PS	167	0.198	0.187	0.191	0.187
RE3s	PS	133	0.157	0.158	0.165	0.146

Notes: The forecast errors correspond to the estimated models in Tables 15 - 19 (inclusive). Each sub-table (A - E) represents a different technology group. Each entry in each sub-table is the RMSPE (root mean square percentage error) associated with a model: AR3 autoregressive model of order 3, RE1 regression model of R&D and one lagged dependent variable, and RE3 regression model of R&D and three lags of the dependent variable. RE1s denotes the regression model with sector-specific R&D and one lagged dependent variable and RE3s the regression model with sector-specific R&D and three lags of the dependent variable. Thus AR3, RE1, RE3, RE1s, and RE3s correspond to the results in columns 1 - 5 respectively in Tables 15 - 19. N denotes the number of observations, EF Euro-direct first filings, ES Euro-direct subsequent filings, PF Euro via PCT first filings, and PS Euro via PCT subsequent filings. Total is the sum (=EF+ES+PF+PS).

B. Joint Cluster Group 2 (Industrial Chemistry, Organic Chemistry, and Biotechnology):

Model	Variable	N	1997	1998	1999	2000
AR3	Total	178	0.164	0.226	0.142	0.086
RE1	Total	219	0.203	0.279	0.146	0.176
RE3	Total	176	0.164	0.222	0.132	0.085
RE1s	Total	174	0.131	0.145	0.108	0.136
RE3s	Total	140	0.120	0.132	0.079	0.093
AR3	EF	169	0.651	0.792	0.278	0.592
RE1	EF	204	1.260	1.29	0.584	0.867
RE3	EF	163	0.655	0.849	0.297	0.697
RE1s	EF	169	0.517	1.431	0.694	0.487
RE3s	EF	139	0.346	1.089	0.289	0.422
AR3	ES	225	0.328	0.254	0.290	0.308
RE1	ES	257	0.418	0.266	0.322	0.299
RE3	ES	216	0.328	0.249	0.297	0.267
RE1s	ES	202	0.393	0.214	0.299	0.394
RE3s	ES	174	0.234	0.216	0.270	0.344
AR3	PF	122	0.336	0.441	0.419	0.463
RE1	PF	169	0.339	0.502	0.437	0.462
RE3	PF	122	0.337	0.441	0.420	0.467
RE1s	PF	131	0.323	0.489	0.413	0.277
RE3s	PF	96	0.379	0.339	0.431	0.225
AR3	PS	209	0.173	0.115	0.205	0.107
RE1	PS	247	0.249	0.118	0.195	0.158
RE3	PS	203	0.157	0.102	0.190	0.093
RE1s	PS	187	0.159	0.134	0.161	0.152
RE3s	PS	159	0.096	0.109	0.146	0.110

C. Joint Cluster Group 3 (Handling & Processing, Human Necessities, and Polymers):

Model	Variable	N	1997	1998	1999	2000
AR3	Total	178	0.170	0.118	0.103	0.183
RE1	Total	219	0.199	0.214	0.109	0.238
RE3	Total	176	0.161	0.103	0.088	0.154
RE1s	Total	154	0.131	0.119	0.135	0.126
RE3s	Total	124	0.097	0.082	0.077	0.079
AR3	EF	176	0.352	0.331	0.327	0.214
RE1	EF	214	0.453	0.311	0.400	0.276
RE3	EF	171	0.345	0.317	0.334	0.233
RE1s	EF	157	0.295	0.375	0.460	0.257
RE3s	EF	130	0.157	0.205	0.304	0.199
AR3	ES	225	0.257	0.159	0.166	0.395
RE1	ES	256	0.229	0.210	0.162	0.402
RE3	ES	216	0.209	0.163	0.151	0.385
RE1s	ES	182	0.114	0.144	0.196	0.474
RE3s	ES	158	0.107	0.115	0.179	0.528
AR3	PF	169	0.436	0.290	0.261	0.376
RE1	PF	217	0.367	0.256	0.258	0.424
RE3	PF	169	0.430	0.272	0.258	0.378
RE1s	PF	145	0.285	0.237	0.207	0.349
RE3s	PF	115	0.363	0.218	0.206	0.291
AR3	PS	216	0.178	0.215	0.139	0.148
RE1	PS	253	0.228	0.244	0.119	0.182
RE3	PS	212	0.178	0.201	0.123	0.114
RE1s	PS	174	0.198	0.111	0.147	0.165
RE3s	PS	148	0.098	0.119	0.144	0.077

D. Joint Cluster Group 4 (Measuring & Optics, Civil Engineering, Thermodynamics, Vehicles & General Technology):

Model	Variable	N	1997	1998	1999	2000
AR3	Total	178	0.135	0.111	0.091	0.087
RE1	Total	219	0.188	0.122	0.158	0.125
RE3	Total	176	0.135	0.113	0.090	0.074
RE1s	Total	219	0.141	0.116	0.145	0.112
RE3s	Total	176	0.098	0.097	0.104	0.058
AR3	EF	175	0.729	0.329	0.328	0.268
RE1	EF	209	0.910	0.431	0.563	0.351
RE3	EF	170	0.792	0.344	0.315	0.255
RE1s	EF	209	0.955	0.409	0.534	0.339
RE3s	EF	170	0.842	0.329	0.320	0.266
AR3	ES	223	0.256	0.227	0.269	0.209
RE1	ES	257	0.231	0.179	0.314	0.211
RE3	ES	214	0.221	0.194	0.224	0.186
RE1s	ES	257	0.204	0.183	0.319	0.211
RE3s	ES	214	0.191	0.193	0.254	0.202
AR3	PF	160	0.387	0.355	0.329	0.597
RE1	PF	204	0.328	0.420	0.406	0.780
RE3	PF	160	0.386	0.354	0.334	0.589
RE1s	PF	204	0.349	0.469	0.407	0.815
RE3s	PF	160	0.448	0.389	0.349	0.608
AR3	PS	212	0.190	0.186	0.163	0.205
RE1	PS	252	0.266	0.202	0.195	0.216
RE3	PS	209	0.174	0.175	0.153	0.196
RE1s	PS	252	0.220	0.206	0.195	0.204
RE3s	PS	209	0.159	0.185	0.158	0.192

E. Joint Cluster Group 5 (Computers and Telecommunications):

Model	Variable	N	1997	1998	1999	2000
AR3	Total	175	0.345	0.285	0.267	0.306
RE1	Total	212	0.286	0.279	0.235	0.394
RE3	Total	173	0.325	0.298	0.255	0.304
RE1s	Total	131	0.282	0.322	0.235	0.174
RE3s	Total	103	0.319	0.294	0.253	0.147
AR3	EF	86	0.396	0.501	0.405	0.304
RE1	EF	117	0.339	0.479	0.552	0.409
RE3	EF	86	0.404	0.528	0.386	0.387
RE1s	EF	68	0.298	0.599	0.737	0.213
RE3s	EF	45	0.339	0.581	0.482	0.773
AR3	ES	173	0.339	0.261	0.351	0.237
RE1	ES	215	0.381	0.299	0.445	0.289
RE3	ES	170	0.321	0.279	0.379	0.249
RE1s	ES	140	0.484	0.331	0.528	0.448
RE3s	ES	110	0.398	0.338	0.496	0.388
AR3	PF	45	0.455	0.413	0.624	0.517
RE1	PF	83	0.558	0.549	0.691	0.605
RE3	PF	45	0.477	0.362	0.596	0.525
RE1s	PF	49	0.726	0.666	1.070	0.370
RE3s	PF	27	0.216	0.296	1.060	0.302
AR3	PS	167	0.342	0.263	0.333	0.296
RE1	PS	208	0.329	0.317	0.387	0.385
RE3	PS	167	0.333	0.259	0.299	0.269
RE1s	PS	129	0.386	0.371	0.399	0.356
RE3s	PS	103	0.375	0.274	0.405	0.034

Table 21. Sample Filing Forecasts by Technological Sectors: Actual vs. Predicted values for 1998

Panel A

Tech			Tech			Tech			Tech			Tech		
Group	TOTAL	TOTAL	Group	TOTAL	TOTAL	Group	TOTAL	TOTAL	Group	TOTAL	TOTAL	Group	TOTAL	TOTAL
G1	Actual	Predicted	G2	Actual	Predicted	G3	Actual	Predicted	G4	Actual	Predicted	G5	Actual	Predicted
Australia	82	73	Australia	275	249	Australia	318	327	Australia	300	262	Australia	54	44
Austria	103	78	Austria	192	166	Austria	335	275	Austria	294	229	Austria	21	18
Belgium	138	131	Belgium	428	326	Belgium	433	380	Belgium	240	213	Belgium	34	35
Canada	192	174	Canada	528	462	Canada	335	341	Canada	331	282	Canada	309	182
Denmark	77	54	Denmark	286	301	Denmark	232	205	Denmark	159	163	Denmark	30	16
Finland	183	148	Finland	159	153	Finland	370	309	Finland	183	163	Finland	517	259
France	1293	1126	France	1702	1541	France	1632	1653	France	1716	1633	France	709	425
Germany	3241	2452	Germany	4063	3617	Germany	5148	4877	Germany	6143	5093	Germany	1249	768
Italy	416	391	Italy	530	506	Italy	1341	1113	Italy	901	805	Italy	108	84
Japan	4894	4321	Japan	3346	3349	Japan	3368	3445	Japan	3101	2940	Japan	1469	1177
Korea	291	303	Korea	164	156	Korea	132	134	Korea	169	123	Korea	76	40
Neth	1202	914	Neth	969	898	Neth	859	895	Neth	638	545	Neth	334	262
Norway	49	40	Norway	139	97	Norway	98	121	Norway	187	164	Norway	10	16
Portugal	1		Portugal	2	6	Portugal	9	13	Portugal	9	8	Portugal	0	
Spain	90	50	Spain	149	129	Spain	256	225	Spain	188	173	Spain	23	24
Sweden	484	324	Sweden	408	362	Sweden	744	684	Sweden	537	600	Sweden	697	341
UK	622	627	UK	1544	1824	UK	1225	1201	UK	1152	1126	UK	482	397
USA	6030	6273	USA	10746	10031	USA	9495	9102	USA	5908	5711	USA	4673	3911
<i>Sum</i>	<i>19388</i>	<i>17479</i>	<i>Sum</i>	<i>25630</i>	<i>24171</i>	<i>Sum</i>	<i>26330</i>	<i>25299</i>	<i>Sum</i>	<i>22156</i>	<i>20233</i>	<i>Sum</i>	<i>10795</i>	<i>8001</i>
<i>RMSPE</i>		<i>0.166</i>	<i>RMSPE</i>		<i>0.222</i>	<i>RMSPE</i>		<i>0.103</i>	<i>RMSPE</i>		<i>0.113</i>	<i>RMSPE</i>		<i>0.298</i>

Notes: G1 is electronics-related, G2 chemicals, G3 manufacturing, G4 physics, and G5 computer-related.

Predictions are from a regression model of R&D and three lags of the dependent variable.

RMSPE denotes root mean squared percentage error.

EF and ES refers to Euro-direct first and subsequent filings respectively

PF and PS refers to Euro via PCT first and subsequent filings respectively

Total = EF + ES + PF + PS.

Panel B

Tech			Tech			Tech			Tech			Tech		
Group	EF	EF	Group	EF	EF	Group	EF	EF	Group	EF	EF	Group	EF	EF
G1	Actual	Predicted	G2	Actual	Predicted	G3	Actual	Predicted	G4	Actual	Predicted	G5	Actual	Predicted
Australia	1		Australia	1		Australia	0		Australia	1		Australia	0	
Austria	8	8	Austria	15	9	Austria	27	19	Austria	14	15	Austria	3	
Belgium	52	58	Belgium	168	122	Belgium	142	129	Belgium	70	53	Belgium	6	17
Canada	3		Canada	1	7	Canada	7	7	Canada	6	3	Canada	2	
Denmark	5		Denmark	23	7	Denmark	18	7	Denmark	8	3	Denmark	2	
Finland	1		Finland	3		Finland	2		Finland	0		Finland	4	4
France	106	83	France	116	65	France	82	65	France	77	55	France	113	45
Germany	211	151	Germany	308	244	Germany	240	168	Germany	247	183	Germany	161	70
Italy	192	226	Italy	76	41	Italy	176	78	Italy	130	89	Italy	33	34
Japan	53	58	Japan	46	42	Japan	64	77	Japan	52	62	Japan	20	9
Korea	9	11	Korea	3	1	Korea	4	3	Korea	4	2	Korea	1	
Neth	499	492	Neth	367	406	Neth	190	175	Neth	150	101	Neth	94	86
Norway	1		Norway	0		Norway	0		Norway	3	2	Norway	0	
Portugal	0		Portugal	1		Portugal	2	4	Portugal	0		Portugal	0	
Spain	7		Spain	10	6	Spain	15	12	Spain	15	10	Spain	0	
Sweden	7	7	Sweden	3		Sweden	11	3	Sweden	11	5	Sweden	37	11
UK	34	38	UK	112	320	UK	48	51	UK	54	35	UK	120	86
USA	159	105	USA	146	141	USA	218	162	USA	136	84	USA	250	129
Sum	1348	1237	Sum	1399	1410	Sum	1246	960	Sum	978	702	Sum	846	491
RMSPE		0.151	RMSPE		0.849	RMSPE		0.317	RMSPE		0.344	RMSPE		0.528

Panel C

Tech			Tech			Tech			Tech			Tech		
Group	ES	ES	Group	ES	ES	Group	ES	ES	Group	ES	ES	Group	ES	ES
G1	Actual	Predicted	G2	Actual	Predicted	G3	Actual	Predicted	G4	Actual	Predicted	G5	Actual	Predicted
Australia	10	3	Australia	3	6	Australia	6	10	Australia	15	10	Australia	1	
Austria	47	30	Austria	72	57	Austria	180	125	Austria	165	115	Austria	5	6
Belgium	49	45	Belgium	64	53	Belgium	162	149	Belgium	97	91	Belgium	10	6
Canada	67	43	Canada	38	33	Canada	73	54	Canada	53	40	Canada	118	39
Denmark	14	10	Denmark	27	35	Denmark	32	38	Denmark	37	30	Denmark	2	2
Finland	74	66	Finland	16	20	Finland	71	70	Finland	45	35	Finland	115	83
France	773	685	France	569	576	France	804	847	France	1004	988	France	359	237
Germany	1594	1211	Germany	1416	1358	Germany	2870	2606	Germany	3397	3002	Germany	438	378
Italy	154	128	Italy	175	195	Italy	828	673	Italy	574	492	Italy	30	27
Japan	3648	3319	Japan	1433	1513	Japan	2122	2205	Japan	2253	2206	Japan	1132	942
Korea	176	212	Korea	15	28	Korea	56	63	Korea	63	57	Korea	46	21
Neth	120	137	Neth	143	143	Neth	291	289	Neth	206	181	Neth	51	49
Norway	14	8	Norway	5	5	Norway	9	10	Norway	11	21	Norway	1	1
Portugal	1		Portugal	1	1	Portugal	5		Portugal	5	4	Portugal	0	
Spain	61	20	Spain	37	25	Spain	99	87	Spain	70	65	Spain	12	10
Sweden	29	33	Sweden	33	27	Sweden	109	106	Sweden	85	83	Sweden	19	29
UK	189	192	UK	216	229	UK	363	318	UK	388	371	UK	97	104
USA	2026	2613	USA	1551	1479	USA	2413	2457	USA	1909	1866	USA	1240	1540
Sum	9046	8757	Sum	5814	5781	Sum	10493	10107	Sum	10377	9656	Sum	3676	3474
RMSPE		0.255	RMSPE		0.249	RMSPE		0.163	RMSPE		0.194	RMSPE		0.279

Panel D

Tech			Tech			Tech			Tech			Tech		
Group	PF	PF	Group	PF	PF	Group	PF	PF		PF	PF	Group	PF	PF
G1	Actual	Predicted	G2	Actual	Predicted	G3	Actual	Predicted	G4	Actual	Predicted	G5	Actual	Predicted
Australia	2	5	Australia	17	9	Australia	20	20	Australia	18	12	Australia	5	
Austria	1		Austria	1	4	Austria	4	3	Austria	8	3	Austria	0	
Belgium	1		Belgium	2	5	Belgium	6	6	Belgium	8	2	Belgium	1	
Canada	7	10	Canada	18	16	Canada	22	23	Canada	20	21	Canada	5	4
Denmark	4	1	Denmark	5	6	Denmark	4	5	Denmark	4	6	Denmark	1	
Finland	8	2	Finland	5	5	Finland	9	6	Finland	11	9	Finland	50	10
France	23	14	France	20	21	France	41	35	France	29	30	France	9	9
Germany	31	20	Germany	43	29	Germany	46	46	Germany	36	44	Germany	26	12
Italy	5	6	Italy	6	6	Italy	18	14	Italy	12	10	Italy	4	1
Japan	163	86	Japan	128	74	Japan	128	87	Japan	133	66	Japan	71	55
Korea	16	8	Korea	13	8	Korea	13	8	Korea	23	7	Korea	3	
Neth	10	6	Neth	16	12	Neth	12	15	Neth	12	13	Neth	2	2
Norway	0		Norway	5		Norway	3	5	Norway	12	4	Norway	0	
Portugal	0		Portugal	0		Portugal	0		Portugal	2		Portugal	0	
Spain	5		Spain	17	11	Spain	23	9	Spain	15	9	Spain	1	
Sweden	9	5	Sweden	9	7	Sweden	16	19	Sweden	15	19	Sweden	8	
UK	10	11	UK	21	20	UK	25	20	UK	22	17	UK	5	8
USA	138	121	USA	355	311	USA	428	326	USA	197	203	USA	107	91
Sum	433	296	Sum	681	543	Sum	818	648	Sum	577	474	Sum	298	191
RMSPE		0.472	RMSPE		0.441	RMSPE		0.272	RMSPE		0.354	RMSPE		0.362

Panel E

Tech			Tech			Tech			Tech			Tech		
Group	PS	PS	Group	PS	PS	Group	PS	PS	Group	PS	PS	Group	PS	PS
G1	Actual	Predicted	G2	Actual	Predicted	G3	Actual	Predicted	G4	Actual	Predicted	G5	Actual	Predicted
Australia	69	66	Australia	254	268	Australia	292	304	Australia	266	259	Australia	48	48
Austria	47	38	Austria	104	106	Austria	124	128	Austria	107	96	Austria	13	13
Belgium	36	29	Belgium	194	158	Belgium	123	97	Belgium	65	63	Belgium	17	11
Canada	115	120	Canada	471	440	Canada	233	268	Canada	252	229	Canada	184	165
Denmark	54	40	Denmark	231	274	Denmark	178	157	Denmark	110	130	Denmark	25	15
Finland	100	79	Finland	135	142	Finland	288	212	Finland	127	120	Finland	348	229
France	391	319	France	997	910	France	705	665	France	606	549	France	228	147
Germany	1405	1082	Germany	2296	2079	Germany	1992	1825	Germany	2463	1947	Germany	624	366
Italy	65	65	Italy	273	260	Italy	319	283	Italy	185	191	Italy	41	27
Japan	1030	737	Japan	1739	1619	Japan	1054	963	Japan	663	556	Japan	246	188
Korea	90	57	Korea	133	114	Korea	59	62	Korea	79	57	Korea	26	14
Neth	573	436	Neth	443	427	Neth	366	417	Neth	270	241	Neth	187	177
Norway	34	31	Norway	129	91	Norway	86	105	Norway	161	141	Norway	9	15
Portugal	0		Portugal	0		Portugal	2	6	Portugal	2	3	Portugal	0	
Spain	17	22	Spain	85	89	Spain	119	115	Spain	88	83	Spain	10	9
Sweden	439	260	Sweden	363	347	Sweden	608	549	Sweden	426	489	Sweden	633	367
UK	389	391	UK	1195	1241	UK	789	844	UK	688	726	UK	260	258
USA	3707	3420	USA	8694	8536	USA	6436	6343	USA	3666	3542	USA	3076	2591
Sum	8561	7191	Sum	17736	17101	Sum	13773	13344	Sum	10224	9423	Sum	5975	4641
RMSPE		0.199	RMSPE		0.102	RMSPE		0.201	RMSPE		0.175	RMSPE		0.259

Table 22. Patent Families Derived from First Filings, by Country of Origin and Selected Years

1) Priority Year 1999

Secondary Filings as a Percentage of First Filings
(% Priority filings claimed in Country of Origin From):

Country Bloc	Cntry of Origin	First Filings	All Other Blocs	Other Trilateral Blocs	EPC	Japan	USA	Other Countries	Trilateral Patent Families
EPC:	Austria	2005	29.6%	12.2%	--	4.5%	10.1%	22.1%	2.4%
	Belgium	568	28.3%	8.5%	--	3.2%	6.9%	24.1%	1.6%
	Denmark	1373	60.3%	15.7%	--	6.3%	11.8%	58.9%	2.4%
	EPO	11535	46.5%	28.0%	--	14.5%	20.3%	33.3%	6.7%
	Finland	2495	51.5%	17.6%	--	6.1%	14.0%	47.4%	2.6%
	France	12972	39.9%	20.8%	--	12.2%	14.4%	28.7%	5.9%
	Germany	65819	24.4%	14.5%	--	7.6%	10.7%	14.5%	3.8%
	Greece	295	12.9%	2.4%	--	1.0%	1.4%	12.9%	0.0%
	Ireland	323	52.9%	21.7%	--	4.6%	20.1%	48.3%	3.1%
	Italy	7333	26.7%	12.1%	--	3.9%	10.3%	19.1%	2.0%
	Luxem.	68	58.8%	13.2%	--	4.4%	8.8%	52.9%	0.0%
	Monaco	17	23.5%	5.9%	--	5.9%	0.0%	17.6%	0.0%
	Netherl.	2438	40.0%	13.6%	--	5.2%	11.0%	34.2%	2.6%
	Portugal	142	15.5%	2.1%	--	0.7%	2.1%	14.8%	0.7%
	Spain	4624	11.1%	3.4%	--	1.0%	2.8%	9.5%	0.4%
	Sweden	3343	71.5%	20.7%	--	8.5%	15.8%	66.0%	3.5%
	Switzerl.	1188	69.8%	28.3%	--	18.7%	20.4%	58.8%	10.8%
	Turkey	472	100.0%	100.0%	--	0.0%	0.6%	12.1%	0.0%
	UK	13989	55.5%	21.5%	--	9.4%	15.1%	46.3%	2.9%
<i>Subtotal for EPC</i>		<i>130999</i>	<i>34.1%</i>	<i>17.1%</i>	--	<i>8.3%</i>	<i>12.3%</i>	<i>24.4%</i>	<i>3.9%</i>
Japan:	Jpn	356397	12.7%	11.6%	8.3%	--	7.1%	5.1%	3.8%
US:	USA	153350	41.6%	30.1%	28.7%	9.2%	--	32.2%	7.8%
Other:	Argen	92	59.8%	59.8%	39.1%	12.0%	27.2%	40.2%	6.5%
	Austral	6856	18.9%	18.9%	15.5%	3.4%	4.5%	17.6%	0.6%
	Brazil	6528	2.4%	2.4%	2.1%	0.3%	0.9%	2.3%	0.2%
	Bulgaria	336	100.0%	100.0%	8.0%	0.0%	0.0%	8.0%	0.0%
	Canada	1122	70.9%	70.9%	35.0%	6.1%	22.4%	39.5%	3.6%
	China	62649	0.6%	0.6%	0.3%	0.1%	0.3%	0.5%	0.0%
	Hungary	876	9.5%	9.5%	9.2%	0.5%	1.3%	9.5%	0.0%
	India	238	75.6%	75.6%	53.8%	9.2%	19.3%	50.8%	3.8%
	Israel	771	64.2%	64.2%	49.4%	7.8%	19.8%	50.8%	3.1%
	Korea	57107	8.5%	8.5%	3.5%	4.8%	4.6%	7.2%	1.1%
	Malay	43	60.5%	60.5%	27.9%	18.6%	16.3%	30.2%	4.7%
	Mexico	68	42.6%	42.6%	30.9%	1.5%	13.2%	32.4%	0.0%
	NewZeal	779	25.2%	25.2%	22.6%	3.1%	5.4%	24.3%	0.4%
	Norway	945	38.8%	38.8%	37.2%	3.0%	5.7%	38.3%	0.8%
	Philipp	7	42.9%	42.9%	28.6%	14.3%	0.0%	28.6%	0.0%
	Poland	3444	1.2%	1.2%	1.1%	0.1%	0.1%	1.1%	0.0%
	Romania	250	2.4%	2.4%	2.4%	0.4%	0.4%	2.4%	0.0%
	Russia	13845	1.6%	1.6%	1.4%	0.2%	0.5%	1.5%	0.1%
	SouthAfr	1595	13.9%	13.9%	10.6%	1.3%	4.1%	12.9%	0.3%
	Singapore	337	43.0%	43.0%	27.3%	8.6%	11.9%	34.1%	0.9%
<i>Subtotal for Other</i>		<i>157888</i>	<i>6.2%</i>	<i>6.2%</i>	<i>3.5%</i>	<i>2.1%</i>	<i>2.5%</i>	<i>5.1%</i>	<i>0.5%</i>

2) Priority Year 1994

Secondary Filings as a Percentage of First Filings
 (% Priority filings claimed in Country of Origin From):

Country Bloc	Cntry of Origin	First Filings	All Other Blocs	Other Trilateral Blocs	EPC	Japan	USA	Other Countries	Trilateral Patent Families
EPC:	Austria	1873	27.7%	19.6%	--	12.5%	17.2%	19.0%	10.1%
	Belgium	794	30.7%	23.2%	--	15.7%	20.4%	20.4%	13.0%
	Denmark	1169	46.8%	30.4%	--	22.1%	25.7%	42.1%	17.5%
	EPO	7068	43.8%	34.7%	--	24.2%	30.4%	26.1%	19.9%
	Finland	2310	37.2%	27.7%	--	18.9%	25.1%	30.4%	16.2%
	France	11513	35.6%	30.7%	--	20.4%	27.5%	21.8%	17.2%
	Germany	51874	23.0%	20.0%	--	14.1%	17.0%	11.8%	11.0%
	Greece	202	9.9%	5.4%	--	1.5%	5.4%	8.4%	1.5%
	Ireland	555	15.9%	9.5%	--	4.5%	8.3%	14.4%	3.2%
	Italy	7961	17.8%	14.1%	--	7.9%	12.4%	12.2%	6.2%
	Luxem.	59	47.5%	28.8%	--	18.6%	25.4%	44.1%	15.3%
	Monaco	27	7.4%	7.4%	--	7.4%	3.7%	7.4%	3.7%
	Netherl.	1659	40.5%	30.7%	--	20.3%	24.7%	26.5%	14.3%
	Portugal	202	3.0%	2.0%	--	0.5%	2.0%	2.5%	0.5%
	Spain	4246	7.4%	4.6%	--	2.3%	3.8%	5.4%	1.5%
	Sweden	2793	57.5%	42.9%	--	31.2%	37.4%	46.5%	25.7%
	Switzerl.	2164	55.0%	46.2%	--	35.8%	39.2%	40.8%	28.8%
	Turkey	74	100.0%	100.0%	--	0.0%	0.0%	2.7%	0.0%
	UK	11118	53.2%	41.2%	--	26.6%	35.5%	38.2%	20.8%
<i>Subtotal for EPC</i>		<i>107661</i>	<i>30.3%</i>	<i>24.8%</i>	--	<i>16.8%</i>	<i>21.3%</i>	<i>18.9%</i>	<i>13.4%</i>
Japan:	Jpn	318261	11.1%	10.6%	6.0%	--	9.8%	4.1%	5.2%
US:	USA	121733	28.1%	21.9%	20.3%	15.8%	--	20.7%	14.2%
Other:	Argen	48	64.6%	64.6%	33.3%	8.3%	27.1%	39.6%	4.2%
	Austral	5363	19.0%	19.0%	13.9%	8.2%	14.2%	18.3%	6.1%
	Brazil	4810	3.0%	3.0%	2.6%	0.9%	1.9%	3.0%	0.5%
	Bulgaria	415	100.0%	100.0%	1.2%	0.2%	0.5%	0.5%	0.0%
	Canada	3064	27.2%	27.2%	11.5%	5.2%	23.2%	26.3%	3.9%
	China	44210	0.4%	0.4%	0.2%	0.1%	0.2%	0.4%	0.1%
	Hungary	1032	5.5%	5.5%	4.7%	2.3%	2.9%	4.8%	1.7%
	India	180	12.2%	12.2%	3.3%	0.0%	2.2%	3.9%	0.0%
	Israel	725	54.6%	54.6%	36.0%	17.1%	39.9%	46.5%	13.8%
	Korea	32317	8.4%	8.4%	3.5%	5.4%	6.3%	7.0%	2.1%
	Malay	23	56.5%	56.5%	34.8%	26.1%	34.8%	26.1%	4.3%
	Mexico	47	63.8%	63.8%	23.4%	2.1%	23.4%	29.8%	0.0%
	NewZeal	731	21.3%	21.3%	16.6%	8.8%	15.0%	21.1%	5.2%
	Norway	643	37.2%	37.2%	35.3%	13.1%	23.8%	35.9%	11.8%
	Philipp	59	3.4%	3.4%	1.7%	0.0%	3.4%	3.4%	0.0%
	Poland	4107	0.9%	0.9%	0.7%	0.1%	0.2%	0.7%	0.0%
	Romania	1029	0.8%	0.8%	0.7%	0.1%	0.3%	0.8%	0.1%
	Russia	12721	1.2%	1.2%	1.0%	0.4%	0.6%	1.1%	0.3%
	SouthAfr	1434	16.9%	16.9%	12.1%	3.3%	9.7%	14.8%	2.6%
	Singapore	1644	14.7%	14.7%	14.5%	11.5%	12.0%	14.7%	10.0%
<i>Subtotal for Other</i>		<i>114602</i>	<i>6.0%</i>	<i>6.0%</i>	<i>3.3%</i>	<i>2.7%</i>	<i>4.1%</i>	<i>5.1%</i>	<i>1.4%</i>

3) Priority Year 1989

Secondary Filings as a Percentage of First Filings
 (% Priority filings claimed in Country of Origin From):

Country Bloc	Cntry of Origin	First Filings	All Other Blocs	Other Trilateral Blocs	EPC	Japan	USA	Other Countries	Trilateral Patent Families
EPC:	Austria	1490	34.0%	26.6%	--	18.5%	22.4%	22.7%	14.2%
	Belgium	535	29.9%	23.9%	--	16.8%	18.9%	19.6%	11.8%
	Denmark	842	49.8%	31.0%	--	22.3%	24.0%	41.9%	15.3%
	EPO	3676	43.1%	33.9%	--	24.2%	26.7%	22.7%	17.0%
	Finland	2076	29.8%	20.2%	--	11.7%	16.4%	23.6%	7.9%
	France	11506	35.2%	31.8%	--	23.1%	27.0%	21.5%	18.3%
	Germany	41034	26.2%	24.0%	--	18.1%	19.2%	12.4%	13.3%
	Greece	246	5.7%	3.3%	--	0.8%	2.4%	4.1%	0.0%
	Ireland	1955	4.5%	2.5%	--	1.4%	1.9%	3.7%	0.8%
	Italy	9349	18.0%	15.3%	--	9.9%	12.8%	10.5%	7.3%
	Luxem.	73	50.7%	46.6%	--	32.9%	38.4%	35.6%	24.7%
	Monaco	18	5.6%	5.6%	--	5.6%	5.6%	5.6%	5.6%
	Netherl.	2042	49.4%	44.8%	--	37.0%	36.6%	31.2%	28.8%
	Portugal	302	1.7%	1.0%	--	0.3%	0.7%	1.3%	0.0%
	Spain	5530	3.8%	3.0%	--	1.8%	2.3%	2.3%	1.1%
	Sweden	2140	58.0%	38.5%	--	28.2%	32.5%	45.0%	22.2%
	Switzerl.	2641	54.4%	50.3%	--	41.8%	40.4%	32.3%	31.9%
	Turkey	87	100.0%	100.0%	--	0.0%	0.0%	0.0%	0.0%
	UK	11142	51.9%	41.3%	--	30.9%	32.2%	37.0%	21.7%
Subtotal for EPC		96684	30.7%	26.3%	--	19.4%	21.2%	18.1%	14.3%
Japan:	Jpn	316186	12.2%	11.8%	8.2%	--	10.5%	4.4%	7.0%
US:	USA	77303	35.8%	28.9%	26.9%	22.5%	--	27.0%	20.6%
Other:	Argen	108	30.6%	30.6%	17.6%	8.3%	13.9%	22.2%	4.6%
	Austral	4323	20.6%	20.6%	12.6%	6.7%	11.9%	17.2%	3.9%
	Brazil	4894	1.2%	1.2%	0.8%	0.5%	0.8%	1.2%	0.2%
	Bulgaria	1226	100.0%	100.0%	0.6%	0.3%	0.1%	0.3%	0.1%
	Canada	3328	42.0%	42.0%	22.1%	13.0%	34.9%	38.9%	9.6%
	China	22641	0.3%	0.3%	0.2%	0.1%	0.2%	0.3%	0.0%
	Hungary	1906	10.2%	10.2%	9.2%	4.7%	5.0%	9.7%	3.5%
	India	1740	3.6%	3.6%	3.2%	2.4%	2.6%	3.5%	2.1%
	Israel	510	55.9%	55.9%	38.6%	20.0%	39.0%	37.8%	15.7%
	Korea	12721	5.7%	5.7%	2.8%	3.3%	3.4%	4.0%	1.3%
	Malay	7	100.0%	100.0%	57.1%	28.6%	28.6%	57.1%	14.3%
	Mexico	283	6.0%	6.0%	2.8%	1.4%	3.9%	4.9%	1.4%
	NewZeal	489	17.8%	17.8%	14.5%	9.0%	8.6%	15.3%	3.9%
	Norway	806	27.9%	27.9%	25.6%	10.3%	14.3%	26.7%	7.9%
	Philipp	311	1.0%	1.0%	0.3%	0.0%	0.6%	1.0%	0.0%
	Poland	3775	0.5%	0.5%	0.4%	0.1%	0.2%	0.4%	0.1%
	Romania	2852	0.1%	0.1%	0.1%	0.0%	0.1%	0.1%	0.0%
	Russia	69743	0.3%	0.3%	0.3%	0.2%	0.1%	0.3%	0.1%
	SouthAfr	2335	8.0%	8.0%	6.9%	1.8%	4.4%	7.4%	1.4%
	Singapore	11	63.6%	63.6%	63.6%	36.4%	27.3%	63.6%	27.3%
Subtotal for Other		134009	4.3%	4.3%	2.1%	1.3%	2.1%	2.9%	0.8%

4) Priority Year 1984

Secondary Filings as a Percentage of First Filings
 (% Priority filings claimed in Country of Origin From):

Country Bloc	Cntry of Origin	First Filings	All Other Blocs	Other Trilateral Blocs	EPC	Japan	USA	Other Countries	Trilateral Patent Families
EPC:	Austria	1642	25.9%	21.2%	--	12.7%	17.7%	16.0%	9.3%
	Belgium	1619	8.0%	6.4%	--	4.0%	5.0%	4.6%	2.6%
	Denmark	744	34.9%	21.1%	--	12.9%	16.3%	27.7%	8.1%
	EPO	2014	45.9%	37.2%	--	18.1%	31.3%	19.9%	12.3%
	Finland	1419	27.6%	18.1%	--	11.0%	15.0%	21.6%	7.9%
	France	10844	28.4%	25.5%	--	17.5%	21.9%	15.9%	13.9%
	Germany	41290	24.1%	22.0%	--	15.6%	17.9%	10.4%	11.5%
	Greece	17	41.2%	29.4%	--	11.8%	23.5%	35.3%	5.9%
	Ireland	1521	2.1%	1.8%	--	0.9%	1.6%	1.1%	0.8%
	Italy	9597	12.8%	11.3%	--	6.7%	9.7%	6.8%	5.1%
	Luxem.	178	43.3%	40.4%	--	34.8%	34.3%	33.7%	28.7%
	Monaco	20	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
	Netherl.	1803	44.5%	42.4%	--	34.0%	36.3%	28.4%	27.8%
	Portugal	270	3.0%	3.0%	--	2.2%	2.2%	2.6%	1.5%
	Spain	6817	2.4%	1.9%	--	1.0%	1.5%	1.2%	0.6%
	Sweden	2609	46.9%	34.5%	--	23.5%	29.5%	33.3%	18.5%
	Switzerl.	2473	51.9%	47.0%	--	36.4%	39.3%	31.3%	28.7%
	Turkey	128	100.0%	100.0%	--	0.0%	0.0%	0.0%	0.0%
	UK	11711	39.5%	32.7%	--	21.4%	26.0%	25.8%	14.8%
Subtotal for EPC		96716	25.6%	22.3%	--	15.2%	18.3%	13.7%	11.2%
Japan:	Jpn	257983	9.9%	9.4%	6.4%	--	8.5%	3.3%	5.5%
US:	USA	48665	40.6%	34.8%	32.0%	25.6%	--	28.9%	22.8%
Other:	Argen	252	6.7%	6.7%	3.2%	1.2%	2.4%	2.4%	1.2%
	Austral	2983	28.4%	28.4%	22.5%	8.8%	14.4%	22.8%	5.9%
	Brazil	4235	1.1%	1.1%	0.8%	0.3%	0.6%	1.1%	0.1%
	Bulgaria	1654	100.0%	100.0%	2.4%	1.3%	1.0%	2.1%	0.7%
	Canada	3138	18.1%	18.1%	8.5%	5.7%	13.5%	16.4%	3.4%
	China	0							
	Hungary	2270	13.4%	13.4%	12.2%	5.2%	4.8%	12.6%	3.0%
	India	822	0.5%	0.5%	0.2%	0.0%	0.0%	0.2%	0.0%
	Israel	433	35.8%	35.8%	24.2%	13.4%	22.4%	27.5%	6.9%
	Korea	7433	1.7%	1.7%	0.8%	0.7%	0.8%	1.3%	0.2%
	Malay	14	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
	Mexico	307	4.6%	4.6%	1.3%	0.3%	2.0%	2.0%	0.3%
	NewZeal	472	17.6%	17.6%	14.8%	5.3%	10.0%	16.1%	3.8%
	Norway	735	27.2%	27.2%	24.4%	6.3%	12.4%	25.4%	4.8%
	Philipp	380	0.8%	0.8%	0.3%	0.3%	0.5%	0.8%	0.0%
	Poland	4990	1.1%	1.1%	1.0%	0.2%	0.2%	1.1%	0.1%
	Romania	1963	0.3%	0.3%	0.1%	0.1%	0.1%	0.1%	0.1%
	Russia	57954	0.7%	0.7%	0.6%	0.1%	0.2%	0.6%	0.1%
	SouthAfr	2140	9.4%	9.4%	7.6%	1.9%	4.8%	8.8%	1.0%
	Singapore	0							
Subtotal for Other		92175	5.1%	5.1%	2.5%	1.0%	1.7%	2.9%	0.6%

5) Priority Year 1979

Secondary Filings as a Percentage of First Filings
 (% Priority filings claimed in Country of Origin From):

Country Bloc	Cntry of Origin	First Filings	All Other Blocs	Other Trilateral Blocs	EPC	Japan	USA	Other Countries	Trilateral Patent Families
EPC:	Austria	2328	17.3%	13.7%	--	9.1%	10.7%	11.3%	6.1%
	Belgium	1974	7.7%	6.2%	--	2.8%	5.2%	4.3%	1.8%
	Denmark	811	19.7%	15.2%	--	9.5%	11.6%	13.6%	5.9%
	EPO	622	23.8%	12.5%	--	9.0%	10.0%	17.8%	6.4%
	Finland	982	23.3%	16.1%	--	11.1%	12.0%	18.5%	7.0%
	France	11790	26.0%	23.1%	--	14.4%	19.6%	14.8%	10.9%
	Germany	38896	23.3%	21.4%	--	15.3%	16.4%	10.4%	10.3%
	Greece	25	44.0%	28.0%	--	16.0%	16.0%	24.0%	4.0%
	Ireland	912	2.6%	2.5%	--	1.1%	2.3%	1.9%	0.9%
	Italy	9557	12.0%	9.9%	--	6.3%	7.9%	6.9%	4.3%
	Luxem.	234	28.6%	26.5%	--	19.7%	21.8%	23.1%	15.0%
	Monaco	24	8.3%	4.2%	--	0.0%	4.2%	8.3%	0.0%
	Netherl.	2194	36.3%	33.6%	--	26.5%	29.3%	24.7%	22.1%
	Portugal	222	3.6%	0.9%	--	0.5%	0.5%	3.6%	0.0%
	Spain	7987	1.6%	0.8%	--	0.5%	0.6%	1.1%	0.2%
	Sweden	2885	37.9%	30.2%	--	21.1%	23.4%	26.2%	14.2%
	Switzerl.	3165	50.4%	46.7%	--	34.8%	36.3%	29.4%	24.4%
	Turkey	16	100.0%	100.0%	--	0.0%	0.0%	0.0%	0.0%
	UK	14145	28.5%	23.8%	--	14.4%	19.2%	17.8%	9.8%
Subtotal for EPC		98769	22.4%	19.7%	--	13.4%	15.6%	12.3%	9.3%
Japan:	Jpn	152164	9.3%	9.1%	6.9%	--	7.9%	2.4%	5.7%
US:	USA	49694	34.2%	28.2%	26.3%	19.6%	--	25.6%	17.7%
Other:	Argen	1346	1.4%	1.4%	1.1%	0.2%	0.6%	1.3%	0.0%
	Austral	3334	18.8%	18.8%	14.5%	6.9%	9.6%	16.0%	4.1%
	Brazil	4204	1.1%	1.1%	0.8%	0.2%	0.4%	1.0%	0.1%
	Bulgaria	1393	100.0%	100.0%	2.2%	1.1%	1.0%	1.8%	0.6%
	Canada	4042	11.7%	11.7%	5.6%	3.4%	8.0%	10.4%	1.6%
	China	0							
	Hungary	1100	20.7%	20.7%	19.1%	9.5%	9.7%	19.2%	7.5%
	India	705	1.3%	1.3%	0.9%	0.3%	0.3%	0.9%	0.3%
	Israel	437	27.0%	27.0%	18.5%	6.4%	17.4%	22.4%	4.3%
	Korea	4975	0.6%	0.6%	0.3%	0.2%	0.3%	0.4%	0.2%
	Malay	2	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
	Mexico	679	1.6%	1.6%	1.2%	1.0%	1.0%	1.5%	0.7%
	NewZeal	306	29.7%	29.7%	23.5%	6.2%	15.0%	27.1%	4.6%
	Norway	661	24.2%	24.2%	21.8%	6.5%	10.0%	23.1%	3.8%
	Philipp	313	2.6%	2.6%	1.3%	0.6%	2.2%	2.2%	0.6%
	Poland	6739	1.6%	1.6%	1.6%	0.2%	0.4%	1.6%	0.1%
	Romania	1550	0.4%	0.4%	0.4%	0.2%	0.1%	0.4%	0.1%
	Russia	67096	0.7%	0.7%	0.6%	0.3%	0.2%	0.6%	0.1%
	SouthAfr	1196	16.7%	16.7%	13.8%	4.0%	6.9%	15.0%	2.5%
	Singapore	0							
Subtotal for Other		100078	4.0%	4.0%	2.0%	0.9%	1.2%	2.3%	0.5%

Table 23. Patent Families Derived from First Filings, by Country of Origin and IPC field, Selected Years

YEAR: 1994

1. IPC = A

Human Necessities

Secondary Filings as a Percentage of First Filings
(% Priority filings claimed in Country of Origin From):

Country	Cntry of	First	All Other	Other						Trilateral
<u>Bloc</u>	<u>Origin</u>	<u>Filings</u>	<u>Blocs</u>	<u>Blocs</u>	<u>EPC</u>	<u>Japan</u>	<u>USA</u>	<u>Other</u>	<u>Patent</u>	<u>Families</u>
		<u>by IPC</u>						<u>Countries</u>		
EPC	Austria	367	27.0%	20.2%	--	11.7%	16.3%	14.7%		7.9%
EPC	Belgium	155	22.6%	13.5%	--	8.4%	11.6%	17.4%		6.5%
EPC	Denmark	344	39.2%	25.3%	--	18.3%	20.6%	36.0%		13.7%
EPC	EPO	1107	42.2%	28.9%	--	18.2%	25.7%	33.7%		14.9%
EPC	Finland	424	19.6%	13.2%	--	6.8%	11.8%	16.7%		5.4%
EPC	France	2276	30.1%	25.6%	--	18.0%	22.7%	21.4%		15.1%
EPC	Germany	7897	15.8%	12.9%	--	7.8%	11.0%	9.5%		6.0%
EPC	Greece	68	7.4%	2.9%	--	2.9%	2.9%	7.4%		2.9%
EPC	Ireland	159	22.6%	14.5%	--	7.5%	12.6%	20.8%		5.7%
EPC	Italy	2113	15.2%	12.0%	--	6.3%	10.6%	11.0%		5.0%
EPC	Luxem.	8	75.0%	12.5%	--	0.0%	12.5%	75.0%		0.0%
EPC	Monaco	6	0.0%	0.0%	--	0.0%	0.0%	0.0%		0.0%
EPC	Netherl.	373	34.9%	27.9%	--	18.2%	24.4%	22.5%		14.7%
EPC	Portugal	57	5.3%	3.5%	--	1.8%	3.5%	5.3%		1.8%
EPC	Spain	1433	5.7%	3.8%	--	2.0%	2.9%	4.0%		1.0%
EPC	Sweden	564	62.6%	47.9%	--	38.1%	42.2%	52.1%		32.4%
EPC	Switzerl.	350	40.9%	31.7%	--	20.6%	26.0%	32.9%		14.9%
EPC	Turkey	16	100.0%	100.0%	--	0.0%	0.0%	6.3%		0.0%
EPC	UK	1969	51.8%	37.1%	--	22.4%	31.9%	43.0%		17.2%
<i>Subtotal for EPC</i>		19686	24.7%	18.9%	--	11.9%	16.3%	18.1%		9.4%
Japan	Jpn	28084	7.3%	6.5%	4.5%	--	5.5%	3.7%		3.5%
US	USA	23170	27.7%	19.7%	19.1%	14.0%	--	24.5%		13.4%
Other	Argen	19	52.6%	52.6%	36.8%	5.3%	31.6%	36.8%		5.3%
Other	Austral	1332	16.3%	16.3%	12.1%	5.9%	11.5%	16.0%		3.9%
Other	Brazil	1318	1.6%	1.6%	1.4%	0.5%	0.8%	1.6%		0.2%
Other	Bulgaria	110	100.0%	100.0%	0.9%	0.9%	0.0%	0.0%		0.0%
Other	Canada	805	22.9%	22.9%	9.6%	4.2%	19.0%	21.7%		3.1%
Other	China	12405	0.2%	0.2%	0.2%	0.1%	0.1%	0.2%		0.0%
Other	Hungary	340	4.1%	4.1%	3.2%	1.8%	2.6%	3.8%		1.5%
Other	India	77	7.8%	7.8%	1.3%	0.0%	1.3%	1.3%		0.0%
Other	Israel	163	59.5%	59.5%	40.5%	19.6%	46.6%	54.6%		17.8%
Other	Korea	3548	5.2%	5.2%	2.3%	3.0%	3.0%	4.1%		1.0%
Other	Malay	1	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%		0.0%
Other	Mexico	5	60.0%	60.0%	0.0%	0.0%	0.0%	0.0%		0.0%
Other	NewZeal	203	24.1%	24.1%	17.7%	9.4%	17.2%	23.2%		6.4%
Other	Norway	121	37.2%	37.2%	33.9%	18.2%	21.5%	31.4%		18.2%
Other	Philipp	21	4.8%	4.8%	4.8%	0.0%	4.8%	4.8%		0.0%
Other	Poland	615	0.8%	0.8%	0.5%	0.0%	0.2%	0.5%		0.0%
Other	Romania	179	0.6%	0.6%	0.0%	0.0%	0.6%	0.6%		0.0%
Other	Russia	2646	0.8%	0.8%	0.7%	0.2%	0.4%	0.7%		0.2%
Other	SouthAfr	333	14.1%	14.1%	10.5%	4.2%	9.0%	13.8%		3.6%
Other	Singapore	193	9.3%	9.3%	9.3%	8.8%	8.8%	9.3%		8.3%
<i>Subtotal for Other</i>		24434	4.3%	4.3%	2.4%	1.5%	2.7%	3.5%		0.9%

2. IPC = B
Performing Operations

Secondary Filings as a Percentage of First Filings
 (% Priority filings claimed in Country of Origin From):

Country	Cntry of	First	All Other	Other					
<u>Bloc</u>	<u>Origin</u>	<u>Filings</u>	<u>Blocs</u>	<u>Trilateral</u>	<u>EPC</u>	<u>Japan</u>	<u>USA</u>	<u>Other</u>	<u>Trilateral</u>
		<u>by IPC</u>		<u>Blocs</u>				<u>Countries</u>	<u>Patent</u>
									<u>Families</u>
EPC	Austria	456	25.9%	18.0%	--	9.6%	15.6%	17.5%	7.2%
EPC	Belgium	195	23.1%	15.4%	--	9.2%	13.3%	15.4%	7.2%
EPC	Denmark	258	37.2%	21.7%	--	14.7%	18.6%	31.4%	11.6%
EPC	EPO	1162	40.2%	29.7%	--	17.1%	26.1%	22.5%	13.6%
EPC	Finland	574	31.9%	22.0%	--	15.0%	19.9%	28.0%	12.9%
EPC	France	2747	26.8%	22.8%	--	14.0%	20.0%	16.7%	11.2%
EPC	Germany	15065	19.9%	17.6%	--	11.9%	15.0%	9.2%	9.3%
EPC	Greece	37	8.1%	5.4%	--	0.0%	5.4%	5.4%	0.0%
EPC	Ireland	101	13.9%	7.9%	--	2.0%	6.9%	12.9%	1.0%
EPC	Italy	2498	16.8%	12.9%	--	6.3%	11.5%	11.0%	4.9%
EPC	Luxem.	22	31.8%	22.7%	--	13.6%	22.7%	31.8%	13.6%
EPC	Monaco	3	33.3%	33.3%	--	33.3%	33.3%	33.3%	33.3%
EPC	Netherl.	479	38.8%	29.2%	--	19.0%	26.3%	26.9%	16.1%
EPC	Portugal	51	3.9%	2.0%	--	0.0%	2.0%	3.9%	0.0%
EPC	Spain	1100	7.5%	4.5%	--	2.3%	3.5%	5.0%	1.3%
EPC	Sweden	697	52.5%	38.6%	--	27.0%	33.3%	41.0%	21.7%
EPC	Switzerl.	539	55.1%	46.6%	--	33.8%	40.3%	37.1%	27.5%
EPC	Turkey	16	100.0%	100.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	UK	2248	44.5%	31.5%	--	16.2%	27.6%	32.6%	12.4%
<i>Subtotal for EPC</i>		28248	24.9%	20.1%	--	12.7%	17.4%	14.8%	10.0%
Japan	Jpn	64084	8.3%	7.8%	5.2%	--	7.0%	3.2%	4.4%
US	USA	22400	27.1%	19.9%	18.5%	13.0%	--	20.3%	11.6%
Other	Argen	9	77.8%	77.8%	0.0%	0.0%	11.1%	11.1%	0.0%
Other	Austral	1152	17.5%	17.5%	12.2%	6.9%	13.2%	16.8%	5.3%
Other	Brazil	1437	2.9%	2.9%	2.4%	0.6%	1.5%	2.9%	0.3%
Other	Bulgaria	78	100.0%	100.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Canada	710	24.8%	24.8%	10.1%	3.9%	21.0%	24.4%	3.1%
Other	China	9871	0.5%	0.5%	0.3%	0.2%	0.3%	0.4%	0.1%
Other	Hungary	157	5.1%	5.1%	3.2%	0.6%	1.3%	3.2%	0.0%
Other	India	16	12.5%	12.5%	12.5%	0.0%	0.0%	12.5%	0.0%
Other	Israel	127	55.9%	55.9%	40.2%	12.6%	40.9%	45.7%	10.2%
Other	Korea	6763	3.1%	3.1%	1.2%	1.8%	1.8%	2.2%	0.7%
Other	Malay	5	60.0%	60.0%	40.0%	20.0%	20.0%	20.0%	0.0%
Other	Mexico	13	53.8%	53.8%	15.4%	0.0%	0.0%	15.4%	0.0%
Other	NewZeal	166	15.7%	15.7%	11.4%	5.4%	13.9%	15.7%	4.2%
Other	Norway	147	37.4%	37.4%	36.7%	14.3%	23.8%	37.4%	11.6%
Other	Philipp	15	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Poland	872	1.9%	1.9%	1.7%	0.1%	0.6%	1.7%	0.1%
Other	Romania	168	0.6%	0.6%	0.6%	0.0%	0.0%	0.6%	0.0%
Other	Russia	2343	1.3%	1.3%	0.9%	0.2%	0.5%	1.0%	0.1%
Other	SouthAfr	372	17.7%	17.7%	14.2%	3.8%	11.0%	16.9%	2.7%
Other	Singapore	250	13.2%	13.2%	13.2%	9.6%	10.4%	13.2%	7.2%
<i>Subtotal for Other</i>		24671	4.4%	4.4%	2.5%	1.4%	2.7%	3.6%	0.9%

3. IPC = C

Chemistry & Metallurgy

Secondary Filings as a Percentage of First Filings
(% Priority filings claimed in Country of Origin From):

Country Bloc	Cntry of Origin	First Filings by IPC	All Other Blocs	Other Trilateral Blocs	EPC	Japan	USA	Other Countries	Trilateral Patent Families
EPC	Austria	174	50.6%	39.7%	--	31.6%	36.2%	43.7%	28.2%
EPC	Belgium	133	55.6%	45.9%	--	35.3%	39.8%	42.9%	29.3%
EPC	Denmark	181	81.2%	62.4%	--	49.2%	57.5%	77.9%	44.2%
EPC	EPO	1514	47.7%	32.9%	--	25.7%	25.8%	38.4%	18.6%
EPC	Finland	201	47.8%	35.3%	--	25.4%	30.3%	40.3%	20.4%
EPC	France	1103	67.8%	59.0%	--	49.2%	49.9%	56.7%	40.1%
EPC	Germany	4691	54.8%	50.2%	--	43.5%	40.7%	37.2%	33.9%
EPC	Greece	14	14.3%	14.3%	--	0.0%	14.3%	14.3%	0.0%
EPC	Ireland	58	22.4%	12.1%	--	5.2%	10.3%	20.7%	3.4%
EPC	Italy	497	44.3%	41.0%	--	32.0%	36.2%	35.0%	27.2%
EPC	Luxem.	15	46.7%	40.0%	--	33.3%	40.0%	46.7%	33.3%
EPC	Monaco	7	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Netherl.	108	65.7%	49.1%	--	37.0%	35.2%	50.0%	23.1%
EPC	Portugal	8	12.5%	12.5%	--	0.0%	12.5%	0.0%	0.0%
EPC	Spain	186	24.2%	14.5%	--	11.3%	11.8%	23.1%	8.6%
EPC	Sweden	201	71.1%	51.7%	--	41.3%	45.8%	60.7%	35.3%
EPC	Switzerl.	371	83.3%	76.5%	--	72.2%	65.8%	70.6%	61.5%
EPC	Turkey	14	100.0%	100.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	UK	1626	83.3%	69.1%	--	56.7%	56.7%	69.7%	44.3%
<i>Subtotal for EPC</i>		11102	59.7%	50.9%	--	42.5%	41.8%	46.1%	33.6%
Japan	Jpn	33551	13.9%	13.2%	10.8%	--	11.7%	7.1%	9.3%
US	USA	17831	33.6%	27.6%	26.8%	21.7%	--	28.4%	20.9%
Other	Argen	4	75.0%	75.0%	50.0%	50.0%	50.0%	75.0%	25.0%
Other	Austral	595	25.7%	25.7%	22.7%	16.0%	19.7%	25.2%	12.8%
Other	Brazil	181	12.7%	12.7%	12.2%	3.3%	7.7%	12.7%	1.7%
Other	Bulgaria	57	100.0%	100.0%	3.5%	0.0%	3.5%	0.0%	0.0%
Other	Canada	341	24.9%	24.9%	12.6%	10.0%	20.2%	23.2%	7.3%
Other	China	3135	0.8%	0.8%	0.5%	0.3%	0.6%	0.7%	0.3%
Other	Hungary	184	8.7%	8.7%	8.2%	4.9%	4.3%	8.2%	3.8%
Other	India	72	12.5%	12.5%	1.4%	0.0%	2.8%	4.2%	0.0%
Other	Israel	123	43.1%	43.1%	30.1%	17.1%	34.1%	39.0%	14.6%
Other	Korea	2299	11.9%	11.9%	6.2%	9.6%	8.2%	10.6%	4.4%
Other	Malay	4	100.0%	100.0%	75.0%	100.0%	50.0%	75.0%	25.0%
Other	Mexico	13	92.3%	92.3%	61.5%	0.0%	69.2%	69.2%	0.0%
Other	NewZeal	100	12.0%	12.0%	12.0%	5.0%	11.0%	12.0%	5.0%
Other	Norway	77	32.5%	32.5%	32.5%	14.3%	23.4%	32.5%	13.0%
Other	Philipp	5	20.0%	20.0%	0.0%	0.0%	20.0%	20.0%	0.0%
Other	Poland	625	1.0%	1.0%	1.0%	0.5%	0.2%	1.0%	0.2%
Other	Romania	272	0.4%	0.4%	0.4%	0.0%	0.4%	0.4%	0.0%
Other	Russia	2254	1.8%	1.8%	1.7%	0.7%	0.9%	1.7%	0.5%
Other	SouthAfr	147	24.5%	24.5%	20.4%	6.8%	15.6%	22.4%	6.1%
Other	Singapore	327	11.3%	11.3%	11.3%	8.6%	8.0%	11.3%	6.7%
<i>Subtotal for Other</i>		10815	8.1%	8.1%	5.3%	4.4%	5.3%	7.0%	2.8%

4. IPC = D
Textiles & Paper

Secondary Filings as a Percentage of First Filings
(% Priority filings claimed in Country of Origin From):

Country Bloc	Cntry of Origin	First Filings by IPC	All Other Blocs	Other Trilateral Blocs	EPC	Japan	USA	Other Countries	Trilateral Patent Families
EPC	Austria	40	55.0%	47.5%	--	40.0%	47.5%	35.0%	40.0%
EPC	Belgium	35	51.4%	48.6%	--	17.1%	48.6%	11.4%	17.1%
EPC	Denmark	27	74.1%	59.3%	--	55.6%	51.9%	55.6%	48.1%
EPC	EPO	131	43.5%	31.3%	--	23.7%	26.7%	26.7%	19.1%
EPC	Finland	138	42.0%	34.1%	--	21.0%	29.0%	34.1%	15.9%
EPC	France	146	40.4%	32.2%	--	22.6%	31.5%	28.8%	21.9%
EPC	Germany	1085	36.3%	33.5%	--	22.6%	27.8%	19.1%	16.9%
EPC	Greece	4	50.0%	25.0%	--	0.0%	25.0%	50.0%	0.0%
EPC	Ireland	17	5.9%	0.0%	--	0.0%	0.0%	5.9%	0.0%
EPC	Italy	296	24.3%	21.6%	--	11.8%	17.9%	12.2%	8.1%
EPC	Luxem.	0			--				
EPC	Monaco	6	16.7%	16.7%	--	16.7%	0.0%	16.7%	0.0%
EPC	Netherl.	16	50.0%	37.5%	--	37.5%	25.0%	43.8%	25.0%
EPC	Portugal	3	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Spain	47	8.5%	4.3%	--	2.1%	4.3%	4.3%	2.1%
EPC	Sweden	96	69.8%	59.4%	--	39.6%	51.0%	63.5%	31.3%
EPC	Switzerl.	77	57.1%	55.8%	--	45.5%	35.1%	31.2%	24.7%
EPC	Turkey	2	100.0%	100.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	UK	153	61.4%	47.7%	--	33.3%	43.8%	45.8%	29.4%
<i>Subtotal for EPC</i>		2319	39.8%	34.5%	--	23.4%	29.2%	24.5%	18.1%
Japan	Jpn	4738	8.2%	7.2%	5.8%	--	5.9%	4.9%	4.6%
US	USA	1300	35.8%	28.3%	27.5%	19.4%	--	29.8%	18.5%
Other	Argen	0							
Other	Austral	54	22.2%	22.2%	20.4%	9.3%	16.7%	20.4%	9.3%
Other	Brazil	69	4.3%	4.3%	4.3%	2.9%	2.9%	4.3%	2.9%
Other	Bulgaria	5	100.0%	100.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Canada	36	38.9%	38.9%	11.1%	13.9%	27.8%	36.1%	2.8%
Other	China	779	0.5%	0.5%	0.0%	0.5%	0.0%	0.5%	0.0%
Other	Hungary	6	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	India	2	50.0%	50.0%	50.0%	0.0%	0.0%	50.0%	0.0%
Other	Israel	7	28.6%	28.6%	0.0%	0.0%	28.6%	28.6%	0.0%
Other	Korea	1180	4.4%	4.4%	1.2%	3.9%	3.2%	4.1%	0.8%
Other	Malay	0							
Other	Mexico	0							
Other	NewZeal	11	36.4%	36.4%	18.2%	36.4%	9.1%	36.4%	9.1%
Other	Norway	10	10.0%	10.0%	10.0%	10.0%	10.0%	10.0%	10.0%
Other	Philipp	4	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Poland	41	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Romania	31	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Russia	116	0.9%	0.9%	0.0%	0.0%	0.9%	0.9%	0.0%
Other	SouthAfr	19	5.3%	5.3%	5.3%	0.0%	5.3%	5.3%	0.0%
Other	Singapore	17	11.8%	11.8%	11.8%	11.8%	11.8%	11.8%	11.8%
<i>Subtotal for Other</i>		2387	4.3%	4.3%	1.6%	2.9%	2.8%	3.8%	0.9%

5. IPC = E
Fixed Constructions

Secondary Filings as a Percentage of First Filings
 (% Priority filings claimed in Country of Origin From):

Country	Cntry of	First	All Other	Other					Trilateral
<u>Bloc</u>	<u>Origin</u>	<u>Filings</u>	<u>Blocs</u>	<u>Blocs</u>	<u>EPC</u>	<u>Japan</u>	<u>USA</u>	<u>Other</u>	<u>Patent</u>
		<u>by IPC</u>						<u>Countries</u>	<u>Families</u>
EPC	Austria	249	24.5%	15.3%	--	8.4%	14.1%	20.5%	7.2%
EPC	Belgium	80	17.5%	12.5%	--	6.3%	11.3%	13.7%	5.0%
EPC	Denmark	99	32.3%	13.1%	--	6.1%	12.1%	31.3%	5.1%
EPC	EPO	204	31.9%	17.6%	--	3.4%	15.7%	23.0%	1.5%
EPC	Finland	243	19.8%	9.1%	--	6.6%	7.0%	18.1%	4.5%
EPC	France	856	12.1%	6.8%	--	2.3%	6.1%	10.4%	1.6%
EPC	Germany	4460	10.0%	5.0%	--	1.9%	4.2%	7.5%	1.1%
EPC	Greece	26	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Ireland	56	3.6%	0.0%	--	0.0%	0.0%	3.6%	0.0%
EPC	Italy	609	8.7%	5.6%	--	2.0%	4.9%	5.3%	1.3%
EPC	Luxem.	3	100.0%	0.0%	--	0.0%	0.0%	100.0%	0.0%
EPC	Monaco	2	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Netherl.	169	22.5%	11.2%	--	3.0%	10.7%	19.5%	2.4%
EPC	Portugal	32	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Spain	479	3.3%	1.5%	--	0.2%	1.3%	2.1%	0.0%
EPC	Sweden	227	45.4%	19.4%	--	11.5%	17.6%	43.6%	9.7%
EPC	Switzerl.	178	28.1%	18.5%	--	11.2%	15.7%	24.2%	8.4%
EPC	Turkey	4	100.0%	100.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	UK	919	28.0%	18.3%	--	4.5%	17.6%	22.4%	3.9%
<i>Subtotal for EPC</i>		8895	14.6%	8.0%	--	3.0%	7.1%	11.6%	2.1%
Japan	Jpn	17437	2.1%	1.6%	1.1%	--	1.4%	1.2%	0.9%
US	USA	3201	31.3%	17.0%	16.2%	4.3%	--	28.0%	3.6%
Other	Argen	3	66.7%	66.7%	33.3%	0.0%	33.3%	66.7%	0.0%
Other	Austral	824	12.0%	12.0%	7.8%	2.5%	9.0%	11.5%	1.8%
Other	Brazil	448	1.3%	1.3%	1.3%	0.0%	0.9%	1.3%	0.0%
Other	Bulgaria	31	100.0%	100.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Canada	302	30.8%	30.8%	9.6%	2.0%	27.8%	30.1%	1.7%
Other	China	3534	0.2%	0.2%	0.1%	0.1%	0.1%	0.2%	0.0%
Other	Hungary	73	2.7%	2.7%	1.4%	0.0%	0.0%	1.4%	0.0%
Other	India	0							
Other	Israel	36	36.1%	36.1%	27.8%	0.0%	16.7%	33.3%	0.0%
Other	Korea	1233	2.4%	2.4%	1.1%	1.9%	0.8%	1.8%	0.6%
Other	Malay	7	42.9%	42.9%	42.9%	0.0%	42.9%	14.3%	0.0%
Other	Mexico	8	62.5%	62.5%	0.0%	12.5%	12.5%	12.5%	0.0%
Other	NewZeal	87	19.5%	19.5%	12.6%	6.9%	16.1%	19.5%	4.6%
Other	Norway	97	25.8%	25.8%	25.8%	2.1%	14.4%	25.8%	2.1%
Other	Philipp	1	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Poland	543	0.4%	0.4%	0.2%	0.0%	0.0%	0.2%	0.0%
Other	Romania	52	1.9%	1.9%	1.9%	0.0%	0.0%	1.9%	0.0%
Other	Russia	949	0.5%	0.5%	0.2%	0.1%	0.0%	0.2%	0.0%
Other	SouthAfr	221	10.0%	10.0%	7.7%	0.5%	6.3%	9.0%	0.0%
Other	Singapore	80	13.8%	13.8%	13.8%	5.0%	10.0%	13.8%	3.8%
<i>Subtotal for Other</i>		8529	4.4%	4.4%	2.3%	0.8%	2.8%	3.7%	0.4%

6. IPC = F
Mechanical Engineering

Secondary Filings as a Percentage of First Filings
 (% Priority filings claimed in Country of Origin From):

Country	Cntry of	First	All Other	Other					Trilateral
<u>Bloc</u>	<u>Origin</u>	<u>by IPC</u>	<u>Blocs</u>	<u>Blocs</u>	<u>EPC</u>	<u>Japan</u>	<u>USA</u>	<u>Other</u>	<u>Patent</u>
								<u>Countries</u>	<u>Families</u>
EPC	Austria	307	15.3%	7.8%	--	5.5%	6.8%	11.1%	4.6%
EPC	Belgium	67	32.8%	19.4%	--	14.9%	14.9%	22.4%	10.4%
EPC	Denmark	127	40.2%	27.6%	--	18.1%	16.5%	29.9%	7.1%
EPC	EPO	384	29.7%	21.4%	--	10.2%	18.5%	19.0%	7.3%
EPC	Finland	180	35.6%	26.7%	--	17.8%	21.7%	25.0%	12.8%
EPC	France	1394	30.8%	27.5%	--	14.3%	25.8%	15.6%	12.6%
EPC	Germany	7295	21.0%	18.2%	--	11.8%	15.7%	9.3%	9.3%
EPC	Greece	32	18.8%	6.3%	--	3.1%	6.3%	15.6%	3.1%
EPC	Ireland	31	12.9%	9.7%	--	6.5%	6.5%	12.9%	3.2%
EPC	Italy	903	12.5%	9.2%	--	4.2%	7.6%	7.4%	2.7%
EPC	Luxem.	9	55.6%	55.6%	--	33.3%	33.3%	33.3%	11.1%
EPC	Monaco	1	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Netherl.	175	36.0%	23.4%	--	13.1%	19.4%	23.4%	9.1%
EPC	Portugal	26	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Spain	410	9.8%	6.8%	--	2.2%	6.3%	6.6%	1.7%
EPC	Sweden	339	49.0%	35.1%	--	22.1%	31.3%	39.8%	18.3%
EPC	Switzerl.	203	50.2%	38.4%	--	27.1%	36.5%	38.4%	25.1%
EPC	Turkey	18	100.0%	100.0%	--	0.0%	0.0%	5.6%	0.0%
EPC	UK	1135	42.8%	32.8%	--	15.3%	29.1%	25.8%	11.6%
<i>Subtotal for EPC</i>		13036	25.0%	20.4%	--	12.0%	17.7%	13.5%	9.5%
Japan	Jpn	28512	9.7%	8.8%	5.3%	--	8.1%	3.8%	4.6%
US	USA	7866	28.7%	21.3%	20.3%	12.7%	--	20.0%	11.7%
Other	Argen	5	60.0%	60.0%	60.0%	0.0%	40.0%	60.0%	0.0%
Other	Austral	556	19.6%	19.6%	14.6%	7.7%	14.7%	18.7%	6.5%
Other	Brazil	526	4.9%	4.9%	3.8%	1.1%	3.4%	4.9%	1.0%
Other	Bulgaria	66	100.0%	100.0%	3.0%	0.0%	0.0%	3.0%	0.0%
Other	Canada	237	30.4%	30.4%	12.7%	3.0%	26.6%	29.1%	2.5%
Other	China	6650	0.3%	0.3%	0.2%	0.0%	0.2%	0.3%	0.0%
Other	Hungary	103	2.9%	2.9%	2.9%	0.0%	1.9%	2.9%	0.0%
Other	India	7	14.3%	14.3%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Israel	40	45.0%	45.0%	22.5%	10.0%	30.0%	27.5%	7.5%
Other	Korea	3769	7.5%	7.5%	4.5%	4.7%	5.9%	6.3%	2.4%
Other	Malay	3	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Mexico	3	66.7%	66.7%	33.3%	0.0%	33.3%	66.7%	0.0%
Other	NewZeal	66	28.8%	28.8%	24.2%	18.2%	12.1%	28.8%	4.5%
Other	Norway	87	44.8%	44.8%	42.5%	11.5%	27.6%	44.8%	10.3%
Other	Philipp	10	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Poland	599	0.5%	0.5%	0.5%	0.0%	0.0%	0.5%	0.0%
Other	Romania	115	0.9%	0.9%	0.9%	0.0%	0.0%	0.9%	0.0%
Other	Russia	1814	0.8%	0.8%	0.7%	0.3%	0.5%	0.8%	0.3%
Other	SouthAfr	163	11.7%	11.7%	8.0%	0.0%	5.5%	11.0%	0.0%
Other	Singapore	98	7.1%	7.1%	7.1%	4.1%	6.1%	7.1%	3.1%
<i>Subtotal for Other</i>		14917	4.7%	4.7%	2.8%	1.8%	3.1%	3.9%	1.1%

7. IPC = G
Physics

Secondary Filings as a Percentage of First Filings
(% Priority filings claimed in Country of Origin From):

Country Bloc	Cntry of Origin	First Filings by IPC	All Other Blocs	Other Trilateral Blocs	EPC	Japan	USA	Other Countries	Trilateral Patent Families
EPC	Austria	141	29.1%	22.7%	--	15.6%	20.6%	14.9%	13.5%
EPC	Belgium	90	21.1%	18.9%	--	12.2%	16.7%	8.9%	10.0%
EPC	Denmark	77	48.1%	26.0%	--	16.9%	22.1%	46.8%	13.0%
EPC	EPO	1265	43.2%	40.9%	--	31.5%	36.3%	14.3%	27.1%
EPC	Finland	171	40.4%	32.2%	--	19.3%	30.4%	24.6%	17.5%
EPC	France	1556	42.9%	38.9%	--	24.3%	36.0%	18.8%	21.4%
EPC	Germany	5973	23.1%	20.8%	--	14.4%	18.2%	8.1%	11.8%
EPC	Greece	15	13.3%	13.3%	--	0.0%	13.3%	6.7%	0.0%
EPC	Ireland	93	8.6%	5.4%	--	3.2%	4.3%	7.5%	2.2%
EPC	Italy	556	19.2%	13.8%	--	5.8%	12.4%	13.1%	4.3%
EPC	Luxem.	1	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Monaco	0			--				
EPC	Netherl.	191	46.1%	37.7%	--	24.6%	30.4%	28.3%	17.3%
EPC	Portugal	19	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Spain	345	4.6%	2.9%	--	0.9%	2.9%	2.9%	0.9%
EPC	Sweden	339	58.4%	47.8%	--	33.9%	41.9%	39.5%	28.0%
EPC	Switzerl.	271	58.7%	45.8%	--	34.3%	38.4%	39.5%	26.9%
EPC	Turkey	1	100.0%	100.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	UK	1688	54.6%	43.7%	--	28.9%	37.7%	31.1%	23.0%
<i>Subtotal for EPC</i>		12792	33.3%	28.8%	--	19.5%	25.4%	15.5%	16.2%
Japan	Jpn	75463	13.8%	13.5%	5.9%	--	12.8%	3.7%	5.3%
US	USA	25834	24.7%	20.8%	18.1%	16.3%	--	13.9%	13.6%
Other	Argen	5	60.0%	60.0%	20.0%	0.0%	20.0%	20.0%	0.0%
Other	Austral	511	30.1%	30.1%	21.1%	16.4%	22.3%	29.2%	11.7%
Other	Brazil	490	2.2%	2.2%	2.2%	0.8%	1.4%	2.2%	0.6%
Other	Bulgaria	32	100.0%	100.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Canada	354	30.5%	30.5%	12.4%	6.2%	26.0%	29.9%	4.5%
Other	China	4228	0.4%	0.4%	0.3%	0.1%	0.2%	0.4%	0.0%
Other	Hungary	91	5.5%	5.5%	5.5%	3.3%	4.4%	5.5%	3.3%
Other	India	5	60.0%	60.0%	20.0%	0.0%	20.0%	0.0%	0.0%
Other	Israel	148	62.8%	62.8%	38.5%	22.3%	43.2%	51.4%	15.5%
Other	Korea	4844	14.5%	14.5%	5.0%	9.4%	11.5%	12.3%	3.3%
Other	Malay	0							
Other	Mexico	3	33.3%	33.3%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	NewZeal	44	40.9%	40.9%	31.8%	15.9%	25.0%	40.9%	6.8%
Other	Norway	67	52.2%	52.2%	47.8%	19.4%	38.8%	52.2%	16.4%
Other	Philipp	0							
Other	Poland	411	0.5%	0.5%	0.5%	0.0%	0.2%	0.5%	0.0%
Other	Romania	104	1.0%	1.0%	1.0%	0.0%	0.0%	1.0%	0.0%
Other	Russia	1582	1.7%	1.7%	1.1%	0.7%	0.9%	1.4%	0.5%
Other	SouthAfr	89	40.4%	40.4%	14.6%	3.4%	14.6%	19.1%	3.4%
Other	Singapore	337	21.7%	21.7%	21.4%	20.2%	19.6%	21.7%	19.0%
<i>Subtotal for Other</i>		13345	9.9%	9.9%	4.7%	5.3%	7.4%	8.4%	2.6%

**8. IPC = H
Electricity**

Secondary Filings as a Percentage of First Filings
(% Priority filings claimed in Country of Origin From):

Country Bloc	Cntry of Origin	First Filings by IPC	All Other Blocs	Other Trilateral Blocs	EPC	Japan	USA	Other Countries	Trilateral Patent Families
EPC	Austria	139	30.2%	21.6%	--	11.5%	18.0%	18.7%	7.9%
EPC	Belgium	39	43.6%	38.5%	--	38.5%	35.9%	25.6%	35.9%
EPC	Denmark	56	51.8%	26.8%	--	19.6%	25.0%	46.4%	17.9%
EPC	EPO	1301	50.6%	47.3%	--	34.4%	44.1%	22.6%	31.3%
EPC	Finland	379	68.1%	56.7%	--	42.2%	54.4%	55.9%	39.8%
EPC	France	1435	46.3%	41.0%	--	26.6%	37.5%	20.4%	23.0%
EPC	Germany	5408	25.4%	22.5%	--	15.1%	19.3%	10.1%	11.9%
EPC	Greece	6	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Ireland	40	25.0%	17.5%	--	7.5%	17.5%	20.0%	7.5%
EPC	Italy	489	21.9%	17.6%	--	11.9%	15.5%	16.0%	9.8%
EPC	Luxem.	1	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Monaco	2	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Netherl.	148	59.5%	50.0%	--	37.8%	27.7%	25.7%	15.5%
EPC	Portugal	6	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Spain	246	12.2%	7.3%	--	3.7%	6.5%	10.2%	2.8%
EPC	Sweden	330	63.9%	52.1%	--	40.0%	43.9%	50.6%	31.8%
EPC	Switzerl.	175	49.7%	43.4%	--	28.6%	36.0%	30.3%	21.1%
EPC	Turkey	3	100.0%	100.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	UK	1380	56.9%	48.8%	--	34.1%	41.7%	31.7%	27.0%
<i>Subtotal for EPC</i>		11583	37.7%	32.9%	--	22.7%	28.8%	19.1%	18.7%
Japan	Jpn	66392	14.0%	13.6%	6.4%	--	12.9%	4.7%	5.8%
US	USA	20131	27.8%	23.4%	21.0%	17.8%	--	17.3%	15.4%
Other	Argen	3	100.0%	100.0%	66.7%	33.3%	0.0%	66.7%	0.0%
Other	Austral	339	21.2%	21.2%	13.9%	9.4%	17.1%	20.1%	7.1%
Other	Brazil	341	4.1%	4.1%	3.5%	2.1%	3.8%	4.1%	1.8%
Other	Bulgaria	36	100.0%	100.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Canada	279	36.2%	36.2%	19.4%	8.2%	32.3%	35.5%	6.5%
Other	China	3608	0.5%	0.5%	0.2%	0.1%	0.4%	0.5%	0.1%
Other	Hungary	78	11.5%	11.5%	10.3%	6.4%	6.4%	10.3%	3.8%
Other	India	1	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Israel	81	60.5%	60.5%	38.3%	22.2%	43.2%	50.6%	17.3%
Other	Korea	8681	11.3%	11.3%	4.4%	7.0%	9.3%	9.6%	2.8%
Other	Malay	3	100.0%	100.0%	0.0%	33.3%	66.7%	33.3%	0.0%
Other	Mexico	2	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	NewZeal	54	20.4%	20.4%	20.4%	3.7%	13.0%	20.4%	3.7%
Other	Norway	37	37.8%	37.8%	32.4%	10.8%	24.3%	35.1%	10.8%
Other	Philipp	3	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Poland	401	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Romania	108	1.9%	1.9%	1.9%	0.9%	0.9%	1.9%	0.9%
Other	Russia	1017	1.9%	1.9%	1.9%	0.8%	0.9%	1.9%	0.3%
Other	SouthAfr	90	16.7%	16.7%	13.3%	6.7%	8.9%	15.6%	4.4%
Other	Singapore	342	17.5%	17.5%	17.3%	12.3%	13.7%	17.5%	10.5%
<i>Subtotal for Other</i>		15504	9.1%	9.1%	4.2%	4.9%	7.1%	7.8%	2.3%

YEAR: 1984

1. IPC = A

Human Necessities

Secondary Filings as a Percentage of First Filings
(% Priority filings claimed in Country of Origin From):

Country	Cntry of	First	All Other	Other					Trilateral
<u>Bloc</u>	<u>Origin</u>	<u>by IPC</u>	<u>Blocs</u>	<u>Blocs</u>	<u>EPC</u>	<u>Japan</u>	<u>USA</u>	<u>Other</u>	<u>Patent</u>
								<u>Countries</u>	<u>Families</u>
EPC	Austria	317	24.9%	20.5%	--	12.6%	16.4%	16.1%	8.5%
EPC	Belgium	414	7.2%	6.8%	--	3.1%	5.6%	3.9%	1.9%
EPC	Denmark	181	39.8%	26.5%	--	18.2%	18.8%	32.6%	10.5%
EPC	EPO	299	48.2%	38.5%	--	16.7%	31.8%	25.4%	10.0%
EPC	Finland	202	19.8%	9.9%	--	6.9%	8.9%	16.3%	5.9%
EPC	France	1884	16.6%	14.6%	--	9.4%	12.1%	9.2%	7.0%
EPC	Germany	6282	14.8%	12.8%	--	7.4%	10.7%	7.9%	5.3%
EPC	Greece	2	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Ireland	263	1.9%	1.5%	--	1.1%	1.1%	1.1%	0.8%
EPC	Italy	2889	10.9%	9.0%	--	5.2%	7.9%	6.4%	4.1%
EPC	Luxem.	41	56.1%	48.8%	--	43.9%	46.3%	51.2%	41.5%
EPC	Monaco	5	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Netherl.	376	26.9%	25.3%	--	13.0%	20.2%	17.6%	8.0%
EPC	Portugal	55	1.8%	1.8%	--	1.8%	1.8%	1.8%	1.8%
EPC	Spain	1961	1.7%	1.7%	--	0.8%	1.4%	0.5%	0.5%
EPC	Sweden	434	43.8%	33.6%	--	21.2%	28.8%	33.2%	16.4%
EPC	Switzerl.	431	45.7%	40.6%	--	28.3%	33.6%	29.9%	21.3%
EPC	Turkey	22	100.0%	100.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	UK	1667	35.5%	27.9%	--	17.5%	21.0%	24.8%	10.6%
<i>Subtotal for EPC</i>		17725	17.4%	14.5%	--	8.7%	11.8%	10.6%	6.1%
Japan	Jpn	19772	12.2%	11.4%	9.2%	--	9.2%	5.5%	7.1%
US	USA	7995	33.3%	27.7%	26.3%	17.8%	--	25.8%	16.4%
Other	Argen	91	4.4%	4.4%	2.2%	1.1%	1.1%	2.2%	1.1%
Other	Austral	729	29.6%	29.6%	23.3%	6.6%	15.1%	23.6%	4.3%
Other	Brazil	1121	1.0%	1.0%	0.2%	0.1%	0.8%	1.0%	0.0%
Other	Bulgaria	244	100.0%	100.0%	1.2%	0.8%	0.8%	1.2%	0.4%
Other	Canada	508	16.1%	16.1%	5.7%	2.8%	12.2%	14.8%	1.4%
Other	China	0							
Other	Hungary	441	16.6%	16.6%	14.3%	6.3%	5.4%	14.5%	3.4%
Other	India	92	1.1%	1.1%	1.1%	0.0%	0.0%	1.1%	0.0%
Other	Israel	91	47.3%	47.3%	28.6%	14.3%	30.8%	38.5%	6.6%
Other	Korea	1751	1.4%	1.4%	0.5%	0.3%	0.7%	1.1%	0.0%
Other	Malay	2	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Mexico	58	1.7%	1.7%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	NewZeal	114	23.7%	23.7%	21.1%	6.1%	13.2%	22.8%	3.5%
Other	Norway	115	28.7%	28.7%	25.2%	6.1%	15.7%	26.1%	5.2%
Other	Philipp	118	0.8%	0.8%	0.0%	0.0%	0.8%	0.8%	0.0%
Other	Poland	324	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Romania	257	0.4%	0.4%	0.4%	0.4%	0.4%	0.4%	0.4%
Other	Russia	5473	0.8%	0.8%	0.7%	0.3%	0.3%	0.8%	0.1%
Other	SouthAfr	474	7.6%	7.6%	6.1%	1.9%	4.9%	7.2%	1.3%
Other	Singapore	0							
<i>Subtotal for Other</i>		12003	7.0%	7.0%	3.6%	1.3%	2.7%	4.3%	0.7%

2. IPC = B
Performing Operations

Secondary Filings as a Percentage of First Filings
 (% Priority filings claimed in Country of Origin From):

Country Bloc	Cntry of Origin	First Filings by IPC	All Other Blocs	Other Trilateral Blocs	EPC	Japan	USA	Other Countries	Trilateral Patent Families
EPC	Austria	344	29.9%	24.4%	--	17.2%	19.5%	20.6%	12.2%
EPC	Belgium	393	7.9%	5.9%	--	4.3%	4.3%	4.8%	2.8%
EPC	Denmark	148	31.8%	17.6%	--	8.8%	13.5%	23.0%	4.7%
EPC	EPO	487	40.7%	33.5%	--	13.8%	28.5%	15.4%	8.8%
EPC	Finland	423	30.3%	19.6%	--	12.1%	15.1%	25.1%	7.6%
EPC	France	2532	22.7%	19.9%	--	12.2%	16.8%	13.7%	9.0%
EPC	Germany	11857	21.6%	20.0%	--	13.4%	16.6%	8.4%	10.0%
EPC	Greece	5	20.0%	20.0%	--	20.0%	0.0%	20.0%	0.0%
EPC	Ireland	148	6.1%	4.7%	--	1.4%	4.7%	3.4%	1.4%
EPC	Italy	2834	10.5%	9.4%	--	4.8%	8.3%	5.2%	3.7%
EPC	Luxem.	33	39.4%	36.4%	--	30.3%	27.3%	24.2%	21.2%
EPC	Monaco	4	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Netherl.	405	31.1%	28.1%	--	20.0%	21.7%	19.0%	13.6%
EPC	Portugal	73	2.7%	2.7%	--	2.7%	2.7%	1.4%	2.7%
EPC	Spain	1794	2.0%	1.5%	--	0.7%	1.2%	1.3%	0.4%
EPC	Sweden	781	44.6%	33.0%	--	22.3%	29.4%	31.4%	18.7%
EPC	Switzerl.	571	47.6%	42.4%	--	30.8%	36.3%	25.6%	24.7%
EPC	Turkey	32	100.0%	100.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	UK	2664	34.3%	27.7%	--	16.3%	22.5%	24.0%	11.2%
<i>Subtotal for EPC</i>		25528	22.3%	19.4%	--	12.3%	16.1%	11.5%	9.1%
Japan	Jpn	58218	9.6%	9.1%	6.3%	--	8.2%	3.2%	5.3%
US	USA	10749	36.2%	29.7%	27.4%	20.9%	--	27.1%	18.6%
Other	Argen	67	1.5%	1.5%	1.5%	0.0%	1.5%	0.0%	0.0%
Other	Austral	779	23.4%	23.4%	17.5%	7.4%	13.2%	19.1%	4.6%
Other	Brazil	1180	0.8%	0.8%	0.8%	0.2%	0.3%	0.8%	0.2%
Other	Bulgaria	333	100.0%	100.0%	3.3%	2.4%	0.9%	2.7%	0.6%
Other	Canada	743	18.6%	18.6%	8.6%	5.4%	14.1%	16.7%	3.5%
Other	China	0							
Other	Hungary	312	10.3%	10.3%	9.0%	1.9%	3.2%	9.3%	1.3%
Other	India	165	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Israel	80	35.0%	35.0%	22.5%	10.0%	26.2%	26.2%	8.7%
Other	Korea	1459	1.1%	1.1%	0.7%	0.4%	0.7%	0.9%	0.2%
Other	Malay	2	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Mexico	68	5.9%	5.9%	2.9%	1.5%	4.4%	4.4%	1.5%
Other	NewZeal	121	13.2%	13.2%	10.7%	4.1%	5.8%	12.4%	3.3%
Other	Norway	165	25.5%	25.5%	23.6%	8.5%	12.1%	24.2%	6.7%
Other	Philipp	67	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Poland	1134	1.0%	1.0%	0.8%	0.4%	0.4%	1.0%	0.2%
Other	Romania	474	0.6%	0.6%	0.2%	0.0%	0.0%	0.2%	0.0%
Other	Russia	15961	0.7%	0.7%	0.6%	0.1%	0.1%	0.6%	0.0%
Other	SouthAfr	586	10.6%	10.6%	8.2%	1.5%	5.5%	9.9%	0.9%
Other	Singapore	0							
<i>Subtotal for Other</i>		23696	4.2%	4.2%	2.0%	0.7%	1.4%	2.4%	0.5%

3. IPC = C
Chemistry & Metallurgy

Secondary Filings as a Percentage of First Filings
 (% Priority filings claimed in Country of Origin From):

Country Bloc	Cntry of Origin	First Filings by IPC	All Other Blocs	Other Trilateral Blocs	EPC	Japan	USA	Other Countries	Trilateral Patent Families
EPC	Austria	234	29.9%	23.5%	--	16.2%	19.7%	24.8%	12.4%
EPC	Belgium	136	16.2%	12.5%	--	8.8%	8.1%	9.6%	4.4%
EPC	Denmark	127	23.6%	16.5%	--	11.0%	14.2%	22.0%	8.7%
EPC	EPO	359	51.5%	39.3%	--	16.7%	30.9%	21.7%	8.4%
EPC	Finland	170	24.1%	18.8%	--	11.2%	16.5%	20.6%	8.8%
EPC	France	1000	60.5%	54.4%	--	46.2%	44.2%	43.8%	36.0%
EPC	Germany	3700	57.2%	53.1%	--	44.9%	41.5%	33.2%	33.3%
EPC	Greece	0			--				
EPC	Ireland	780	0.1%	0.1%	--	0.1%	0.1%	0.0%	0.1%
EPC	Italy	569	29.2%	27.2%	--	20.9%	23.0%	18.8%	16.7%
EPC	Luxem.	63	52.4%	52.4%	--	47.6%	44.4%	42.9%	39.7%
EPC	Monaco	0			--				
EPC	Netherl.	196	65.3%	61.2%	--	55.6%	49.5%	44.9%	43.9%
EPC	Portugal	43	7.0%	7.0%	--	4.7%	4.7%	7.0%	2.3%
EPC	Spain	521	6.7%	5.4%	--	2.9%	4.0%	5.4%	1.5%
EPC	Sweden	171	62.6%	46.8%	--	37.4%	36.3%	43.9%	26.9%
EPC	Switzerl.	411	80.3%	73.5%	--	65.2%	60.6%	59.6%	52.3%
EPC	Turkey	23	100.0%	100.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	UK	1604	69.8%	59.4%	--	48.4%	43.4%	52.5%	32.4%
<i>Subtotal for EPC</i>		10107	49.6%	44.2%	--	36.1%	34.4%	32.6%	26.5%
Japan	Jpn	26426	12.0%	11.6%	9.3%	--	10.0%	4.8%	7.7%
US	USA	8562	49.5%	44.5%	42.2%	35.0%	--	38.6%	32.7%
Other	Argen	8	12.5%	12.5%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Austral	246	40.2%	40.2%	32.9%	15.9%	16.7%	33.3%	11.8%
Other	Brazil	277	5.1%	5.1%	4.3%	2.2%	2.5%	5.1%	0.7%
Other	Bulgaria	446	100.0%	100.0%	1.3%	0.9%	0.4%	1.3%	0.4%
Other	Canada	550	14.9%	14.9%	10.0%	7.5%	10.4%	13.5%	4.2%
Other	China	0							
Other	Hungary	503	18.1%	18.1%	17.3%	11.1%	10.1%	17.9%	7.8%
Other	India	278	0.7%	0.7%	0.4%	0.0%	0.0%	0.4%	0.0%
Other	Israel	107	21.5%	21.5%	17.8%	14.0%	10.3%	18.7%	6.5%
Other	Korea	566	3.7%	3.7%	1.8%	2.5%	1.1%	3.2%	0.0%
Other	Malay	3	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Mexico	84	8.3%	8.3%	2.4%	0.0%	3.6%	3.6%	0.0%
Other	NewZeal	71	8.5%	8.5%	8.5%	4.2%	4.2%	7.0%	1.4%
Other	Norway	128	16.4%	16.4%	14.1%	6.2%	9.4%	15.6%	4.7%
Other	Philipp	41	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Poland	1143	1.4%	1.4%	1.3%	0.3%	0.4%	1.4%	0.3%
Other	Romania	550	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Russia	5089	1.0%	1.0%	0.8%	0.2%	0.2%	0.8%	0.1%
Other	SouthAfr	160	12.5%	12.5%	8.8%	3.8%	7.5%	11.9%	1.9%
Other	Singapore	0							
<i>Subtotal for Other</i>		10250	8.8%	8.8%	3.6%	2.0%	2.2%	4.0%	1.2%

4. IPC = D
Textiles & Paper

Secondary Filings as a Percentage of First Filings
(% Priority filings claimed in Country of Origin From):

Country Bloc	Cntry of Origin	First Filings by IPC	All Other Blocs	Other Trilateral Blocs	EPC	Japan	USA	Other Countries	Trilateral Patent Families
EPC	Austria	43	27.9%	27.9%	--	9.3%	25.6%	9.3%	7.0%
EPC	Belgium	57	8.8%	8.8%	--	0.0%	8.8%	0.0%	0.0%
EPC	Denmark	12	25.0%	16.7%	--	8.3%	16.7%	25.0%	8.3%
EPC	EPO	59	49.2%	44.1%	--	23.7%	37.3%	22.0%	16.9%
EPC	Finland	99	41.4%	36.4%	--	17.2%	32.3%	34.3%	13.1%
EPC	France	159	31.4%	27.0%	--	21.4%	20.8%	18.2%	15.1%
EPC	Germany	865	34.9%	33.3%	--	23.1%	29.2%	12.3%	19.1%
EPC	Greece	1	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Ireland	10	10.0%	10.0%	--	0.0%	10.0%	10.0%	0.0%
EPC	Italy	307	20.8%	19.9%	--	14.7%	15.0%	7.8%	9.8%
EPC	Luxem.	2	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Monaco	0			--				
EPC	Netherl.	19	47.4%	47.4%	--	31.6%	47.4%	31.6%	31.6%
EPC	Portugal	6	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Spain	138	4.3%	3.6%	--	2.2%	3.6%	0.7%	2.2%
EPC	Sweden	60	61.7%	48.3%	--	41.7%	38.3%	53.3%	31.7%
EPC	Switzerl.	91	75.8%	73.6%	--	59.3%	65.9%	48.4%	51.6%
EPC	Turkey	3	100.0%	100.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	UK	173	38.7%	31.8%	--	17.9%	28.9%	22.0%	15.0%
<i>Subtotal for EPC</i>		2104	33.2%	30.5%	--	20.6%	26.2%	15.9%	16.5%
Japan	Jpn	4400	8.7%	8.3%	6.1%	--	7.3%	3.1%	5.1%
US	USA	565	45.8%	40.5%	38.4%	27.4%	--	35.8%	25.3%
Other	Argen	11	9.1%	9.1%	9.1%	0.0%	0.0%	9.1%	0.0%
Other	Austral	27	37.0%	37.0%	33.3%	25.9%	14.8%	37.0%	11.1%
Other	Brazil	69	1.4%	1.4%	1.4%	0.0%	0.0%	1.4%	0.0%
Other	Bulgaria	52	100.0%	100.0%	1.9%	0.0%	0.0%	1.9%	0.0%
Other	Canada	46	19.6%	19.6%	10.9%	6.5%	13.0%	15.2%	4.3%
Other	China	0							
Other	Hungary	10	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	India	34	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Israel	3	33.3%	33.3%	33.3%	33.3%	0.0%	33.3%	0.0%
Other	Korea	301	1.3%	1.3%	0.3%	1.0%	0.3%	1.0%	0.3%
Other	Malay	0							
Other	Mexico	5	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	NewZeal	14	50.0%	50.0%	35.7%	28.6%	28.6%	35.7%	21.4%
Other	Norway	8	12.5%	12.5%	12.5%	0.0%	0.0%	12.5%	0.0%
Other	Philipp	6	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Poland	96	5.2%	5.2%	5.2%	1.0%	0.0%	5.2%	0.0%
Other	Romania	61	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Russia	738	1.5%	1.5%	1.2%	0.1%	0.3%	1.2%	0.1%
Other	SouthAfr	17	17.6%	17.6%	17.6%	5.9%	5.9%	17.6%	5.9%
Other	Singapore	0							
<i>Subtotal for Other</i>		1498	7.0%	7.0%	2.8%	1.4%	1.2%	3.1%	0.7%

5. IPC = E
Fixed Constructions

Secondary Filings as a Percentage of First Filings
 (% Priority filings claimed in Country of Origin From):

Country Bloc	Cntry of Origin	First Filings by IPC	All Other Blocs	Other Trilateral Blocs	EPC	Japan	USA	Other Countries	Trilateral Patent Families
EPC	Austria	247	17.4%	14.6%	--	5.3%	12.6%	9.7%	3.2%
EPC	Belgium	190	3.2%	1.1%	--	0.0%	1.1%	2.6%	0.0%
EPC	Denmark	59	35.6%	15.3%	--	3.4%	15.3%	32.2%	3.4%
EPC	EPO	102	34.3%	20.6%	--	10.8%	17.6%	23.5%	7.8%
EPC	Finland	171	22.2%	7.6%	--	2.9%	7.0%	20.5%	2.3%
EPC	France	1036	14.7%	11.6%	--	5.3%	10.5%	10.2%	4.2%
EPC	Germany	3809	9.6%	7.6%	--	3.5%	6.6%	5.5%	2.5%
EPC	Greece	5	40.0%	20.0%	--	0.0%	20.0%	20.0%	0.0%
EPC	Ireland	43	4.7%	2.3%	--	0.0%	2.3%	2.3%	0.0%
EPC	Italy	749	5.7%	4.3%	--	1.2%	4.1%	4.0%	1.1%
EPC	Luxem.	10	10.0%	10.0%	--	10.0%	10.0%	10.0%	10.0%
EPC	Monaco	0			--				
EPC	Netherl.	172	10.5%	8.7%	--	2.9%	8.7%	8.1%	2.9%
EPC	Portugal	26	3.8%	3.8%	--	0.0%	3.8%	3.8%	0.0%
EPC	Spain	601	1.3%	1.0%	--	0.2%	0.8%	0.8%	0.0%
EPC	Sweden	322	40.7%	21.4%	--	10.6%	19.3%	36.0%	8.4%
EPC	Switzerl.	197	18.8%	14.2%	--	7.1%	12.7%	10.2%	5.6%
EPC	Turkey	6	100.0%	100.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	UK	986	21.9%	13.8%	--	4.1%	12.6%	17.5%	2.8%
<i>Subtotal for EPC</i>		8731	12.9%	9.0%	--	3.7%	8.0%	9.0%	2.7%
Japan	Jpn	6599	4.3%	3.6%	2.8%	--	3.0%	2.0%	2.1%
US	USA	2080	31.2%	20.1%	19.4%	6.8%	--	27.2%	6.1%
Other	Argen	23	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Austral	390	23.6%	23.6%	18.7%	4.4%	13.3%	21.0%	3.3%
Other	Brazil	332	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Bulgaria	56	100.0%	100.0%	1.8%	0.0%	1.8%	1.8%	0.0%
Other	Canada	262	20.2%	20.2%	5.7%	2.7%	15.3%	18.3%	1.1%
Other	China	0							
Other	Hungary	247	6.1%	6.1%	4.5%	1.2%	1.6%	5.3%	0.0%
Other	India	36	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Israel	16	12.5%	12.5%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Korea	420	1.2%	1.2%	1.0%	0.2%	0.5%	1.2%	0.0%
Other	Malay	2	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Mexico	29	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	NewZeal	37	10.8%	10.8%	10.8%	5.4%	5.4%	10.8%	5.4%
Other	Norway	113	31.9%	31.9%	27.4%	6.2%	11.5%	30.1%	3.5%
Other	Philipp	26	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Poland	437	1.8%	1.8%	1.8%	0.0%	0.0%	1.8%	0.0%
Other	Romania	61	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Russia	4851	0.4%	0.4%	0.4%	0.1%	0.1%	0.4%	0.1%
Other	SouthAfr	419	6.4%	6.4%	5.0%	0.7%	2.6%	6.0%	0.2%
Other	Singapore	0							
<i>Subtotal for Other</i>		7757	4.1%	4.1%	2.4%	0.6%	1.7%	3.1%	0.3%

6. IPC = F

Mechanical Engineering

Secondary Filings as a Percentage of First Filings
(% Priority filings claimed in Country of Origin From):

Country Bloc	Cntry of Origin	First Filings by IPC	All Other Blocs	Other Trilateral Blocs	EPC	Japan	USA	Other Countries	Trilateral Patent Families
EPC	Austria	235	22.6%	17.4%	--	9.4%	15.7%	12.8%	7.7%
EPC	Belgium	224	4.5%	2.7%	--	0.9%	2.2%	3.1%	0.4%
EPC	Denmark	114	41.2%	23.7%	--	14.0%	16.7%	30.7%	7.0%
EPC	EPO	236	40.7%	32.2%	--	11.4%	29.2%	16.1%	8.5%
EPC	Finland	182	25.3%	14.8%	--	7.7%	11.5%	17.6%	4.4%
EPC	France	1559	26.4%	24.1%	--	16.4%	21.5%	12.8%	13.8%
EPC	Germany	5935	22.5%	20.7%	--	13.7%	16.7%	8.2%	9.7%
EPC	Greece	0			--				
EPC	Ireland	72	1.4%	1.4%	--	0.0%	1.4%	1.4%	0.0%
EPC	Italy	1127	11.4%	9.5%	--	4.8%	7.3%	5.5%	2.6%
EPC	Luxem.	16	37.5%	31.2%	--	12.5%	18.7%	12.5%	0.0%
EPC	Monaco	6	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Netherl.	153	36.6%	35.9%	--	25.5%	31.4%	19.6%	20.9%
EPC	Portugal	42	2.4%	2.4%	--	2.4%	0.0%	2.4%	0.0%
EPC	Spain	863	2.9%	2.2%	--	1.3%	1.5%	1.4%	0.6%
EPC	Sweden	417	46.0%	33.8%	--	22.5%	29.3%	31.2%	18.0%
EPC	Switzerl.	290	42.4%	39.7%	--	30.7%	31.7%	21.0%	22.8%
EPC	Turkey	29	100.0%	100.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	UK	1597	30.8%	25.9%	--	14.8%	22.5%	18.1%	11.4%
<i>Subtotal for EPC</i>		13097	23.3%	20.4%	--	12.8%	16.8%	10.8%	9.4%
Japan	Jpn	24635	9.7%	8.9%	5.5%	--	8.1%	3.5%	4.7%
US	USA	5028	36.1%	30.0%	27.7%	21.3%	--	25.9%	19.0%
Other	Argen	39	7.7%	7.7%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Austral	365	27.4%	27.4%	22.2%	11.0%	13.4%	20.8%	7.9%
Other	Brazil	538	0.9%	0.9%	0.9%	0.0%	0.6%	0.9%	0.0%
Other	Bulgaria	101	100.0%	100.0%	3.0%	0.0%	1.0%	3.0%	0.0%
Other	Canada	373	15.0%	15.0%	4.0%	2.1%	12.1%	13.4%	1.1%
Other	China	0							
Other	Hungary	262	13.7%	13.7%	13.4%	2.7%	2.7%	13.4%	1.5%
Other	India	111	0.9%	0.9%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Israel	58	46.6%	46.6%	39.7%	17.2%	32.8%	37.9%	10.3%
Other	Korea	792	1.9%	1.9%	1.0%	0.8%	0.9%	1.1%	0.3%
Other	Malay	2	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Mexico	34	5.9%	5.9%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	NewZeal	58	19.0%	19.0%	13.8%	0.0%	12.1%	17.2%	0.0%
Other	Norway	94	33.0%	33.0%	29.8%	6.4%	11.7%	29.8%	5.3%
Other	Philipp	88	2.3%	2.3%	1.1%	1.1%	1.1%	2.3%	0.0%
Other	Poland	552	1.4%	1.4%	1.3%	0.2%	0.2%	1.4%	0.0%
Other	Romania	139	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Russia	6013	1.1%	1.1%	0.9%	0.0%	0.6%	0.9%	0.0%
Other	SouthAfr	283	9.5%	9.5%	8.5%	2.5%	3.2%	8.8%	1.1%
Other	Singapore	0							
<i>Subtotal for Other</i>		9902	5.0%	5.0%	3.0%	0.9%	2.0%	3.3%	0.5%

7. IPC = G
Physics

Secondary Filings as a Percentage of First Filings
(% Priority filings claimed in Country of Origin From):

Country Bloc	Cntry of Origin	First Filings by IPC	All Other Blocs	Other Trilateral Blocs	EPC	Japan	USA	Other Countries	Trilateral Patent Families
EPC	Austria	109	35.8%	32.1%	--	17.4%	29.4%	13.8%	14.7%
EPC	Belgium	133	11.3%	10.5%	--	9.8%	7.5%	3.0%	6.8%
EPC	Denmark	65	38.5%	24.6%	--	15.4%	23.1%	24.6%	13.8%
EPC	EPO	282	50.0%	43.3%	--	28.7%	35.5%	18.4%	21.3%
EPC	Finland	118	37.3%	30.5%	--	24.6%	24.6%	21.2%	18.6%
EPC	France	1468	34.9%	32.7%	--	22.1%	28.6%	16.1%	18.1%
EPC	Germany	4363	26.6%	25.5%	--	18.9%	20.9%	7.4%	14.2%
EPC	Greece	1	100.0%	100.0%	--	100.0%	100.0%	100.0%	100.0%
EPC	Ireland	106	8.5%	7.5%	--	3.8%	6.6%	3.8%	2.8%
EPC	Italy	634	16.6%	15.6%	--	10.3%	13.4%	7.9%	8.0%
EPC	Luxem.	11	9.1%	9.1%	--	9.1%	9.1%	9.1%	9.1%
EPC	Monaco	2	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Netherl.	225	68.0%	65.3%	--	57.3%	59.1%	37.3%	51.1%
EPC	Portugal	8	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Spain	564	0.9%	0.9%	--	0.5%	0.4%	0.2%	0.0%
EPC	Sweden	252	55.2%	45.6%	--	34.5%	38.1%	29.0%	27.0%
EPC	Switzerl.	267	56.6%	54.7%	--	42.7%	43.8%	27.0%	31.8%
EPC	Turkey	9	100.0%	100.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	UK	1675	40.6%	35.3%	--	22.6%	28.8%	21.9%	16.1%
<i>Subtotal for EPC</i>		10292	31.0%	28.5%	--	20.2%	23.7%	12.9%	15.5%
Japan	Jpn	65030	11.1%	10.7%	6.5%	--	9.9%	2.5%	5.7%
US	USA	7230	46.3%	41.9%	37.2%	33.2%	--	27.4%	28.5%
Other	Argen	9	55.6%	55.6%	44.4%	22.2%	44.4%	33.3%	22.2%
Other	Austral	291	35.7%	35.7%	28.5%	12.7%	16.2%	26.1%	8.6%
Other	Brazil	383	1.0%	1.0%	0.8%	0.3%	0.5%	1.0%	0.3%
Other	Bulgaria	268	100.0%	100.0%	2.2%	0.4%	1.1%	1.5%	0.4%
Other	Canada	339	18.0%	18.0%	9.4%	8.3%	13.3%	16.2%	5.0%
Other	China	0							
Other	Hungary	288	9.4%	9.4%	8.3%	1.7%	2.1%	8.7%	1.0%
Other	India	49	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Israel	55	41.8%	41.8%	23.6%	14.5%	25.5%	25.5%	7.3%
Other	Korea	1052	1.9%	1.9%	0.8%	1.1%	1.3%	1.8%	0.6%
Other	Malay	1	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Mexico	14	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	NewZeal	31	16.1%	16.1%	9.7%	3.2%	12.9%	12.9%	3.2%
Other	Norway	71	31.0%	31.0%	26.8%	1.4%	12.7%	28.2%	1.4%
Other	Philipp	16	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Poland	764	0.1%	0.1%	0.1%	0.0%	0.0%	0.1%	0.0%
Other	Romania	237	0.4%	0.4%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Russia	12963	0.5%	0.5%	0.4%	0.0%	0.0%	0.4%	0.0%
Other	SouthAfr	78	15.4%	15.4%	12.8%	1.3%	9.0%	14.1%	0.0%
Other	Singapore	0							
<i>Subtotal for Other</i>		16909	3.6%	3.6%	1.5%	0.6%	1.0%	1.7%	0.4%

**8. IPC = H
Electricity**

Secondary Filings as a Percentage of First Filings
(% Priority filings claimed in Country of Origin From):

Country Bloc	Cntry of Origin	First Filings by IPC	All Other Blocs	Other Trilateral Blocs	EPC	Japan	USA	Other Countries	Trilateral Patent Families
EPC	Austria	113	23.0%	17.7%	--	12.4%	13.3%	8.0%	8.0%
EPC	Belgium	72	13.9%	12.5%	--	11.1%	11.1%	13.9%	9.7%
EPC	Denmark	38	39.5%	21.1%	--	18.4%	10.5%	31.6%	7.9%
EPC	EPO	190	51.1%	44.7%	--	28.4%	40.5%	23.2%	24.2%
EPC	Finland	54	24.1%	18.5%	--	13.0%	16.7%	13.0%	11.1%
EPC	France	1206	38.1%	35.3%	--	23.3%	31.8%	16.5%	19.7%
EPC	Germany	4479	25.9%	23.0%	--	17.0%	17.7%	9.9%	11.7%
EPC	Greece	3	100.0%	66.7%	--	0.0%	66.7%	100.0%	0.0%
EPC	Ireland	99	4.0%	4.0%	--	4.0%	4.0%	2.0%	4.0%
EPC	Italy	488	22.3%	20.9%	--	13.7%	18.2%	9.4%	11.1%
EPC	Luxem.	2	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Monaco	3	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Netherl.	257	82.5%	81.7%	--	75.9%	73.2%	57.2%	67.3%
EPC	Portugal	17	0.0%	0.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	Spain	375	3.2%	2.4%	--	2.4%	1.1%	1.3%	1.1%
EPC	Sweden	172	46.5%	36.0%	--	25.0%	29.1%	32.0%	18.0%
EPC	Switzerl.	215	48.8%	40.9%	--	29.3%	36.3%	26.0%	24.7%
EPC	Turkey	4	100.0%	100.0%	--	0.0%	0.0%	0.0%	0.0%
EPC	UK	1345	40.2%	35.2%	--	23.4%	28.8%	19.6%	17.0%
<i>Subtotal for EPC</i>		9132	31.2%	27.8%	--	20.0%	22.9%	14.3%	15.1%
Japan	Jpn	52903	7.9%	7.5%	5.0%	--	7.0%	2.9%	4.4%
US	USA	6456	44.5%	39.2%	34.2%	31.6%	--	26.5%	26.5%
Other	Argen	4	50.0%	50.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Austral	156	28.8%	28.8%	23.7%	10.3%	15.4%	20.5%	7.1%
Other	Brazil	335	0.9%	0.9%	0.6%	0.3%	0.3%	0.9%	0.0%
Other	Bulgaria	154	100.0%	100.0%	5.2%	4.5%	3.2%	5.2%	3.2%
Other	Canada	317	27.8%	27.8%	16.4%	12.3%	20.5%	25.9%	7.6%
Other	China	0							
Other	Hungary	207	15.0%	15.0%	14.0%	6.8%	2.9%	14.5%	1.9%
Other	India	57	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Israel	23	34.8%	34.8%	21.7%	13.0%	17.4%	26.1%	0.0%
Other	Korea	1092	1.8%	1.8%	0.6%	0.5%	1.0%	1.2%	0.1%
Other	Malay	2	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Mexico	15	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	NewZeal	26	26.9%	26.9%	26.9%	11.5%	19.2%	26.9%	11.5%
Other	Norway	41	34.1%	34.1%	34.1%	7.3%	19.5%	34.1%	4.9%
Other	Philipp	18	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Poland	540	0.7%	0.7%	0.7%	0.0%	0.0%	0.7%	0.0%
Other	Romania	184	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Other	Russia	6866	0.6%	0.6%	0.5%	0.1%	0.1%	0.5%	0.0%
Other	SouthAfr	123	12.2%	12.2%	10.6%	3.3%	6.5%	10.6%	2.4%
Other	Singapore	0							
<i>Subtotal for Other</i>		10160	4.2%	4.2%	2.1%	1.0%	1.4%	2.4%	0.5%

Table 24. Growth Rates of Priority (First) Filings, by Selected Countries and IPC code**1. IPC = A
Human Necessities**

Priority Country or Bloc	Growth Rate of 1st filings Between: 1984 and 1999	1984 and 1994
France	23.87%	18.90%
Germany	46.90%	22.88%
UK	44.83%	16.65%
<i>EPC As a Whole</i>	24.92%	10.49%
Japan	57.78%	35.09%
USA	133.53%	106.40%
Austral	79.97%	60.28%
Canada	-82.78%	46.04%
Korea	156.98%	70.62%

**2. IPC = B
Performing Operations**

Priority Country or Bloc	Growth Rate of 1st filings Between: 1984 and 1999	1984 and 1994
France	20.11%	8.15%
Germany	40.52%	23.95%
UK	-5.52%	-16.98%
<i>EPC As a Whole</i>	22.37%	10.12%
Japan	14.22%	9.60%
USA	80.66%	73.42%
Austral	56.09%	39.12%
Canada	-124.47%	-4.54%
Korea	186.09%	153.37%

**5. IPC = E
Fixed Constructions**

Priority Country or Bloc	Growth Rate of 1st filings Between: 1984 and 1999	1984 and 1994
France	-22.90%	-19.09%
Germany	28.18%	15.78%
UK	6.38%	-7.04%
<i>EPC As a Whole</i>	10.49%	1.86%
Japan	101.33%	97.17%
USA	68.76%	43.11%
Austral	76.73%	74.80%
Canada	-85.88%	14.21%
Korea	183.49%	107.70%

**6. IPC = F
Mechanical Engineering**

Priority Country or Bloc	Growth Rate of 1st filings Between: 1984 and 1999	1984 and 1994
France	-1.36%	-11.19%
Germany	49.11%	20.63%
UK	-27.92%	-34.15%
<i>EPC As a Whole</i>	18.69%	-0.47%
Japan	21.23%	14.62%
USA	59.65%	44.75%
Austral	56.31%	42.09%
Canada	-161.75%	-45.35%
Korea	200.56%	156.00%

3. IPC = C

Chemistry & Metallurgy

Priority Country or Bloc	Growth Rate of 1st filings Between:	
	<u>1984 and 1999</u>	<u>1984 and 1994</u>
France	23.19%	9.80%
Germany	40.53%	23.73%
UK	24.73%	1.36%
<i>EPC As a Whole</i>	27.07%	9.39%
Japan	33.63%	23.87%
USA	91.52%	73.36%
Austral	120.28%	88.32%
Canada	-134.71%	-47.80%
Korea	196.22%	140.16%

4. IPC = D

Textiles & Paper

Priority Country or Bloc	Growth Rate of 1st filings Between:	
	<u>1984 and 1999</u>	<u>1984 and 1994</u>
France	-9.91%	-8.53%
Germany	27.87%	22.66%
UK	-19.75%	-12.29%
<i>EPC As a Whole</i>	14.98%	9.73%
Japan	4.62%	7.40%
USA	76.81%	83.33%
Austral	51.08%	69.31%
Canada	-143.07%	-24.51%
Korea	113.98%	136.62%

7. IPC = G

Physics

Priority Country or Bloc	Growth Rate of 1st filings Between:	
	<u>1984 and 1999</u>	<u>1984 and 1994</u>
France	16.31%	5.82%
Germany	63.26%	31.41%
UK	34.24%	0.77%
<i>EPC As a Whole</i>	51.51%	21.75%
Japan	25.36%	14.88%
USA	156.05%	127.35%
Austral	122.84%	56.30%
Canada	-57.90%	4.33%
Korea	214.63%	152.70%

8. IPC = H

Electricity

Priority Country or Bloc	Growth Rate of 1st filings Between:	
	<u>1984 and 1999</u>	<u>1984 and 1994</u>
France	50.03%	17.39%
Germany	62.23%	18.85%
UK	41.85%	2.57%
<i>EPC As a Whole</i>	67.44%	23.78%
Japan	41.37%	22.71%
USA	148.81%	113.73%
Austral	122.30%	77.61%
Canada	-70.27%	-12.77%
Korea	273.51%	207.31%

Table 25. EPO Filings and Transfer Rate as a Percentage of Previous Year's First Filing:

<i>EPC Contracting States</i>		EPO	Transfer	<i>EPC Contracting States</i>		EPO	Transfer
<u>Year</u>	<u>Cntry</u>	<u>Filings</u>	<u>Rate</u>	<u>Year</u>	<u>Cntry</u>	<u>Filings</u>	<u>Rate</u>
1980	Austria	293	12.6%	1980	Luxem	59	25.2%
1985	Austria	534	32.5%	1985	Luxem	98	55.1%
1990	Austria	549	36.8%	1990	Luxem	31	42.5%
1995	Austria	501	26.7%	1995	Luxem	25	42.4%
2000	Austria	491	24.5%	2000	Luxem	39	57.4%
1980	Bel	118	6.0%	1980	Monacc	2	8.3%
1985	Bel	155	9.6%	1985	Monacc	0	0.0%
1990	Bel	202	37.8%	1990	Monacc	1	5.6%
1995	Bel	234	29.5%	1995	Monacc	2	7.4%
2000	Bel	113	19.9%	2000	Monacc	0	0.0%
1980	Den	131	16.2%	1980	Neth	561	25.6%
1985	Den	221	29.7%	1985	Neth	976	54.1%
1990	Den	297	35.3%	1990	Neth	1083	53.0%
1995	Den	479	41.0%	1995	Neth	821	49.5%
2000	Den	285	20.8%	2000	Neth	568	23.3%
1980	Fin	85	8.7%	1980	Port	1	0.5%
1985	Fin	171	12.1%	1985	Port	8	3.0%
1990	Fin	395	19.0%	1990	Port	5	1.7%
1995	Fin	646	28.0%	1995	Port	8	4.0%
2000	Fin	516	20.7%	2000	Port	10	7.0%
1980	Fra	2495	21.2%	1980	Spa	45	0.6%
1985	Fra	3793	35.0%	1985	Spa	135	2.0%
1990	Fra	4867	42.3%	1990	Spa	226	4.1%
1995	Fra	4947	43.0%	1995	Spa	337	7.9%
2000	Fra	3990	30.8%	2000	Spa	364	7.9%
1980	Ger	6372	16.4%	1980	Swe	606	21.0%
1985	Ger	9920	24.0%	1985	Swe	941	36.1%
1990	Ger	11662	28.4%	1990	Swe	914	42.7%
1995	Ger	13851	26.7%	1995	Swe	1424	51.0%
2000	Ger	14127	21.5%	2000	Swe	585	17.5%
1980	Greece	4	16.0%	1980	Swit	1021	32.3%
1985	Greece	5	29.4%	1985	Swit	1319	53.3%
1990	Greece	16	6.5%	1990	Swit	1480	56.0%
1995	Greece	20	9.9%	1995	Swit	1192	55.1%
2000	Greece	22	7.5%	2000	Swit	549	46.2%
1980	Ire	30	3.3%	1980	Tur	0	0.0%
1985	Ire	36	2.4%	1985	Tur	1	0.8%
1990	Ire	65	3.3%	1990	Tur	1	1.1%
1995	Ire	95	17.1%	1995	Tur	1	1.4%
2000	Ire	44	13.6%	2000	Tur	4	0.8%
1980	Ita	601	6.3%	1980	UK	2658	18.8%
1985	Ita	1384	14.4%	1985	UK	4287	36.6%
1990	Ita	2154	23.0%	1990	UK	4506	40.4%
1995	Ita	2056	25.8%	1995	UK	4704	42.3%
2000	Ita	2157	29.4%	2000	UK	3375	24.1%
				<i>Sum</i>		<i>126102</i>	
				<i>Mean</i>		<i>1401</i>	<i>22.6%</i>
				<i>Std Dev</i>		<i>2779</i>	<i>16.8%</i>
<i>Japan</i>		EPO	Transfer	<i>U.S.A.</i>		EPO	Transfer
<u>Year</u>		<u>Filings</u>	<u>Rate</u>	<u>Year</u>		<u>Filings</u>	<u>Rate</u>
1980	Japan	3636	2.4%	1980	USA	7080	14.2%
1985	Japan	10276	4.0%	1985	USA	12866	26.4%
1990	Japan	19217	6.1%	1990	USA	19442	25.2%
1995	Japan	16793	5.3%	1995	USA	24990	20.5%
2000	Japan	21653	6.1%	2000	USA	20358	13.3%
				<i>Sum</i>		<i>84736</i>	
				<i>Mean</i>		<i>16947</i>	<i>19.9%</i>
				<i>Std Dev</i>		<i>7010</i>	<i>6.1%</i>

Other Countries		EPO	Transfer
<u>Year</u>	<u>Cntry</u>	<u>Filings</u>	<u>Rate</u>
1980	Argen		0.5%
1985	Argen	5	2.0%
1990	Argen	16	14.8%
1995	Argen	13	27.1%
2000	Argen	23	25.0%
1980	Austral	290	8.7%
1985	Austral	586	19.6%
1990	Austral	545	12.6%
1995	Austral	668	12.5%
2000	Austral	304	4.4%
1980	Braz	17	0.4%
1985	Braz	29	0.7%
1990	Braz	37	0.8%
1995	Braz	60	1.2%
2000	Braz	31	0.5%
1980	Bulgaria	8	0.6%
1985	Bulgaria	13	0.8%
1990	Bulgaria	2	0.2%
1995	Bulgaria	7	1.7%
2000	Bulgaria	0	0.0%
1980	Canada	94	2.3%
1985	Canada	200	6.4%
1990	Canada	386	11.6%
1995	Canada	378	12.3%
2000	Canada	174	15.5%
1980	China	0	0.0%
1985	China	55	0.0%
1990	China	35	0.2%
1995	China	79	0.2%
2000	China	112	0.2%
1980	Hung	76	6.9%
1985	Hung	109	4.8%
1990	Hung	65	3.4%
1995	Hung	57	5.5%
2000	Hung	21	2.4%
1980	India	0	0.0%
1985	India	3	0.4%
1990	India	6	0.3%
1995	India	4	2.2%
2000	India	31	13.0%
1980	Israel	25	5.7%
1985	Israel	89	20.6%
1990	Israel	168	32.9%
1995	Israel	261	36.0%
2000	Israel	168	21.8%
1980	Korea	2	0.0%
1985	Korea	24	0.3%
1990	Korea	151	1.2%
1995	Korea	527	1.6%
2000	Korea	1123	2.0%

Other Countries		EPO	Transfer
<u>Year</u>	<u>Cntry</u>	<u>Filings</u>	<u>Rate</u>
1980	Malay	0	0.0%
1985	Malay	0	0.0%
1990	Malay	0	0.0%
1995	Malay	5	21.7%
2000	Malay	16	37.2%
1980	Mex	3	0.4%
1985	Mex	2	0.7%
1990	Mex	3	1.1%
1995	Mex	6	12.8%
2000	Mex	3	4.4%
1980	NewZ	19	6.2%
1985	NewZ	48	10.2%
1990	NewZ	25	5.1%
1995	NewZ	75	10.3%
2000	NewZ	68	8.7%
1980	Nor	48	7.3%
1985	Nor	112	15.2%
1990	Nor	104	12.9%
1995	Nor	201	31.3%
2000	Nor	102	10.8%
1980	Philip	5	1.6%
1985	Philip	2	0.5%
1990	Philip	0	0.0%
1995	Philip	4	6.8%
2000	Philip	1	14.3%
1980	Poland	18	0.3%
1985	Poland	19	0.4%
1990	Poland	12	0.3%
1995	Poland	16	0.4%
2000	Poland	12	0.3%
1980	Roman	8	0.5%
1985	Roman	1	0.1%
1990	Roman	5	0.2%
1995	Roman	6	0.6%
2000	Roman	2	0.8%
1980	Rus	1	0.0%
1985	Rus	61	0.1%
1990	Rus	81	0.1%
1995	Rus	125	1.0%
2000	Rus	70	0.5%
1980	Saf	82	6.9%
1985	Saf	90	4.2%
1990	Saf	82	3.5%
1995	Saf	115	8.0%
2000	Saf	45	2.8%
1980	Sing	0	0.0%
1985	Sing	0	0.0%
1990	Sing	9	81.8%
1995	Sing	26	1.6%
2000	Sing	40	11.9%
Sum		8855	
Mean		89	6.9%
Std Dev		168	11.5%

Table 26. EPC Patent Family Counts by Bloc Combinations & Routes

Blocs Involved	Routes Involved	Average 1979-1999	1984	1994
E	National	64009	66998	61659
E	EPO	8752	6150	11048
E	PCT	1032	117	1839
E	PCT/EPO	1187	658	1049
E&J	National	1714	1707	894
E&J	EPO	2436	1990	1710
E&J	PCT	42	27	37
E&J	PCT/EPO	642	284	519
E&O	National	1558	2048	957
E&O	EPO	1244	1140	1542
E&O	PCT	1496	116	2381
E&O	PCT/EPO	2830	747	1733
E&U	National	2888	2638	2756
E&U	EPO	3252	2819	3669
E&U	PCT	114	31	225
E&U	PCT/EPO	1140	580	1240
E,J,&O	National	673	660	383
E,J,&O	EPO	1834	1532	1252
E,J,&O	PCT	80	32	94
E,J,&O	PCT/EPO	1500	344	1649
E,U,&J	National	5487	6321	3814
E,U,&J	EPO	9908	8711	11674
E,U,&J	PCT	102	89	132
E,U,&J	PCT/EPO	1792	1010	3303
E,U,&O	National	1597	1642	1267
E,U,&O	EPO	2507	2974	2650
E,U,&O	PCT	423	105	638
E,U,&O	PCT/EPO	4877	1025	4496
E,U,J,&O	National	3716	3890	2221
E,U,J,&O	EPO	12929	13990	12706
E,U,J,&O	PCT	195	87	316
E,U,J,&O	PCT/EPO	7102	2469	15722

Notation:

E denotes EPC Bloc, U USA, J Japan, and O Other Countries

Table 27. Patent Family Regressions: Priority FilingsDependent Variable: $\ln(\text{Priority_Filings/Labor})$

	(1)	(2)	(3)	(4)
Constant	-0.555*** (0.094)	-3.404*** (0.334)	-3.754*** (0.489)	-2.843*** (0.573)
Lag 1	0.935*** (0.011)	0.719*** (0.024)	0.756*** (0.022)	0.724*** (0.024)
$\ln$ RD		0.187*** (0.031)		0.246*** (0.064)
$\ln$ GDPC			0.183*** (0.038)	-0.091 (0.084)
No. of Obs.	490	444	480	444
Adj. R-sq	0.94	0.93	0.93	0.93
Method of Estimation	OLS	GLS	GLS	GLS

Notes: The equations are estimated over the period 1980-1993. OLS refers to ordinary least squares and GLS to generalized least squares. The dependent variable here is priority (or first) filings. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. RD refers to research and development expenditures per source country worker (in real 1995 US dollars) and GDPC to gross domestic product per capita (in real 1995 US dollars).

Table 28a. Patent Family Regressions: Subsequent Filings in EPO

Dependent Variable: ln (EPO_Filings/Labor)

	(1)	(2)	(3)
Constant	-0.122 (0.086)	-0.012 (0.089)	-2.705*** (0.427)
Lag 1	0.984*** (0.007)	0.702*** (0.049)	0.829*** (0.021)
Lag 2		0.258** (0.058)	
Lag 3		0.032 (0.046)	
ln FF			0.062*** (0.021)
ln RD			0.250*** (0.036)
No. of Obs.	492	405	469
Adj. R-sq	0.97	0.97	0.97
Method of Estimation	OLS	OLS	GLS

Notes: The equations are estimated over the period 1980-1994. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. EP_Filings are patent families involving a filing at least the EPO. RD refers to research and development expenditures per source country worker (in real 1995 US dollars) and FF the log of first (or priority) filings per source country worker.

Table 28b. Patent Family Regressions: Subsequent Filings in Other BlocDependent Variable: $\ln(\text{Oth_Bloc/Labor})$

	(1)	(2)	(3)	(4)
Constant	-0.095 (0.065)	-0.062 (0.066)	-1.738*** (0.339)	-15.09*** (0.484)
Lag 1	0.989*** (0.006)	0.776*** (0.050)	0.899*** (0.018)	
Lag 2		0.153** (0.059)		
Lag 3		0.063 (0.047)		
$\ln \text{FF}$			0.005 (0.018)	0.115*** (0.030)
$\ln \text{RD}$			0.139*** (0.028)	0.995*** (0.062)
No. of Obs.	520	438	497	513
Adj. R-sq	0.98	0.98	0.98	0.81
Method of Estimation	OLS	OLS	GLS	GLS

Notes: The equations are estimated over the period 1980-1994. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. Oth_Bloc filings are patent families involving a filing at least a country outside the trilateral area (US, Japan, and EPO). RD refers to research and development expenditures per source country worker (in real 1995 US dollars) and FF the log of first (or priority) filings per source country worker.

Table 28c. Patent Family Regressions: Subsequent Filings in Other TrilateralDependent Variable: $\ln(\text{Oth_Tril}/\text{Labor})$

	(1)	(2)	(3)	(4)
Constant	-0.126* (0.072)	-0.089 (0.074)	-1.702*** (0.342)	-14.77*** (0.498)
Lag 1	0.989*** (0.006)	0.713*** (0.049)	0.902*** (0.018)	
Lag 2		0.204*** (0.058)		
Lag 3		0.072 (0.047)		
$\ln \text{FF}$			0.005 (0.018)	0.129*** (0.031)
$\ln \text{RD}$			0.133*** (0.027)	0.932*** (0.063)
No. of Obs.	520	438	497	513
Adj. R-sq	0.97	0.98	0.98	0.79
Method of Estimation	OLS	OLS	GLS	GLS

Notes: The equations are estimated over the period 1980-1994. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. Oth_Tril filings are patent families involving a filing at least a country in the trilateral area (US, Japan, and EPO). RD refers to research and development expenditures per source country worker (in real 1995 US dollars) and FF the log of first (or priority) filings per source country worker.

Table 28d. Patent Family Regressions: Subsequent Filings in Trilateral Bloc

Dependent Variable: ln (Tril/Labor)

	(1)	(2)	(3)	(4)
Constant	-0.128 (0.107)	-0.044 (0.113)	-3.238*** (0.520)	-17.67*** (0.636)
Lag 1	0.986*** (0.009)	0.719*** (0.056)	0.828*** (0.026)	
Lag 2		0.277*** (0.066)		
Lag 3		-0.004 (0.055)		
ln FF			0.037* (0.023)	0.139*** (0.043)
ln RD			0.279*** (0.043)	1.242*** (0.075)
No. of Obs.	452	366	429	461
Adj. R-sq	0.96	0.97	0.97	0.87
Method of Estimation	OLS	OLS	GLS	GLS

Notes: The equations are estimated over the period 1980-1994. OLS denotes ordinary least squares and GLS generalized least squares. Lag n refers to the dependent variable lagged n period(s), and ***, **, * denote significance levels of 1%, 5%, and 10% respectively. Standard errors are shown in parentheses. Tril filings are patent families involving filing in trilateral states (US, Japan, and at least one EPO state). RD refers to research and development expenditures per source country worker (in real 1995 US dollars) and FF the log of first (or priority) filings per source country worker.

Table 29. Summary of Forecast Errors: Patent Family Filings

Model	Variable	N	<i>Root Mean Square Percentage Error</i>					
			1996	1996	1997	1998	1999	2000
AR1	EP Filing	492	0.439	0.196	0.179	0.275	0.178	1.388
AR3	EP Filing	405	0.319	0.180	0.185	0.257	0.157	1.330
RE1	EP Filing	469	0.420	0.197	0.196	0.250	0.168	1.312
AR1	Oth_Bloc	520	0.236	0.376	0.173	0.268	0.125	0.352
AR3	Oth_Bloc	438	0.195	0.334	0.176	0.245	0.119	0.353
RE1	Oth_Bloc	497	0.243	0.352	0.183	0.267	0.149	0.369
AR1	Oth_Tril	520	0.252	0.389	0.186	0.295	0.478	0.412
AR3	Oth_Tril	438	0.209	0.335	0.177	0.264	0.492	0.484
RE1	Oth_Tril	497	0.264	0.365	0.192	0.297	0.542	0.446
AR1	Tril	452	0.542	0.394	0.225	0.790	2.082	0.538
AR3	Tril	366	0.414	0.193	0.162	0.812	2.600	0.850
RE1	Tril	429	0.530	0.342	0.292	0.859	2.505	0.854

Notes: The forecast errors correspond to the estimated models in Table 28 A - D. Each entry is the RMSPE (root mean square percentage error) associated with each model. AR1 denotes the autoregressive model of order 1, AR3 autoregressive model of order 3, and RE1 regression model of R&D and one lagged dependent variable. AR1, AR3, and RE1 correspond to the results in columns 1 - 3 in Table 28 A- D. N denotes the number of observations, EP Filing are subsequent filings at the EPO, Oth_Bloc subsequent filings in at least one other bloc, Oth_Tril subsequent filings in at least one other trilateral, and Tril subsequent filings in the trilateral area.

Table 30. Sample Patent Family Forecasts: Actual vs. Predicted Values for 1998

Priority	Actual	AR1	RE1	EP Filing	Actual	AR3	RE1
Argen	90	165	197	Argen	29	26	24
Aus	5489	4331	4958	Aus	854	846	934
Aut	1985	1609	1894	Aut	710	694	683
Bel	720	764	1043	Bel	203	282	315
Braz	6321	6909	6338	Braz	59	91	101
Bulg	504	446	290	Bulg	10	5	4
Can	2762	3048	3918	Can	396	535	622
China	52663	60243	36009	China	131	159	143
Den	1298	1063	1285	Den	675	560	523
Fin	2424	1691	2014	Fin	1062	940	946
Fra	12673	10267	12843	Fra	6087	5983	5809
Ger	61416	39884	44383	Ger	20088	18558	17314
Gree	439	571	574	Gree	41	36	39
Hung	1469	1045	852	Hung	55	88	69
India	106	241	246	India	35	9	10
Ire	550	470	564	Ire	165	129	139
Israel	878	704	920	Israel	439	392	453
Italy	11355	9209	9967	Italy	2428	2556	2310
Japan	348327	183884	190515	Japan	21000	20428	23258
Korea	38476	30334	31369	Korea	993	759	1039
Malay	24	100	101	Malay	3	6	5
Mex	68	130	146	Mex	15	15	16
Neth	2433	2117	2711	Neth	1010	1017	1034
NewZ	766	650	687	NewZ	125	115	112
Nor	798	687	849	Nor	246	234	236
Poland	3881	4074	3102	Poland	21	27	27
Port	131	225	261	Port	16	12	13
Romania	902	1224	860	Romania	2	6	6
Russia	12250	13768	10398	Russia	225	158	131
SAfrica	1905	1911	1746	SAfrica	160	154	129
Sing	316	347	426	Sing	74	54	60
Spain	4606	4257	4461	Spain	491	473	499
Swe	3343	2517	3205	Swe	1785	1881	1860
Switzer	1586	1466	1920	Switzer	878	1066	1015
Tur	327	527	568	Tur	19	12	13
UK	12397	10780	12659	UK	6151	5748	5106
USA	151622	98363	122495	USA	35627	32785	37152
Sum	747300	500020	516773	Sum	102308	96838	102150
RMSPE		0.606	0.591	RMSPE		0.257	0.250

Notes: Priority refers to priority (or first) filings by country of origin
EP Filings are subsequent filings at the EPO by source country
Oth_Bloc are subsequent filings in at least one other bloc
Oth_Tril are subsequent filings that involve at least one other trilateral bloc
Tril are subsequent filings that involve all the trilateral blocs
AR3 are forecasts from an autoregressive model of order 3.
RE1 are forecasts from a regression model of R&D and a lagged dependent variable.

Table 30 cont'd ../

Oth Bloc	Actual	AR3	RE1	Oth Tril	Actual	AR3	RE1
Argen	55	51	55	Argen	55	51	54
Aus	1300	1339	1421	Aus	1300	1307	1394
Aut	586	589	621	Aut	380	402	427
Bel	179	237	273	Bel	107	166	193
Braz	120	150	168	Braz	120	149	166
Bulg	328	489	307	Bulg	328	479	307
Can	851	1271	1406	Can	851	1257	1377
China	338	319	320	China	338	312	318
Den	818	685	687	Den	459	428	432
Fin	1309	1138	1169	Fin	767	765	799
Fra	5247	5167	5430	Fra	4114	4273	4478
Ger	16949	16023	15894	Ger	13967	13622	13609
Gree	32	35	39	Gree	12	16	18
Hung	72	111	101	Hung	72	109	101
India	110	88	91	India	110	85	91
Ire	190	156	165	Ire	107	89	96
Israel	570	563	623	Israel	570	548	610
Italy	1794	1875	1913	Italy	1177	1364	1392
Japan	44538	45411	44333	Japan	41400	42476	41463
Korea	4683	5243	5157	Korea	4683	5175	5066
Malay	14	19	16	Malay	14	20	16
Mex	35	31	33	Mex	35	31	32
Neth	936	919	1004	Neth	520	603	670
NewZ	195	193	191	NewZ	195	189	188
Nor	351	313	326	Nor	351	310	320
Poland	30	36	36	Poland	30	37	35
Port	19	15	19	Port	12	10	13
Romania	4	7	7	Romania	4	7	7
Russia	283	225	213	Russia	283	226	211
SAfrica	291	263	250	SAfrica	291	259	248
Sing	146	115	127	Sing	146	114	125
Spain	465	455	497	Spain	231	241	272
Swe	2196	2269	2335	Swe	1381	1637	1709
Switzer	843	992	1014	Switzer	609	801	827
Tur	398	331	313	Tur	398	327	310
UK	7564	6994	6773	UK	5197	5236	5097
USA	53982	50037	53144	USA	42636	38201	41378
Sum	147821	144154	146470	Sum	123250	121321	123848
RMSPE		0.245	0.267	RMSPE		0.264	0.297

Table 30 cont'd ../

Tril	Actual	AR3	RE1
Argen	6	6	6
Aus	186	360	384
Aut	123	174	179
Bel	46	75	91
Braz	16	37	42
Bulg	0	1	1
Can	97	176	225
China	54	80	68
Den	170	209	195
Fin	214	345	350
Fra	1537	2137	2122
Ger	5267	6582	6040
Gree	2	6	6
Hung	6	34	27
India	10	6	8
Ire	25	24	28
Israel	91	146	168
Italy	314	530	529
Japan	17119	20770	19574
Korea	834	1245	1210
Malay	2		
Mex	3	2	3
Neth	172	243	274
NewZ	24	43	44
Nor	33	62	69
Poland	1	6	7
Port	2	3	4
Romania	0	1	1
Russia	38	56	48
SAfrica	25	46	38
Sing	13	23	23
Spain	60	80	95
Swe	496	862	833
Switzer	274	472	460
Tur	1		2
UK	1549	2290	1990
USA	24058	23800	24875
<i>Sum</i>	<i>52868</i>	<i>60933</i>	<i>60019</i>
<i>RMSPE</i>		<i>0.812</i>	<i>0.859</i>